Decentralized Crypto World Bank

A Blockchain-Based Lending Platform
with AI-Enhanced Security

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by

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A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science

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Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at BRAC University.
2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Approval

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Ethics Statement

This project is designed to be functional on public blockchain testnets, with no intention for monetary profit, nor to create institutional leverage over decentralized financing. No personal banking data, or private financial data, or codes of existing projects, are used in the development of the prototype platform and training AI/ML data. The full project prototype build is to be demonstrated in the final defence of the project; the platform is expected to be ready for testing and troubleshooting by pre-thesis 2. All the development workflow can be traced by the official GitHub repository for the project, and the planning, coding, testing, and usage of different dependencies for this project is considered open-source with MIT licensing, and will be open for contribution after the submission of the final build. For the training of the AI models, AI-assisted financial decision-making, risk assessment, privacy features, on-chain and off-chain data processing and data storing, we acknowledge the ethical considerations while making decisions for the development of the project.

Abstract

In the modern world, global financing relies on institutional structures that are multilayered and complex that extend throughout the borders of nations, adapting complex protocols to maintain balance in the economy; however, these systems often struggle to maintain transparency and equal access to credit across the world. Decentralized financing has improved transparency in financing by adapting programmable infrastructures that can automate exchanges or structured governance, but the existing models use mostly flat architecture that does not match with the traditional institutional hierarchies or governance that is well structured based on global economic contexts. This creates a scope to study and explore how tier-based financial systems that mimic the functionalities, but in a decentralized environment with the flexibility and adaptability. As an attempt to address this opportunity and to demonstrate what is possible, we present our research-based project *Crypto World Bank*, a formally specified prototype which is based on blockchain-based institutional finance, that includes coordinating capital allocation across multi-tiered hierarchical governance structures. The primary focus of the project is a smart-contract that includes a hierarchy with a pipeline of oracle-based risk-analytics (Random Forest, Isolation Forest, SHAP, which will be trained and benchmarked with existing similar systems); an optional intelligent MCP agent as a feature to interact with the UI with more comfort, while all the core banking systems remain intact by using wallets. The banking architecture includes risk intermediation, liquidity management, deposit mobilization, credit allocation, payment settlement, and other adjacent financial services. The development workflow including frontend, smart contract, oracle integration and overall system architecture and design is specified in this paper; implementation and testnet deployment are planned across four phases in the final stages. The goal of this project paper is to illustrate how a hybrid system of design of decentralized system can converge with institutional systems, to set a foundation for research on how a hierarchically governed crypto financial system can be developed.

Keywords: Blockchain, Decentralized Finance, Deposit Mobilization, Explainable AI, Financial Sustainability, Group Lending, Hierarchical Lending, Institutional DeFi, Multi-Entity Co-Lending, Oracle-Mediated Risk Analytics, Reserve Management, Smart Contracts

Dedication

Dedicated to our families for their unwavering support throughout our academic journey.

Acknowledgment

We would like to express our sincere gratitude to our panel members for their guidance and support throughout this project. We also thank our supervisor, Mr. Annajiat Alim Rasel, Senior Lecturer at the Department of Computer Science and Engineering, BRAC University, for his guidance and support. Finally, we thank the Department of Computer Science and Engineering at BRAC University for providing us with the resources and academic environment to pursue this work.

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Utilization Rate (U): $U = \frac{L}{L+B}$

Collateral Ratio (CR): $CR = \frac{C}{D}$

Loan-to-Value (LTV): $LTV = \frac{\text{Loan Amount}}{\text{Collateral Value}}$

APR (simple): $APR = r_{\text{period}} \times N$

Compound Interest (Savings): $A = P\left(1 + \frac{r}{n}\right)^{nt}$

EMI (Installment Payment): $EMI = \frac{P \cdot r \cdot (1+r)^n}{(1+r)^n - 1}$

Health Factor (HF): $HF = \frac{C \times LT}{D}$

Reserve Ratio (RR): $RR = \frac{\text{Reserves}}{\text{Total Deposits}}$

Platform Solvency Ratio (PSR): $PSR = \frac{\text{TotalReserveBalance}}{\text{TotalOutstandingLoans}}$

Credit Velocity (CV): $CV = \frac{\text{Capital Recycled per Period}}{\text{Total Reserve}}$

FX Conversion: $\text{Amount}_B = \text{Amount}_A \times \left(\frac{P_A}{P_B}\right)$

Group Loan Individual Share: $\text{Share}_i = \frac{\text{LoanTotal}}{N_{\text{members}}}$

On-Chain Credit Score: $CS = \sum_i w_i \cdot \text{feature}_i$

Isolation Forest Anomaly Score: $s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$

Random Forest Fraud Probability: $P(\text{fraud} \mid x) = \frac{1}{T} \sum_{t=1}^{T} f_t(x)$

Platform Net Interest Allocation: $\text{NetInterest} = \text{InterestCollected} - \text{DepositorYield} - \text{InsuranceAllocation}$

SHAP (Shapley value): $\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[f(S \cup \{i\}) - f(S)\right]$

List of Abbreviations

AA: Account Abstraction (ERC-4337)

ALM: Asset-Liability Management

AI: Artificial Intelligence

AI/ML: Artificial Intelligence / Machine Learning

AML: Anti-Money Laundering

APR: Annual Percentage Rate

API: Application Programming Interface

BRAC: Bangladesh Rural Advancement Committee

CAR: Capital Adequacy Ratio (Basel III; not directly applicable to this platform — see Platform Solvency Ratio, PSR, in List of Formulas)

CBDC: Central Bank Digital Currency

CCIP: Cross-Chain Interoperability Protocol (Chainlink)

CCS: ACM Computing Classification System

CR: Collateral Ratio

CVL: Certora Verification Language

DeFi: Decentralized Finance

DID: Decentralized Identifier (W3C)

DON: Decentralised Oracle Network (Chainlink)

EIP: Ethereum Improvement Proposal

EIP-7702: Ethereum Improvement Proposal 7702 — Session Key standard enabling scoped, time-bound wallet authorisations for an autonomous agent

EMI: Equated Monthly Installment

ERC: Ethereum Request for Comments (token / interface standard)

EVM: Ethereum Virtual Machine

FL: Federated Learning

FATF: Financial Action Task Force

FX: Foreign Exchange

GNN: Graph Neural Network

HF: Health Factor

KYC: Know Your Customer

LLM: Large Language Model

L2: Layer 2

LTV: Loan-to-Value Ratio

MFI: Microfinance Institution

MiCA: Markets in Crypto-Assets Regulation (EU)

ML: Machine Learning

MCP: Model Context Protocol — tool-calling interface used by the AI agent to interact with CWB banking operations

NIM: Net Interest Margin

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PoS: Proof of Stake

PoR: Proof of Reserve (Chainlink on-chain reserve verification)

PSR: Platform Solvency Ratio (on-chain capital adequacy analogue; see List of Formulas)

QRC: QR code

RBAC: Role-Based Access Control

RLS: Row-Level Security (PostgreSQL policy feature enforcing append-only access on audit tables)

RR: Reserve Ratio

SAR: Suspicious Activity Report

SBT: Soulbound Token (non-transferable ERC standard)

SHAP: SHapley Additive exPlanations

SOFR: Secured Overnight Financing Rate

SSE: Server-Sent Events

SSI: Self-Sovereign Identity

SoK: Systematization of Knowledge

SWC: Smart Contract Weakness Classification

TVL: Total Value Locked

U: Utilization

UUPS: Universal Upgradeable Proxy Standard (ERC-1822)

VC: Verifiable Credential (W3C)

XAI: Explainable Artificial Intelligence

ZKP: Zero-Knowledge Proof

zkAML: Zero-Knowledge Anti-Money-Laundering proof

zkEVM: Zero-Knowledge Ethereum Virtual Machine — a ZK rollup that inherits Ethereum L1 security via cryptographic validity proofs

zk-SNARK: Zero-Knowledge Succinct Non-Interactive Argument of Knowledge

Chapter 1

Introduction

The *Crypto World Bank* is a research-based project for exploring how blockchain technology can be shaped to fit into institutional financial systems with all the added benefits that come with a decentralized financing system. Based on our findings that had similar motivation to our study, existing DeFi protocols show isolated functionalities, such as Aave for lending, Uniswap for exchange, and MakerDAO for stablecoins. This project specifies a tier-based, hierarchically governed platform that includes a mixture of institutional financing with DeFi, accommodating deposit mobilization, credit allocation, interbank liquidity, installment repayment, FX facilitation, solidarity group lending, reserve enforcement, syndicated co-lending, and risk-based governance covering all four tiers that systematically mirrors real-world institutional financing. At the top, Tier 1 represents the World Bank reserve; the core banking hierarchy is specified in Solidity and being improved until final deployment. Another significant feature—the multi-entity contract suite—is fully specified in Chapter 3, which is scheduled for practical development in Phases II–III.

The system is designed for three categories of participant: a programmable World Bank reserve as Tier 1, national and local development banks serving in Tiers 2–3, and retail clients in Tier 4. Retail clients get access to savings, group loans, and installment lending features. This hierarchical structure in this project was motivated by unbanked and MSME financing gaps [9–10]. The project is not built with any motivation to deploy only in regions with access to advanced technologies; rather, proper distribution of access to finance and fairness is the leading cause of this attempt to find suitable ground to adapt DeFi among the general population. The whole project is open source and available in our GitHub repository. The added benefits of blockchain: it provides state visibility, tamper-evident audit trails, and programmable rule enforcement among institutions having competition, as demonstrated by the renowned World Bank FundsChain programme and JPMorgan Kinexys. From the study of institutions, open ledger infrastructure and institutional hierarchy are not mutually exclusive.

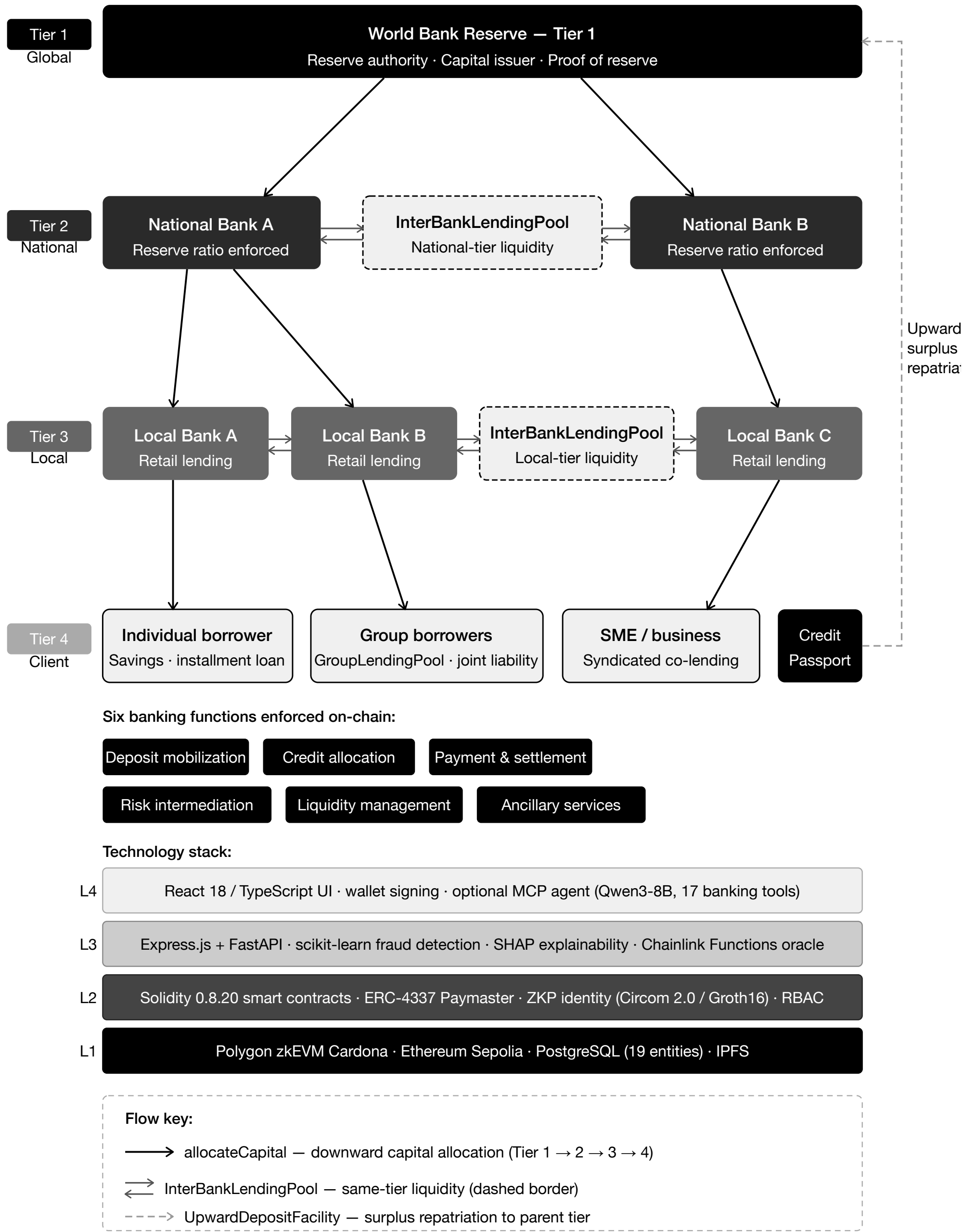

Figure 1.1: Crypto World Bank system overview

1.1 Background

Local lenders, or our target users, for this aspect of lending through layered institutions—supranational bodies that is provided by development finance channels have structural costs. Idle capital is usually locked by correspondent banking in pre-funded nostro/vostro accounts [114], cross-border transactions that take around two to five days cost roughly \$42 per transaction [19], charges for remittance corridors average 6.49%, which is above the UN SDG target of 3% [20]. Due to unequal access to traditional banking systems, 1.4 billion adults remain unbanked [9] and MSMEs face a \$4.5 trillion annual financing gap [15].

In decentralized financing, lending, collateral, and interest can be automated by enabling rules on-chain transparently. However, institutions like Aave v3 that holds $26.3 billion TVL [21] and the broader DeFi lending market exceeds $55 billion [8], use flat pool design that ignore institutional hierarchy, inter-tier exchange, and accessing clients via local bank lend path which is more generalized for normal people with less knowledge about DeFi.

Contribution of smart-contract and blockchain. Smart-contracts can develop lending rules as immutable, atomic, with auditable on-chain logic. This technology replaces the bilateral ledger reconciliation with a system of shared record, when a local bank initiates a loan, the World Bank's reserve ratio updates on the same transaction status [19]. DeFi extends this feature to financial services without licensed custodians. Here in The *Crypto World Bank* adds three dimensional feature that is missing from the existing institutional protocols: institutional hierarchy (four-tier capital flows), multi-entity operations (solidarity groups, syndicated loans, tranched pools), and a connection to the evolving development finance framework identified as absent in the literature [1].

Optional conversational interface. Every user interacts with this system primarily through the React UI and wallet credential signing. For development purposes and demonstration, a locally hosted agent (Qwen3-8B + MCP, 17 banking tools) provides an alternate path to get the essential tasks done, including all the customer services that require system interaction information guidance and assistance in acquiring a loan or exchange finances with the system. In the final build it is planned to be replaced with secured API-based services.

Technical foundations. **Consensus.** The chain is an append-only ledger: nodes agree on history, and no one can rewrite past blocks without controlling most of the network [63].

EVM execution [50]. Loan logic runs as deterministic bytecode on Ethereum-compatible chains. Storage writes are the main cost driver (Section 5.1).

State machines. Each loan moves through clear stages—`PENDING` $\rightarrow$ `APPROVED` $\rightarrow$ `ACTIVE` $\rightarrow$ `REPAYING` $\rightarrow$ `COMPLETED/DEFAULTED`—with only authorised roles allowed to advance it.

Oracles [55, 95]. Contracts cannot read off-chain data on their own. ML risk scores reach the chain via a commit-reveal oracle (Section 3.3.2); Chainlink Feeds, Automation, and Proof of Reserve handle prices, due-date checks, and reserve attestation. Live ML wiring starts after pre-thesis 2.

Polygon zkEVM. We deploy on Cardona testnet (Section 3.2) for low fees and L1-backed validity proofs.

1.2 Rationale and Motivation

Real development finance depends on layered institutions, yet settlement friction, opaque reserves, and correspondent-banking idle capital leave both intermediaries and unbanked clients underserved [20, 114]. Flat DeFi protocols automate lending at scale but ignore institutional hierarchy, cross-tier flows, and the Local-Bank-to-client path that microfinance and development banks use in practice [1]. This study is motivated by the opportunity to encode that hierarchy on a programmable ledger—with enforced reserves, role-based governance, and auditable capital movement—while supporting (not replacing) human credit judgment through lightweight analytics and an optional conversational interface. The thesis specifies a reusable architecture; it does not claim a particular national deployment target.

1.3 Problem Statement

Three primary forces of motivation for this study: development-finance has restricting transparency and settlement; the absence of multi-tiered banking functions in DeFi (deposits, solidarity and lending); the keen interest in combining blockchain auditability with ML computational analytics. There are six structural inefficiencies noticeable in development finance routes for individual borrowers from national and local banks:

1. **Lack of transparency:** there is very little chance of the lower tier users to verify capital management, reserve levels or lending decisions are made by the higher tiers, very less transparency on how money flows through the different levels of systems. Only the top tiers get the information and records of reserves and audited at most quarterly.
2. **Sluggish settlements and idle capital:** cross-border transactions are quite expensive, average $42 per transaction [19], and time consuming taking two to five days [114].
3. **Inconsistent risk evaluation:** fraud and credit assessment are completely relied on human judgement, borrowers are generally re-evaluated at each tier or institutions based on information available at that tier, no unified credit history.
4. **Access and trust barriers:** the cost of inter-institutional trust is quite expensive for legal and audit process. It creates congestions in developing countries where investors earn returns roughly 3% below compared to developed markets [22].
5. **No programmable savings:** depositors in developing economies have no real-time access to visibility into how savings are processed and deployed.
6. **No on-chain group credit:** without individual collateral, borrowers cannot access programmable on-chain mutual group lending with shared responsibility. Real-world organizations like BRAC exist but does not function like DeFi.

DeFi enabled automatic lending transparency at scale such as Aave v3 with $26.3 billion TVL, $17.7 billion borrows [21], Compound v3 ($1.4 billion) [23], MakerDAO with $10.5 billion stablecoin issuance [24]. Despite large market coverage they exhibit four limitations for institutional development finance:

- **Flat lending models.** Aave/Compound uses one protocol for many asset pools (USDC, ETH, etc.). There is no cross-tier allocation, no institutional hierarchy, no difference in role-based governance. Here our CWB models institutional relationships that go deep into what kind of services to provide to what kind of roles/institutions to increase liquidity in overall lending and exchange situations.
- **No AI risk analytics.** DeFi relies on collateral ratio, utilization curves. There is no behavioral fraud detection or explainable credit support system.
- **No tiered governance.** DeFi protocol treats tokens as a measurement of voting, not based on what is the role of the client or user or institution in the bigger to smaller picture as it lacks relation to a deeper level like institutions.
- **Single-level institutional DeFi.** Maple & Goldfinch [19–20] with $2.6–3.8 billion TVL and $680 million originations in 18+ countries respectively serve institutions, but

none models a full multi-tier banking system with interbank flows, lack hierarchical capital flows.

1.4 Objectives

The objectives of this project are:

1. Design and specify all features and constraints of a four-tier lending architecture on an EVM-compatible blockchain and complete a full testnet deployment (World Bank → National Bank → Local Bank → Client). The whole development cycle is planned for Phases I–IV.
2. Design not only pool lending, but specify an extended banking product suite on the hierarchical system (deposits, savings, and solidarity group lending).
3. Investigate a lightweight off-chain analytics layer using ML, training and testing the models in practical testnet for fraud detection, anomaly detection, explainable review, and an optional MCP-based conversational interactive agent for ease of use.
4. Specify programmable on-chain lending with borrowing limits based on financial behaviour analysis, installments, and role-based access control and information via smart contracts.
5. Plan and execute testnet deployment using Polygon zkEVM Cardona, Ethereum Sepolia, and evaluate hierarchical controls in the final thesis phase.
6. Document governance for future improvements, security analysis and feature improvements, role-based regulatory considerations, and a controlled rollout toward sandbox piloting.

These objectives are tested through five research questions: **RQ1:** Does a hierarchical blockchain lending architecture reflect development-finance capital flows more faithfully than flat DeFi protocols? **RQ2:** Can on-chain transparency, reserve controls, and role-based governance reduce opacity and friction across tiers? **RQ3:** Does a lightweight off-chain analytics layer offer practical, auditable fraud and lending support? **RQ4:** Can the architecture deploy on public testnets with the RBAC, capital-flow, and reserve properties required for institutional banking? *Scope: prototype viability only—not production readiness or regulatory compliance.* **RQ5:** Can deposit-funded and solidarity group lending be represented as programmable, auditable on-chain mechanisms feasible for retail users?

1.5 Research Contribution

Four things this thesis actually adds—each with honest limits on what is built vs. specified. **C1—Four-tier lending architecture:** Most DeFi lending is flat; we model development-finance-style capital flow across four tiers with reserve checks and on-chain roles for World, National, Local, and retail clients [1, 37, 90] (*built now:* contract interfaces and workflow spec; *later:* testnet deploy, Phases I–II). **C2—Group lending on-chain:** Solidarity-style group loans with shared consent, split installments, and over-borrow caps are specified as EVM contracts (Section 3.16.3; PoC in Phase II). **C3—Explainable ML through an oracle:** Fraud and anomaly scores leave the ML service, get explained with SHAP, and land on-chain before an approver can sign (Section 4.3.2; training after pre-thesis 2). **C4—Privacy-preserving KYC path:** Prove eligibility with zk-SNARKs instead of posting identity on-chain—KYC circuit specified now; fuller AML circuit left for Future Work. We are not pitching a bank killer: institutions care about legal enforceability, not just clever contracts [65]; CWB is a composable layer for development finance and MFIs (Section 1.8), not a replacement for correspondent banking overnight. Section 1.9 maps where each topic introduced in this chapter is expanded across Chapters 2–6 and the appendices.

1.6 Proposed Solution

The Crypto World Bank will implement a four-tier on-chain lending model: Tier 1 World Bank reserve custodian → Tier 2 National Banks → Tier 3 Local Banks processing retail loans → Tier 4 clients building on-chain credit history (Figure 1.1), with six standard banking functions enforced on-chain—deposit mobilization, credit allocation, payment and settlement, risk intermediation, liquidity management, and ancillary services—rather than discretionary audit alone. Banks also borrow from peers, park surplus with parent tiers, and syndicate large loans; the platform replicates four-tier downward flow, same-tier pools, upward deposits, and multi-bank co-funding on-chain (Figure 1.2).

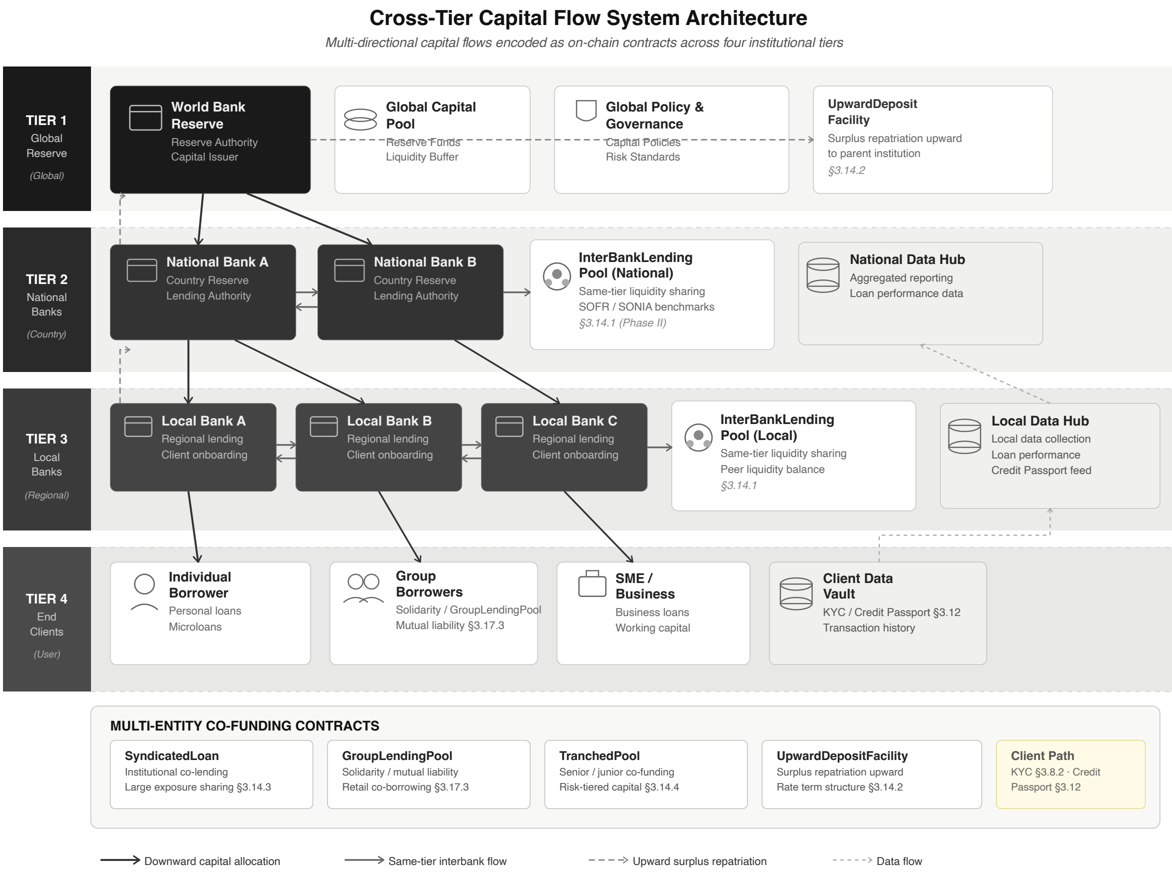

Figure 1.2: Cross-tier lending flow directions

Can a group of entities lend or fund together? Syndicated loans, solidarity groups, tranched pools, and upward surplus deposits (Section 3.14) coexist with same-tier peer liquidity pools (Section 3.14); retail clients enter through Local Banks via KYC, Credit Passport, and borrowing limits, with syndication reserved for large loans.

1.7 Methodology in Brief

There are four Agile phases to this methodology for this project. This report is designed with the architecture specification. The development phase will start from Phase II after pre-thesis 1. In this report there is architecture, contracts, database, and ML pipeline. The Phase II development phase will demonstrate the loan cycle. Phase III will start the implementation of ML scoring and the Chainlink oracle (Section 3.3.2). Phase IV will show the evaluation, metrics, and the overall feasibility of the project in the real world. Stack: Solidity, React, Express/FastAPI, scikit-learn on Polygon zkEVM Cardona. There is also an AI chat agent that will be built as an optional feature that will run alongside the project to help the clients navigate better with the system.

1.8 Scopes and Challenges

In scope now. In our development phases, we have calculated that we can implement a limited amount of our full vision: the four-tier lending architecture, loan lifecycle workflow, borrowing limits for the clients and the banks, the design of Oracle fraud analytics, and the training and the live demonstration of the AI/ML workflow. It can be shown to a limited amount on the prototype builds only.

Out of scope for now. Mainnet deployment, live retail payments in the real world, production KYC/AML at the banking level and client interaction, stress testing the whole project, and providing an MVT checklist before the product is ready is not quite possible yet.

Main risks. There are a couple of risks that we have noticed for our project, starting with the fraud labels may not be completely accurate based on the limited amount of data we have for our system architecture. We need more synthetic data that will be collected over time. If regulations have changed—mitigated by testnet-only builds—then we might have to change the architectural elements. The gas toll on micro-loans may not be sustainable. The ML layer with SHAP may keep borrowers in the loop in the system since it is not 100% accurate and accuracy of the system depends on the training process. Rest of the economics and trade-offs are discussed in Chapter 5.

Economic sustainability. On Cardona, 10–15 txs on a typical loan cost under $0.15—small next to interest earned. Mainnet micro-loans would need tighter gas work.

Institutional capture. Another huge issue that needs to be addressed is that Tier 1 and the National Banks, which have huge reserves and influence over the rates, reserves, and who gets credit, might influence the overall system and cost-to-performance ratio. It might temper the trust and relationship with the retail customers and smaller banks.

Stablecoin-first lending. Retail clients are not to be carrying high-priced positions like ETH, since the prices of Ethereum might fluctuate. MiCA and the U.S. GENIUS Act support the idea that stablecoins should be more generalized for authorized cryptocurrency. There are many settlement projects like *mBridge* and *Agora* [103, 110], moving the value of cryptocurrencies across borders, but it is not a replacement for our lending system. The architectural system for stablecoin pools and L2 deployment (Section 3.13) gives the liquidity move fluidly from the top tier to the bottom, and the surplus to go up from the bottom.

1.9 Thesis Organization

- **Ch. 1 — Introduction:** problem, objectives, approach (**CO1**).
- **Ch. 2 — Literature:** PRISMA review and protocol comparison (**CO9**).

- **Ch. 3 — Architecture:** tiers, contracts, data model, identity, products, security (**CO2–4**, **CO12**).
- **Ch. 4 — Methodology:** build phases, MVT scope, ML oracle, optional agent (**CO5**, **CO10**).
- **Ch. 5 — Evaluation:** feasibility, benchmarks, limits (**CO6–7**, **CO11**).
- **Ch. 6 — Conclusion:** contributions and future work; appendices hold contracts and deployment notes.

Chapters 3–5 already carry the detailed architecture, methodology, and feasibility analysis for this pre-thesis 1 submission; richer implementation evidence, measured benchmarks, and updated evaluation will be added to those chapters after pre-thesis 1, in the final thesis stage. Hard numbers—testnet deploy, ML benchmarks, formal proofs—are scheduled for pre-thesis 2.

Chapter 2

Literature Review

2.1 Preliminaries

Core terms—DeFi, smart contracts, over-collateralization, XAI, PoS, CBDC, solidarity/group lending, ALM, flash loans, ZKP, and Health Factor—are defined in the **Glossary** appendix (Appendix D in the final thesis assembly; the current pre-thesis build temporarily carries the WorldBankReserve interface as Appendix D). Platform-specific terms (*Platform Solvency Ratio*, *Provisional Credit Tier*, *Cold-Start Credit Pathway*) appear there as well. Readers may skip to Section 2.2 if already familiar with blockchain fundamentals (Section 1.1).

2.1.1 Review Methodology (PRISMA-Style Frame)

PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) is a standard way to search, screen, and report which studies made it into a review—so readers can see what was included, what was dropped, and why. We use a PRISMA-*style* frame here: the same transparent screening steps, applied to design literature rather than a clinical meta-analysis [66].

We followed that frame, not a narrative summary alone. Sources had to be peer-reviewed or institutional (2017–2026), relevant to at least one CWB topic (lending, ML fraud, identity, microfinance, security, regulation, or payments), and clear enough to reuse (metrics, reserve rules, gas figures, F1 scores).

We searched ACM, IEEE, Elsevier, MDPI, arXiv, and BIS/World Bank with keywords around blockchain, DeFi, lending, fraud, ZKP, and development finance.

Table 2.1 ranks the top ten; Figure 2.1 shows screening counts. PRISMA-based banking reviews [17] use the same standard.

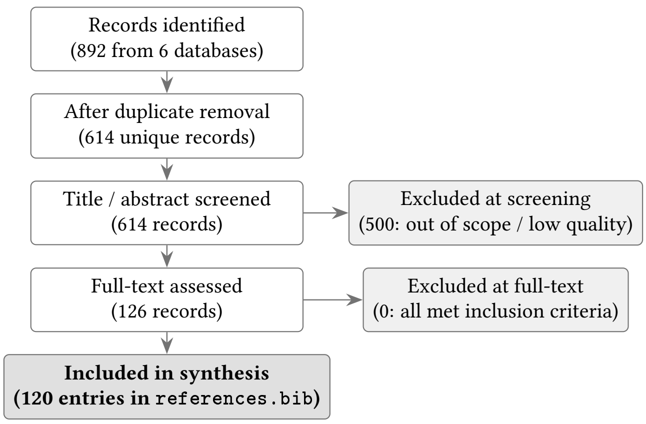

Figure 2.1: PRISMA-style literature screening flow

Table 2.1: Top-10 ranked literature sources

#	Source / Module / Operationalized In	Headline Finding
1	Werner et al. 2022 [1] · DeFi mechanics · Ch. 3	DeFi lending is uniformly flat, pool-based, overcollateralized
2	Tan 2023 (IMF) [5] · CBDC & hierarchy · Ch. 3	Two-tier CBDC distribution (central bank → commercial banks → users); hierarchical infrastructure is the key design choice
3	Palaiokrassas et al. 2024 [3] · Fraud detection · Sec. 4.3	Multi-chain ML on 54M tx, F1 0.76–0.85 with behavioral features
4	Adom et al. 2022 [4] · Explainable AI · Sec. 4.3	SHAP outperforms LIME on lending datasets
5	Weber et al. 2019 / Elliptic [85] · Graph fraud · Sec. 4.5	Graph-structured fraud signal in BTC transactions
6	McMahan et al. 2017 / FedAvg [68] · Federated learning · Sec. 4.6	Privacy-preserving model averaging across institutions
7	BIS WP 905 [61] · Stablecoin risk · Secs. 1.8, 5.4.4	Algorithmic stablecoins unsuitable for EM DeFi; USDC/USDT recommended
8	Atzei et al. 2017 [7] · Smart contract security · Sec. 3.18	SoK of EVM attack vectors
9	EU MiCA + GENIUS Act [75, 77] · Regulation · Sec. 5.4.4	Stablecoin issuer authorization, reserve disclosure
10	Bangladesh Bank SP2025-02 [80] · MFI precedent · Sec. 3.16.3	Female-to-male account parity 49%/49% in 2025; 89% BRAC loans to women

2.2 Review of Existing Research

This part of the paper explores all the literature chosen through a PRISMA-style screening flow to select the papers that best fit our project. We surveyed existing work related to DeFi lending, ML fraud detection and explainable AI, hierarchical development finance and cross-tier finance flows, group or microfinance, smart contracts and gas economics, retail banking and payments, remittance, monetary-policy distributional effects, real-world asset tokenization, stablecoin regulation and institutional settlement, and how inference can be achieved in systems like ours.

2.2.1 Decentralized Lending and DeFi Protocol Design

This is one of the most important parts of the paper: how decentralized financing works for lending, and how DeFi protocols are designed as public smart contracts. Utilization curves set interest rates; oracles trigger liquidations on collateral; settlement stays atomic across currencies. The model's real strength is real-time auditability—per-block interest and transparent on-chain state. It also comes with structural limits for development finance: capital still flows one way through flat pools, with no inter-tier allocation or multi-entity co-funding—a crucial gap for our project.

Werner et al. [1] summarize DeFi as homogeneous, pool-based, and over-collateralized, with no institutional hierarchy. Bastankhah et al. [2] show adaptive fast/slow rate control outperforming static Aave/Compound curves [51], which informs our kinked-rate model (Section 3.9). Chen et al. [39] test multi-chain lending layouts to handle higher throughput. Yang and Cui [40] propose practical metrics—utilization stability, liquidation efficiency, and rate responsiveness—for judging how well a lending stack performs. Hartmann and Hasan [41] look at gas cost and digital identity in peer-to-peer lending. Dao et al. [52] map traditional credit scoring onto DeFi borrowing limits.

2.2.2 Machine Learning, Explainable AI, and Extended AI Security

Palaiokrassas et al. [3] achieve F1 0.76–0.85 on 54M multi-chain DeFi transactions using behavioral features versus 0.08 with raw transaction data alone—direct input to Random Forest feature engineering. Liu et al. [6] supply Isolation Forest for unsupervised detection when labeled fraud is scarce. Hasan et al. [42] train fraud classifiers via federated learning on encrypted features, informing cross-bank intelligence without raw data sharing.

Adom et al. [4] find SHAP more consistent than LIME for loan explanations; Yuan [49] reaches AUC >0.92 with SHAP on credit-default models—motivating SHAP waterfall review before approvals. Beyond the core pipeline, graph features from Palaiokrassas et al. [3] support GNN ring detection; federated learning [42] suits National/Local Bank boundaries; ContractWard [43] and SoliAudit [44] enable real-time exploit monitoring; RL rate tuning and NLP income extraction remain future-work options.

2.2.3 Hierarchical Distribution, Cross-Tier Flows, and Group Lending

Tan [5] models two-tier CBDC distribution (central bank $\rightarrow$ commercial banks $\rightarrow$ users), paralleling the four-tier hierarchy. BIS [46] prefers two-level retail CBDC with interoperability constraints, informing the dual-currency facility. Asaduzzaman et al. [47] report up to 60% admin-cost reduction from smart-contract microcredit—literature precedent for `GroupLendingPool`, not a deployment-jurisdiction claim.

Policy literature establishes hierarchical distribution but sparse programmable *multi-directional* coordination. Kiff et al. [106] distinguish multi-tier CBDC with supervised intermediaries; Bindseil [105] analyses tiered remuneration for `UpwardDepositFacility` rate spreads (Section 3.14); Auer and Böhme [107] hybrid CBDC mirrors client-via-Local-Bank access with Tier 1 reserve authority. *Project Pine* [103] validates layered contract factories for reserve and collateral dependencies extended by `InterBankLendingPool` and `SyndicatedLoan`. Dong et al. [104] model cross-tier direct financing in supply chains—parallel to downward allocation to capital-constrained retail clients (Figure 1.2).

Grameen/BRAC solidarity groups achieve $>95\%$ repayment through social monitoring that on-chain mutual liability cannot fully replicate; `GroupLendingPool` encodes multi-signature consent and pool claims at pre-thesis stage. Güçük et al. [53] review privacy and trust in blockchain microlending.

2.2.4 Smart Contract Security and Layer 2 Scalability

Atzei et al. [7] catalogue reentrancy, access-control, and overflow vectors—motivating OpenZeppelin guards, Solidity 0.8.20, RBAC, Slither, Mythril, and Phase IV formal verification. Wang et al. [43] (ContractWard) reach F1 0.96/0.93 on bytecode vulnerability classes; Liao et al. [44] (SoliAudit) cut audit time ≈70% via ML-prioritized fuzzing; So et al. [45] (VERISMART) report 98.4% precision with zero false positives on arithmetic safety—candidate for reserve-ratio invariants.

Kim and Kim [48] show batching and calldata compression save 30–50% mainnet gas, supporting Polygon zkEVM Cardona deployment where multi-step four-tier lifecycles multiply savings.

2.2.5 Correspondent Banking, Retail Payments, and Settlement Rails

BIS [16] and FSB [27] document nostro/vostro idle capital, MT103 routing, and 2–5-day settlement. World Bank remittance data [20] average 6.49% on a $200 transfer (Sub-Saharan Africa >8%). Cerutti et al. [116] and Du et al. [110] show digital and stablecoin rails can undercut legacy costs but depend on corridor off-ramp depth—scoped separately (Section 6.1).

Retail literature examines stablecoin remittance, merchant checkout, and on/off-ramps for post-MVT modules (`CurrentAccount`, `P2PTransfer`). Ante [108] finds 26% U.S. remittance-user stablecoin adoption, supporting tiered KYC gating. Du et al. [110] document the fiat–stablecoin–fiat "sandwich" pattern. Li et al. [109] (**CLEAR** framework) find programmable settlement advantages but weak open-loop dispute recourse; Zhang et al. [112] warn on refund volatility; Fröhlich et al. [111] show on-ramp friction dominates POS UX. Lin et al. [113] (*SecurePay*) and BIS *Project Sela* [119] validate Local-Bank–sponsored multi-intermediary flows; Cimaszewski et al. [118] blueprint jurisdictional DLT settlement. CPMI [114], BIS Papers 170 [115], and FATF [117] define ramp and VASP compliance for platform-mediated `P2PTransfer`—not permissionless unhosted-wallet remittance. No prior work composes four-tier development-finance lending with closed-loop retail rails in one architecture.

2.2.6 Monetary Policy, Tokenization, and Institutional Landscape

Cross-country evidence links QE to wealth inequality [28]; Federal Reserve metro analysis [29] shows low-income harm from tightening—motivating algorithmic, transparent rate and reserve parameters rather than opaque committee discretion. Tier-uniform utilization rates reduce loan-officer price discrimination; the Cantillon effect does not apply where USDC is fully reserved and the platform creates no new money [30].

R3 Corda reports $17B tokenized RWAs [31]; Centrifuge exceeds $1B across eight chains [32]; World Bank FundsChain tracks disbursements on 13 projects scaling to 250 [33]—institutional precedent for hierarchical on-chain finance beyond fund tracking alone.

Sygnum [65] notes institutional allocators require legal enforceability (Aave Arc ≈$50k TVL). mBridge [78] and *Project Agora* [79] are settlement rails, not retail lending hierarchies; 94% of central banks research CBDCs [71]. CWB positions as a lending-layer specification compatible with these rails.

2.2.7 Graph ML, Federated Learning, Regulation, and Identity Primitives

DeFi fraud is relational: Cheng et al. [93], Chang et al. [93], and DeFiGuard [67] outperform flat ML on graph structure—motivating a GNN extension with hierarchical tier edges (Section 4.5). Federated learning [42, 97] and PrivChain-AI (94.7% accuracy) suit banks that cannot share raw records; FedAvg aggregation at National tier is architecturally natural (Section 4.6).

MiCA [75, 76] and the U.S. GENIUS Act [77] anchor stablecoin-first retail design (Section 1.8; mapping in Section 5.4.4). Buterin et al. [69] introduce non-transferable SBTs for portable credit passports [70] (Section 3.12).

2.2.8 Oracle-Mediated ML and Optional LLM Interfaces

Luo et al. [94] survey AI fraud detection across the DeFi lifecycle; Caldarelli [95] identifies ML-score oracle delivery as under-researched beyond price feeds. BCCC-DeFiFraudTrans-2025 [83] (1,026,867 labeled transactions) is the planned primary training corpus after pre-thesis 2 (Section 4.3.1); Fazliani et al. [96] and Alhasawi et al. [97] supply semi-supervised and federated baselines.

The optional conversational add-on uses an 8B open-weight LLM with QLoRA, RAG, and refusal layers. Finance LLM studies [72, 73, 74] report productivity gains but document regulatory-hallucination risks—motivating the red-team protocol in Section 4.13.1.

2.3 Literature Review Summary

Tables 2.2 and 2.4 consolidate the reviewed works by methodology, findings, and project relevance. Extended six-part synthesis tables are maintained in the thesis repository [17].

Table 2.2: Literature review summary (Part 1 of 2)

Author & Focus	Methodology	Key Findings	Relevance to Our Project
Palaiokrassas et al. (2024) [3] · DeFi fraud	XGBoost, NN on 54M+ txs	F1: 0.76–0.85 vs. 0.08 with features alone	Random Forest fraud modeling
Adom et al. (2022) [4] · XAI lending	LIME / SHAP comparison	SHAP deeper, more consistent than LIME	Primary explainability method
Tan (2023) [5] · CBDC hierarchy	Developing-nation model	Two-tier CBDC distribution; hierarchy decisive	Four-tier architecture
Liu et al. (2008) [6] · Anomaly detection	Isolation Forest	Recursive partitioning; shorter paths	Unsupervised wallet detection
Atzei et al. (2017) [7] · SC security	SoK of Ethereum attacks	Reentrancy, access-control gaps	Security primitives, formal verification
Werner et al. (2022) [1] · DeFi SoK	Survey of 12+ protocol types	Pool-based, overcollateralised; no hierarchy	Core gap CWB addresses
Bastankhah et al. (2024) [2] · Adaptive lending	Dual fast/slow control	Dynamic rates beat static curves	Kinked-rate validation
Hartmann & Hasan (2021) [41] · P2P lending	Smart-contract origination	Gas minimisation; no intermediary	Four-tier P2P design
Hasan et al. (2022) [42] · Federated fraud	FL on encrypted features	Comparable to centralised ML	Cross-bank intelligence
Wang et al. (2020) [43] · SC vulnerabilities	ML on bytecode	F1: 0.96 access control; 0.93 leakage	Security verification models
Liao et al. (2019) [44] · Audit automation	Classification + fuzzing	70% faster than manual audit	Security audit strategy

Table 2.4: Literature review summary (Part 2 of 2)

Author & Focus	Methodology	Key Findings	Relevance to Our Project
BIS et al. (2020) [46] · CBDC design	Retail distribution analysis	Two-tier preferred; interoperability gaps	Dual-currency facility
Kiff et al. (2020) [106] · CBDC models	IMF architecture survey	Multi-tier via supervised intermediaries	Wholesale-to-retail path
Bindseil (2020) [105] · CBDC rates	ECB tiered remuneration	Preserves bank intermediation	Asymmetric rate spread
Dong et al. (2023) [104] · Supply-chain finance	Three-tier game model	Cross-tier direct financing	Downward client lending
Project Pine (2025) [103] · Hierarchical SCs	NY Fed / BIS prototype	Factory-to-facility hierarchy	`IBLP` / `SyndicatedLoan`
BIS/FSB [16] · Cross-border settlement	Nostro/vostro cost analysis	2–5 days; $42/transfer avg.	Near-instant four-tier settlement
Beyer et al. (2025) [28] · QE inequality	Cross-country 1999–2019	QE raises wealth inequality	Algorithmic monetary parameters
World Bank (2025) [33] · Dev. finance blockchain	13-project pilot	Reduces reporting burden	Multi-chain deployment
Chen et al. (2024) [39] · Multi-chain lending	Cross-chain design	Higher throughput, less congestion	Multi-chain strategy
Yang & Cui (2023) [40] · Protocol evaluation	Quantitative risk metrics	Utilisation, liquidation, responsiveness	Tier benchmarks
Asaduzzaman et al. (2020) [47] · Smart micro-credit	Smart-contract MFI	60% admin overhead reduction	`GroupLendingPool`
Kim & Kim (2024) [48] · Gas optimisation	Batching, calldata compression	30–50% gas savings	L2 deployment
Yuan (2024) [49] · SHAP credit default	RF, GBM + SHAP	AUC >0.92, interpretable	Per-client risk explanation

Table 2.6: Literature review summary (Part 2 of 2, continued)

Author & Focus	Methodology	Key Findings	Relevance to Our Project
Ante (2025) [108] · Stablecoin remittance	U.S. survey (n=866)	26% adoption; literacy drives uptake	KYC ladder, `P2PTransfer`
Du et al. (2026) [110] · Payment rails	Corridor cost comparison	Stablecoin sandwich undercuts banks	PSP on/off-ramp work
Li et al. (2026) [109] · Retail payments SoK	CLEAR framework	Closed-loop advantage; weak disputes	Merchant checkout design
Zhang et al. (2024) [112] · Stablecoin checkout	Operations model	Refund volatility disutility	`FXModule`, auto-conversion
Fröhlich et al. (2022) [111] · Crypto POS	Lightning usability	On-ramp friction dominates	Fiat-funded `CurrentAccount`
Lin et al. (2025) [113] · Platform payments	Permissioned chain prototype	≈256 TPS; multi-intermediary escrow	Local-Bank merchant flows
CPMI / BIS [114, 115] · Stablecoin policy	Cross-border analysis	Ramps gate acceptance	Ramp integration
FATF (2021) [117] · VASP / P2P	Risk-based AML	Platform transfers need originator data	Sponsored `P2PTransfer`

2.4 Comparative Protocol Analysis

No surveyed protocol combines multi-tier hierarchy, oracle-mediated explainable ML, and compliance-aware identity. Table 2.7 compares CWB against Aave, Compound, Maker, Maple, and Goldfinch on eleven architectural dimensions.

Feature	Aave	Comp.	Maker	Maple	Gfinch	CWB
Institutional hierarchy	✗	✗	✗	✗	✗	●
Cross-tier capital flow	✗	✗	✗	✗	✗	⧗
Same-tier interbank lending	✗	✗	✗	✗	✗	⧗
Syndicated / co-lending	✗	✗	✗	Part.	Part.	⧗
Upward capital repatriation	✗	✗	✗	✗	✗	⧗
Solidarity group lending	✗	✗	✗	✗	Part.	●
AI/ML fraud detection	✗	✗	✗	Man.	Man.	●
SHAP explainability	✗	✗	✗	✗	✗	●
ZKP KYC compliance	✗	✗	✗	✗	✗	●
Kinked interest rate curve	✓	✓	Gov.	✓	✗	⧗
Role-based access control	Part.	Part.	Gov.	✓	Part.	●
Institutional / L2 focus	✗	✗	✗	✓	Part.	●
TVL / Status (2026)	$26B	$1.4B	$10B	$2.6B	$680M	Planning

Table 2.7: Protocol comparison on eleven architectural dimensions

This answers RQ1: flat-pool protocols differentiate assets, not institutional relationships or inter-bank flows; Maple, Goldfinch, and Morpho V2 [117, 90] lack multilateral development-finance capital allocation. CWB's compound novelty (Section 1.5) spans hierarchy, explainable ML governance, and group lending. Tiers 1–3 use bespoke RBAC (Section 3.18) functionally equivalent to ERC-3643 token-level KYC gating [82]; enforced per-tier reserve ratios address opaque custody failures such as FTX's 2022 collapse.

Six design decisions follow from the survey:

- Gudgeon et al. [56] → kinked-rate model (Section 3.9)
- Piper et al. [58] → ZKP identity path (Section 3.7.1)
- Tan [5], BIS [46], Kiff [106], Auer and Böhme [107] → four-tier capital flows
- Bindseil [105], Project Pine [103], Dong et al. [104] → multi-directional lending (Section 1.6)
- Asaduzzaman et al. [47] → `GroupLendingPool`
- Ante [108], Du et al. [110], Li et al. [109], Lin et al. [113], CPMI [114], FATF [117] → post-MVT retail stack (Section 6.1)

2.5 Summary of Key Findings

The literature review yields the following design-relevant conclusions:

- **Flat DeFi vs. tiered finance.** Pool-based protocols exceed $55B TVL [8] but lack institutional hierarchy; CBDC research confirms tiered distribution as a workable pattern [5]. No prior work models four-tier reserve-enforced lending.

- **Settlement and retail rails.** Correspondent banking traps idle capital [18], settles in 2–5 days at ≈$42/transfer [19], and costs remittance users $48–56B annually [20]. Stablecoin rails can compress costs [116, 110] but need ramp infrastructure; retail studies [109, 112, 111, 113, 119] favour Local-Bank–sponsored closed-loop flows over open permissionless checkout.
- **ML, XAI, and security.** Behavioral DeFi features lift fraud F1 to 0.76–0.85 [3]; Isolation Forest [6] covers unlabeled cases; SHAP beats LIME for regulatory explainability [4]. Contract vulnerabilities are well catalogued [7]; ML audit tools reach F1 $>$0.93 [43], cut review time 70% [44], and formal verifiers such as VERISMART reach 98.4% precision [45].
- **Hierarchy, groups, and institutions.** Cross-tier CBDC and Project Pine literature [106, 105, 107, 103, 104] supports multi-directional programmable flows; analogue MFIs report $>$95% group repayment [47] with on-chain mutual liability still an open gap. FundsChain [33], Corda RWAs [31], and Kinexys [34] show rising institutional blockchain adoption.
- **Scalability, privacy, and regulation.** Multi-chain lending [39] and L2 gas optimization [48] cut costs 30–50%; federated learning [42, 97] enables cross-bank fraud intelligence without raw data sharing. MiCA and GENIUS [75, 77] anchor stablecoin-first design; algorithmic parameters mitigate distributional opacity from discretionary monetary policy [28, 29].
- **Deposit-funded sustainability.** Sustainable MFIs rely on mobilized deposits, not donor capital alone—motivating savings vaults at the Local Bank tier alongside `GroupLendingPool`.

These findings justify the four-tier architecture, cross-tier lending modules, Random Forest / Isolation Forest / SHAP analytics, governance structure, and phased security verification pipeline developed in later chapters.

Chapter 3

System Architecture and Design

3.1 Prototype Scope

Table 3.1: Prototype scope and implementation status

Feature	Status
Four-tier role system (RBAC)	⌛ Specified
World Bank Reserve contract (Tier 1)	⌛ Specified (Phase I)
Tier 2 National Bank contracts	⌛ Specified (Phase I)
Tier 3 Local Bank contracts	⌛ Specified (Phase I)
Cross-tier fund transfer	● Planned (Phase II)
Loan request / approval workflow	⌛ Specified (Phase II)
Installment EMI auto-generation	● Planned (Phase II)
SavingsVault contract	● Planned (Phase II)
FixedDeposit contract	● Planned (Phase II)
GroupLendingPool contract	● Planned (Phase II)
InterBankLendingPool	● Planned (Phase II)
AI/ML fraud detection (Random Forest)	● Planned (Phase III)
SHAP + anomaly explainability	● Planned (Phase III)
Oracle integration (Chainlink Functions)	● Planned (Phase III)
ZKP KYC compliance layer	● Planned (Phase IV)
Testnet deployment (Polygon zkEVM Cardona)	● Planned (Phase IV)
AI Agent / MCP tool server (17 tools)	● Optional (Phase III–IV)
EIP-7702 session key management	● Optional (Phase III)
Authority Brief UI (SHAP for approvers)	● Planned (Phase III)
Chainlink Automation (installment triggers)	● Planned (Phase II)
Chainlink Proof of Reserve	● Planned (Phase I)
The Graph subgraph (event indexing)	● Planned (Phase III)
SAR workflow (AML → freeze queue)	● Planned (Phase III)
`sessions` table schema	⌛ Specified (Phase I)
`agent_action_log` table schema	● Optional (Phase III)
300-client Foundry simulation	● Planned (Phase IV)

3.2 High-Level Architecture

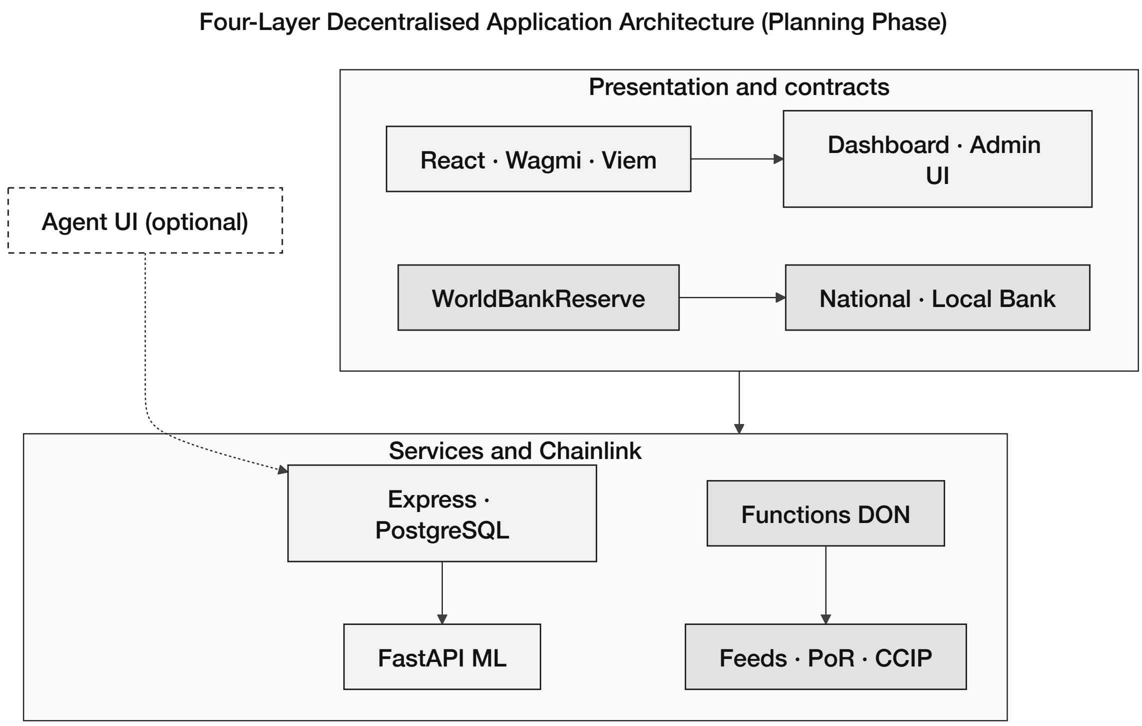

Figure 3.1: Four-layer application architecture

Figure 3.2 shows component interactions. Core lending contracts are specified in Phase I; extended modules are planned extensions documented in Appendix B.

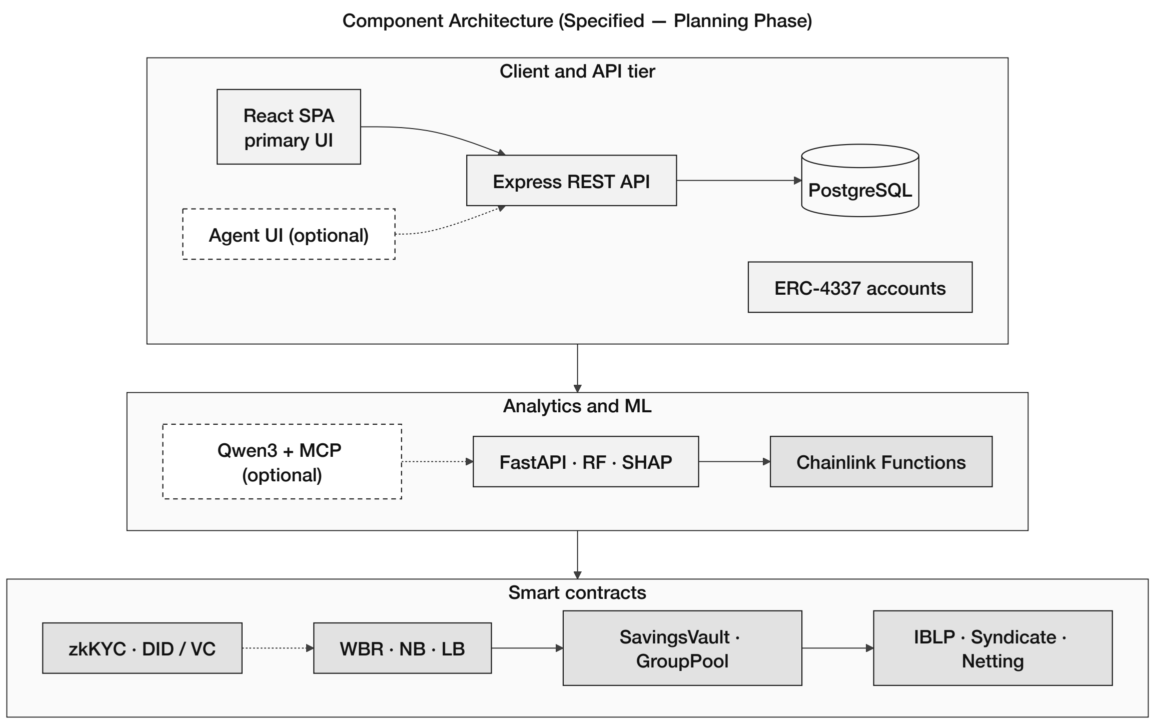

Figure 3.2: Component diagram (planning phase)

Contract	Module	Status	Primary responsibility
WorldBankReserve	Tier 1 core	Specified (Phase I)	Global reserve custody, national-bank registration, downward capital allocation.
NationalBank	Tier 2 core	Specified (Phase I)	Local-bank registration, upward borrowing, downward allocation.
LocalBank	Tier 3 core	Specified (Phase I)	Client onboarding, loan lifecycle, installments, `freezeAccount`.
SavingsVault	Deposits	Specified (Phase II)	Variable-yield savings, reserve-gated withdrawals.
FixedDeposit	Deposits	Specified (Phase II)	Term deposits with deterministic early-withdrawal penalty.
GroupLendingPool	Group credit	Specified (Phase II)	Solidarity-group formation, consent, mutual liability.
FXModule	Treasury / FX	Specified (Phase II)	Oracle-priced retail FX with disclosed spread.
InsuranceFund	Risk buffer	Specified (Phase II)	Default coverage buffer; Chainlink Proof of Reserve.
CurrentAccount	Payments	Specified (Phase II)	Transactional accounts and atomic peer transfers.
InterBankLendingPool	Multi-entity	Specified (Phase II)	Same-tier interbank liquidity pools per tier.
UpwardDepositFacilit	Multi-entity	Specified (Phase II)	Upward surplus repatriation with yield spread.
SyndicatedLoan	Multi-entity	Planned (Phase III)	Multi-lender syndicate consent and disbursement.
TranchedPool	Multi-entity	Planned (Phase III)	Senior/junior tranching and waterfall recovery.
TreasurySwap	Multi-entity	Specified (Phase II)	Institutional oracle-priced asset swaps.
NettingEngine	Multi-entity	Planned (Phase III)	Multilateral settlement netting batches.

3.3 Blockchain Platform Selection

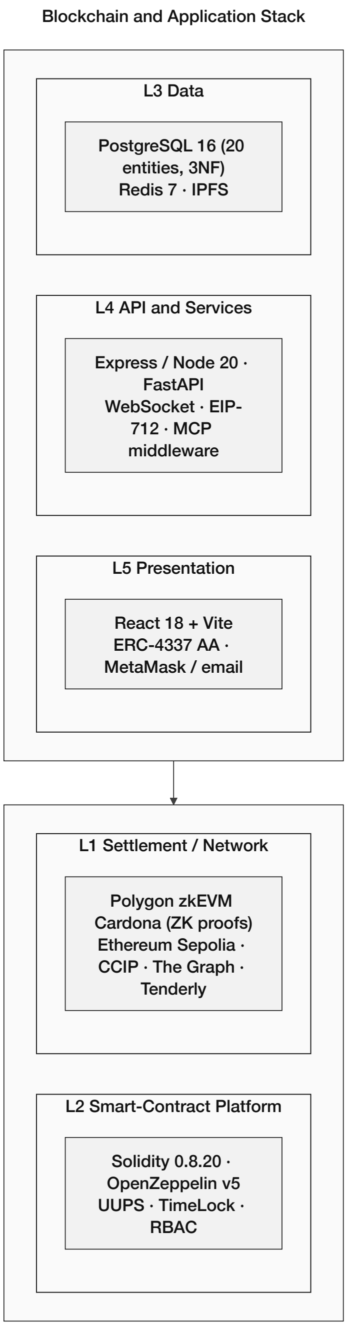

Figure 3.3: Layered blockchain and application stack

Table 3.3: Blockchain platform selection criteria and justification

Criterion	Selection	Justification
Platform	Ethereum Virtual Machine (EVM)	Largest developer ecosystem; battle-tested security model; extensive tooling.
Network	Polygon zkEVM Cardona / Ethereum Sepolia	Zero-cost deployment; production-equivalent behaviour; free faucet access.
Consensus	ZK validity proofs (Polygon zkEVM L2, Ethereum L1 settlement)	Every batch of transactions is accompanied by a zero-knowledge proof verified on Ethereum L1, so security derives from cryptographic validity rather than a validator set assumption.
Smart contract language	Solidity 0.8.20	Industry standard; mature compiler with overflow protection; rich library ecosystem.

Table 3.5: Blockchain platform selection (continued)

Criterion	Selection	Justification
Gas cost	\$0.001–\$0.01 per transaction	Orders of magnitude cheaper than Ethereum mainnet (\$5–\$50); enables micro-loan economics.
Block finality	≈2 seconds	Sub-second practical finality for retail UX; checkpointed to Ethereum for security.
Developer tooling	Hardhat + OpenZeppelin	Automated test suite, deployment scripts, and audited security primitives.
Testnet availability	Cardona (Polygon zkEVM), Sepolia (Ethereum)	Free faucets; EVM-identical behaviour; no real cryptocurrency required for prototype.
Security model	ZK validity proofs	Every batch verified by a ZK proof anchored to Ethereum L1 (vs. PoS validator assumption on Amoy).
Migration path	EVM-compatible	Same Solidity contracts deploy to any EVM chain; reduces future L2 migration cost.

3.3.1 Transaction Verification and Consensus

- **Polygon zkEVM Cardona:** ZK validity proofs anchor batches to Ethereum L1 (cryptographic security vs. PoS validator assumption on Amoy).

- **On prototype testnets:** Cardona and Sepolia use equivalent consensus models at no financial cost, enabling incremental development and testing without exposure to real-asset risk.

Gas-cost sustainability. The hierarchical loan lifecycle generates multiple on-chain state transitions per client; on Polygon zkEVM Cardona the **projected** cost remains well below 0.01% of interest earned on a representative retail loan (Section 5.1), to be validated by the Phase IV Foundry simulation. This margin underpins the micro-loan economics that motivate Layer 2 deployment in the platform selection above.

3.3.2 Oracle Architecture: Off-Chain AI to On-Chain Decision

Contracts cannot read ML scores on their own—the classic oracle problem [55, 95]. Publishing non-price ML output on-chain is still thinly studied [95]. CWB pipes off-chain risk into loan approval via Chainlink [122–124], with a lighter commit-reveal relay for early prototyping [125].

Chainlink Functions oracle (primary). Chainlink Functions [122] is the target path: DON nodes fetch the score independently and only commit after consensus [55], so one bad node cannot move a decision. The DON runs:

```
// Chainlink Functions source (runs in decentralised DON)
const clientId = args[0];
const response = await Functions.makeHttpRequest({
 url: `https://your-ml-service.com/score/${clientId}`,
 method: "POST"
});
const score = response.data.risk_score;
// 6 decimal precision
return Functions.encodeUint256(Math.round(score * 1e6));
```

Expect 30–60 s latency and roughly \$0.10–\$1.00 per call [122]—fine for loan approval, which is not latency-critical.

Commit-reveal relay (prototype fallback). Until Functions is wired (Phase III), FastAPI commits $h = \text{keccak256}(s \,\|\, \text{nonce})$ to `LoanController`, then reveals s and the nonce in-window—the standard commit-reveal pattern [125]. The contract checks the hash and stores the score.

Chainlink Automation: trustless installment triggers. Chainlink Automation [123] flags overdue installments without a central cron; `checkUpkeep`/`performUpkeep` follow the keeper interface [123]:

```
// In LocalBankPool.sol
function checkUpkeep(bytes calldata) external view override
 returns (bool upkeepNeeded, bytes memory performData) {
 uint256[] memory overdueLoans = getOverdueLoans;
 upkeepNeeded = overdueLoans.length > 0;
 performData = abi.encode(overdueLoans);
}
```

```
function performUpkeep(bytes calldata performData) external override {
 uint256[] memory overdueLoans =
 abi.decode(performData, (uint256[]));
 for (uint i = 0; i < overdueLoans.length; i++) {
 _markInstallmentOverdue(overdueLoans[i]);
 }
}
```

Chainlink Price Feeds: fiat and crypto pairs. `FXModule` reads `AggregatorV3Interface` feeds (ETH/USD, USDC/USD, selected fiat pairs) [124]:

```
// In LocalBankPool.sol
AggregatorV3Interface internal fiatUsdFeed; // e.g. EUR/USD
AggregatorV3Interface internal ethUsdFeed;

function getFiatEquivalent(uint256 usdcAmount)
 public view returns (uint256) {
 (, int fiatRate,,,) = fiatUsdFeed.latestRoundData;
 // USDC tracks USD 1:1; Chainlink uses 8 decimals
 return usdcAmount * uint256(fiatRate) / 1e8;
}
```

Chainlink's 8-decimal feed format [124]; Phase I adds optional fiat display in the UI.

Chainlink Proof of Reserve. `WorldBankReserve` can publish to **Chainlink Proof of Reserve (PoR)** [103] so outsiders verify solvency without trusting the admin:

```
// WorldBankReserve.sol
function getReserveSummary external view returns (
 uint256 totalDeposited,
 uint256 totalLoaned,
 uint256 reserveRatio, // scaled 1e4 (e.g. 5000 = 50%)
 uint256 insuranceFundBalance
) {
 return (
 totalDeposited,
 totalLoaned,
 (totalDeposited - totalLoaned) * 1e4 / totalDeposited,
 insuranceFund.balance
 );
}
```

Same reserve logic as Section 3.18; PoR makes it externally auditable (Appendix D).

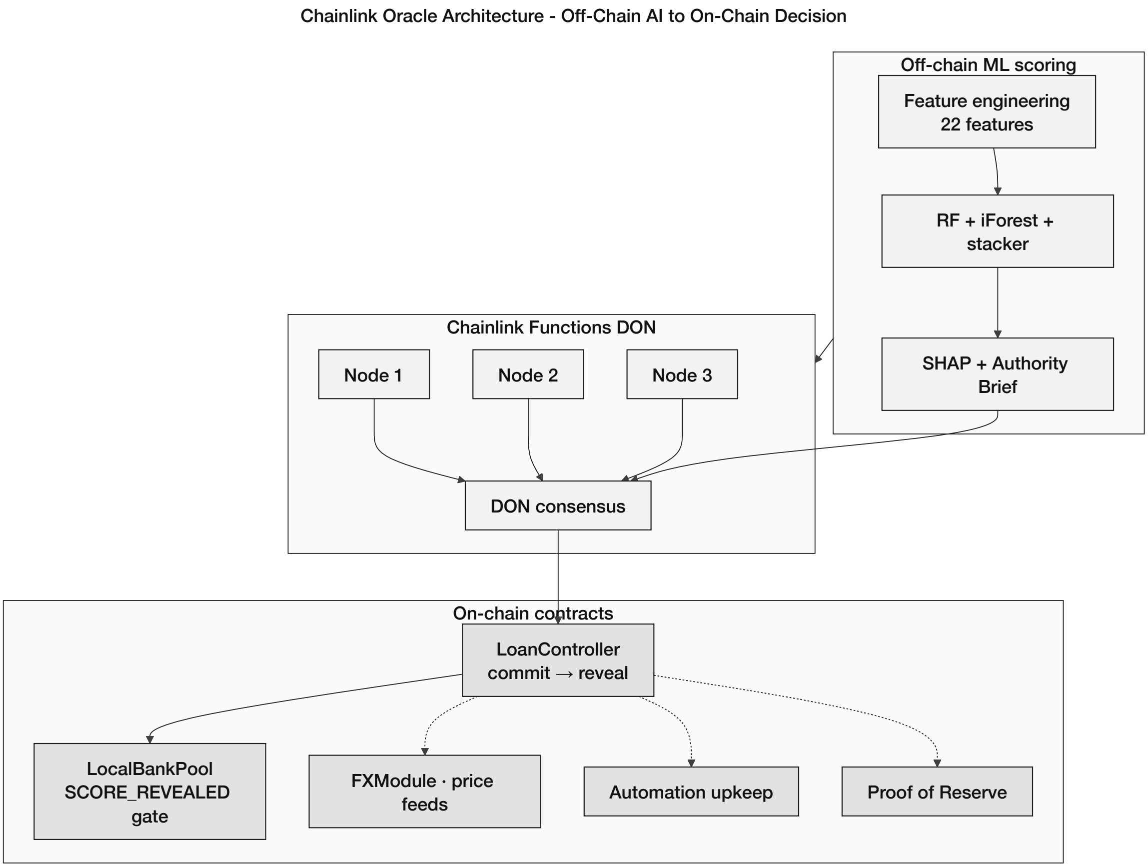

Figure 3.4: Chainlink oracle architecture

3.4 Data Model and Database Design

The relational database schema comprises 20 normalized entities in Third Normal Form (3NF). Figure 3.5 presents the core entity-relationship overview, Figure 3.6 presents extended product and multi-entity entities, and Figure 3.7 presents the Enhanced Entity-Relationship (EER) diagram with specialisation and weak-entity constructs.

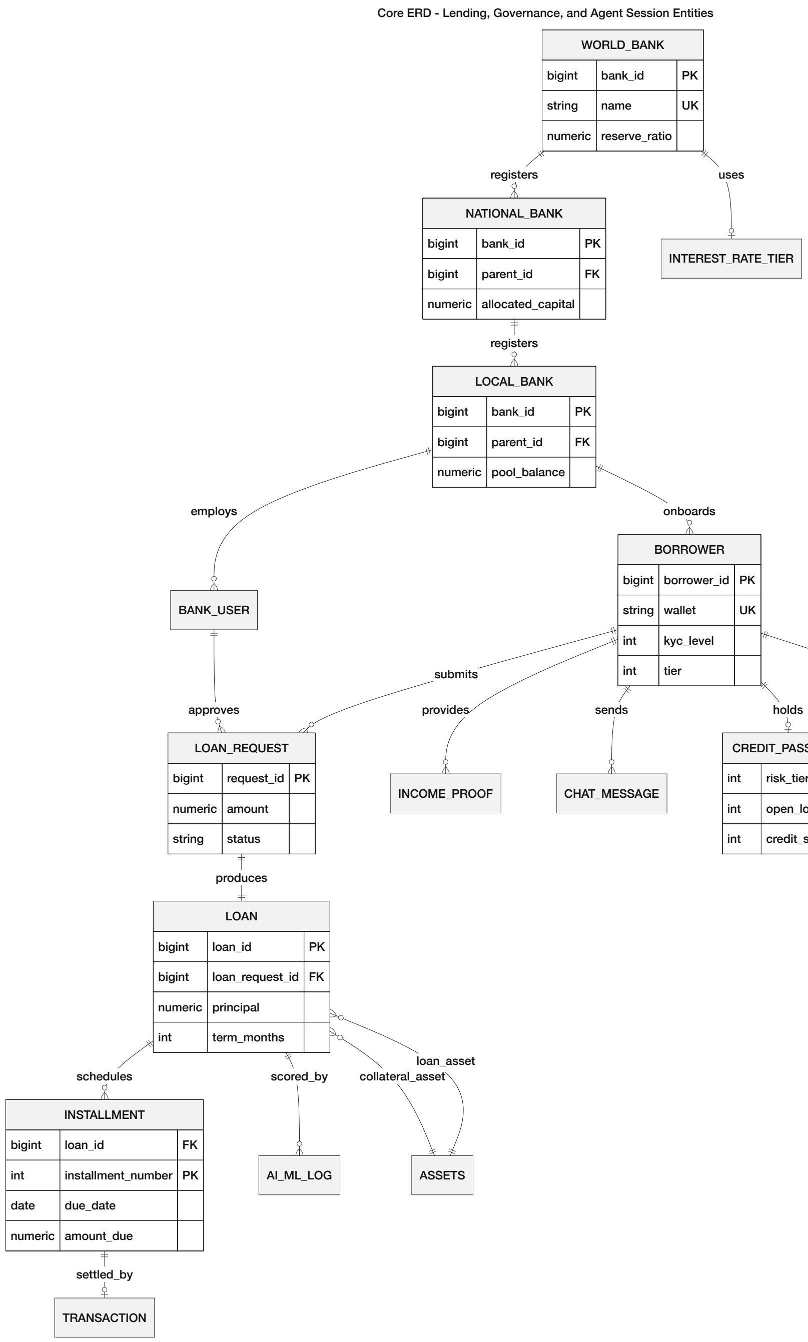

Figure 3.5: Core system entity graph

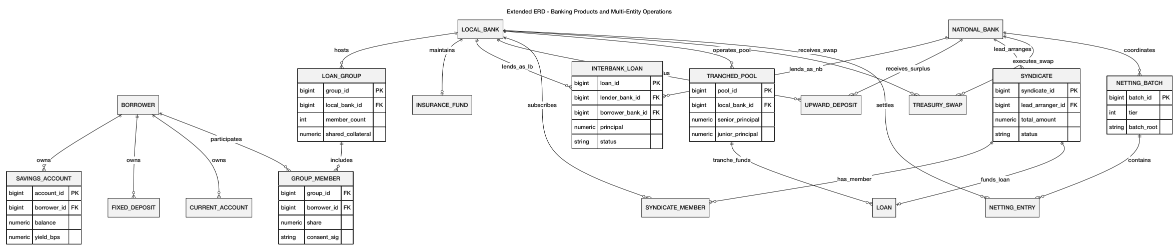

Figure 3.6: Extended ERD entities

Figure 3.6 adds the remaining product and cross-tier tables on top of the core model. Full relationship detail is in Section 3.14; the schema matches the contract layout in the phase plan.

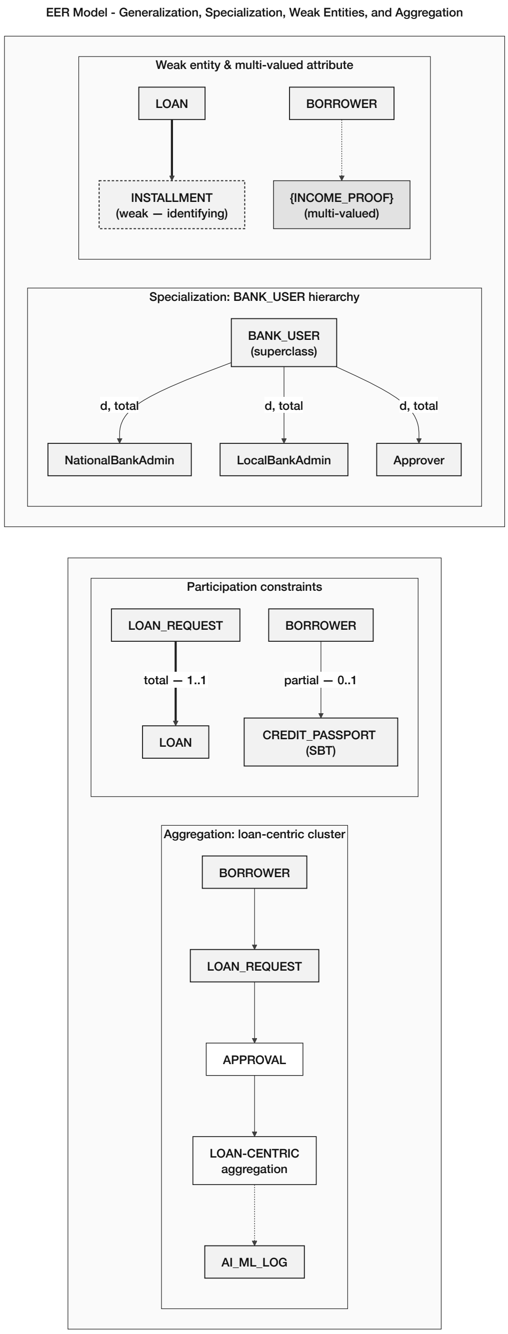

Figure 3.7: Enhanced EER model

3.4.1 Entity Summary

Table 3.7: Database entity summary (20 entities)

Entity	Primary key	Role / Description
WORLD_BANK	`world_bank_id`	Top-level reserve holder; global lending parameters.
NATIONAL_BANK	`national_bank_id`	Country-level banks; borrow from World Bank.
LOCAL_BANK	`local_bank_id`	City-level banks; borrow from National Bank and lend to users.
BANK_USER	`bank_user_id`	Bank staff with role-based permissions (approve/reject loans).
BORROWER (CLIENT)	`borrower_id`	End clients requesting and repaying loans (UK: `wallet_address`). Renamed `CLIENT` in the final migration; prototype retains `BORROWER`.
LOAN_REQUEST	`request_id`	Loan applications and approval workflow tracking.
LOAN	`loan_id`	Active loan after disbursement; FK `loan_request_id` → `LOAN_REQUEST`; collateral and loan asset references.
INSTALLMENT	`loan_id`, `installment_number`	Repayment schedule (weak entity on `LOAN`; composite PK).
TRANSACTION	`transaction_id`	Financial transaction records.
BORROWING_LIMIT	`limit_id`	Per-client limits with 6-month and 1-year rolling windows (UK: `borrower_id`).

Table 3.9: Database entity summary (continued)

Entity	Primary key	Role / Description
INCOME_PROOF	`proof_id`	Income verification documents (multi-valued per client).
CHAT_MESSAGE	`message_id`	Client–bank communication; FK to `LOAN_REQUEST` and/or `LOAN`.
AI_CHATBOT_L	`log_id`	AI chatbot interaction records.
AI_ML_SECURITY	`security_log_id`	Two logically distinct domains aggregated for the prototype: (1) `LOAN_RISK_ASSESSMENT` per-loan scores and SHAP vectors; (2) `SECURITY_EVENT_LOG` for system-wide events. Final thesis splits into separate tables.
MARKET_DATA	`market_data_id`	Cryptocurrency price feed cache (Express.js cron). Read-only auxiliary; no FK into core lending tables. Key attributes: `symbol`, `price_usd`, `source`, `recorded_at`.
PROFILE_SETTING	`profile_id`	Client and bank-user interface preferences. FK `client_id` → `BORROWER` (1:1). Attributes: `language_setting`, `display_currency`, `notification_enabled`.

Table 3.10: Database entity summary (continued)

Entity	Primary key	Role / Description
SESSIONS	`session_id`	Wallet session registry: device/IP hash, EIP-7702 session-key scope and TTL, parent lineage, compression summary, end reason, and delivery source. Audit FK for agent write tools.
AGENT_ACTION_LOG	`action_id`	Append-only agent write-tool log (tool, parameters, confirmation-message FK, tx hash, status). Separate from `AI_CHATBOT_LOG`; INSERT-only RLS policy.
INTEREST_RATE_T	`tier_id`	Normalised kinked-rate parameters (`base_rate`, `kink_utilisation`, `rate_above_kink`, `max_rate`); Section 3.4.3.
ASSETS	`asset_id`	Collateral and loan asset registry (UK: `symbol`); referenced by `LOAN` via `collateral_asset_id` and `loan_asset_id`.

3.4.2 EER Constructs Applied

Table 3.12: EER constructs applied

EER Construct	Applied To	Notes
Specialisation (disjoint)	BANK_USER → NationalBankUser, LocalBankUser	A bank user belongs to exactly one bank type; enforced by CHECK constraint.
Generalisation	WORLD_BANK, NATIONAL_BANK, LOCAL_BANK share institution attributes	Avoids attribute duplication across institution types.
Weak entity	INSTALLMENT depends on LOAN	Existence and identity determined by parent loan.
Multi-valued attribute	INCOME_PROOF per BORROWER	Modelled as separate entity to maintain 1NF.
Aggregation	TRANSACTION, CHAT_MESSAGE → AI_ML_SECURITY_L	AI_ML_SECURITY_LOG aggregates monitoring signals from both sources.
Association entity	LOAN_REQUEST links BORROWER + LOCAL_BANK	LOAN_REQUEST is a strong entity with primary key `request_id` and foreign keys to BORROWER and LOCAL_BANK, capturing the many-to-many relationship between borrowers and lending banks. It is *not* a weak entity and is not an aggregation in the ER-theoretic sense; "association entity" or "junction entity" is the correct ER classification.
Participation constraint	BORROWER must have $\geq$ 1 LOAN_REQUEST to access services	Enforced at application layer; architecturally planned for on-chain.
Append-only table (policy)	AGENT_ACTION_LOG, AUDIT_LOGS	DB-level INSERT-only role + Row-Level Security policy enforces that no UPDATE or DELETE is permitted on audit records. Provides immutable tamper-evident trail for regulatory review and agent action accountability.

3.4.3 Normalization

The schema has been normalized to Third Normal Form (3NF) and checked against Boyce-Codd Normal Form (BCNF):

- **1NF:** There are no repeating groups in any of the attributes. Separate entities store multi-valued attributes, such as proof of income.
- **2NF:** There are no partial dependencies. The full composite key (`loan_id`, `installment_number`) determines the non-key attributes of INSTALLMENT.
- **3NF:** No dependencies that go through other dependencies. Instead of storing redundant values like `total_borrowed` in bank entities, they are calculated at query time.
- **BCNF:** All determinants are candidate keys. A CHECK constraint on BANK_USER enforces that the generalization hierarchy's specializations are not the same.

3.4.4 Relational Integrity Constraints

Table 3.14: Relational integrity constraints

Constraint Type	Examples
Primary Key	One surrogate or composite key per Table 3.7: e.g. `world_bank_id`, `national_bank_id`, `local_bank_id`, `bank_user_id`, `borrower_id`, `request_id` (`LOAN_REQUEST`), `loan_id` (`LOAN`), (`loan_id`, `installment_number`) (`INSTALLMENT`), `transaction_id`, `limit_id`, `proof_id`, `message_id`, `log_id`, `security_log_id`, `market_data_id`, `profile_id`, `session_id`, `action_id`, `tier_id`, `asset_id`.
Foreign Key	`local_bank_id` in `LOAN_REQUEST` references `LOCAL_BANK`; `loan_request_id` in `LOAN` references `LOAN_REQUEST`; `loan_id` in `INSTALLMENT` references `LOAN`.
UNIQUE	`wallet_address` in `BORROWER`; `blockchain_tx_hash` in `LOAN_REQUEST`; `borrower_id` in `BORROWING_LIMIT`; `symbol` in `ASSETS`.
CHECK	BANK_USER: `(bank_type='national' AND national_bank_id IS NOT NULL) OR (bank_type='local' AND local_bank_id IS NOT NULL)`.
NOT NULL	Core attributes: `name`, `wallet_address`, `status`.
APPEND-ONLY (DB policy)	`AGENT_ACTION_LOG`: `CREATE ROLE audit_writer; GRANT INSERT ON agent_action_log TO audit_writer; REVOKE UPDATE, DELETE FROM PUBLIC.` `AUDIT_LOGS`: same policy applied via Row-Level Security (`ALTER TABLE audit_logs ENABLE ROW LEVEL SECURITY; CREATE POLICY audit_insert_only ON audit_logs FOR INSERT WITH CHECK (TRUE)`).
FOREIGN KEY (new)	`LOAN.collateral_asset_id REFERENCES assets(asset_id)`; `LOAN.loan_asset_id REFERENCES assets(asset_id)`; `AGENT_ACTION_LOG.session_id REFERENCES sessions(session_id)`; `AGENT_ACTION_LOG.confirmation_turn_id REFERENCES chat_message(message_id)`.
UNIQUE (new)	`assets.symbol` (asset ticker is globally unique across the platform asset registry).

3.5 On-Chain and Off-Chain Data Partitioning

Table 3.16: Data partitioning between on-chain and off-chain storage

Data Category	Storage	Rationale
Reserve balances, loan requests, approval/rejection events, repayment transactions	On-chain	Immutability, public auditability, trustless verification.
User profiles, income verification documents, chat messages, AI/ML inference logs	Off-chain (database)	Data privacy, query flexibility, storage cost optimisation.
Borrowing limit computations	Off-chain with on-chain enforcement	Complex temporal aggregation; results committed as on-chain constraints.
Cryptocurrency market data	Off-chain (cached)	High-frequency updates; external API dependency.
Agent action execution log	Off-chain (append-only DB)	Immutable audit trail of every agent write-tool call, linked to on-chain tx hash; not stored on-chain to save gas.
Session key material	Off-chain (`sessions` table)	EIP-7702 session key hash, scope JSON, and TTL stored off-chain; the scope restrictions are enforced on-chain by the session key contract at signing time.
Chainlink oracle price data	Off-chain (`MARKET_DATA`)	Chainlink Price Feed results are cached in `MARKET_DATA` with `source` = Chainlink feed address; the on-chain `AggregatorV3Interface` is the authoritative source.
Session lineage chains (SESSIONS, message history, compression summaries)	Off-chain (PostgreSQL)	Full transcript history is too large for on-chain storage. The lineage chain enables searchable long-term memory without gas cost. `AGENT_ACTION_LOG` provides the on-chain audit link via confirmed transaction hashes.

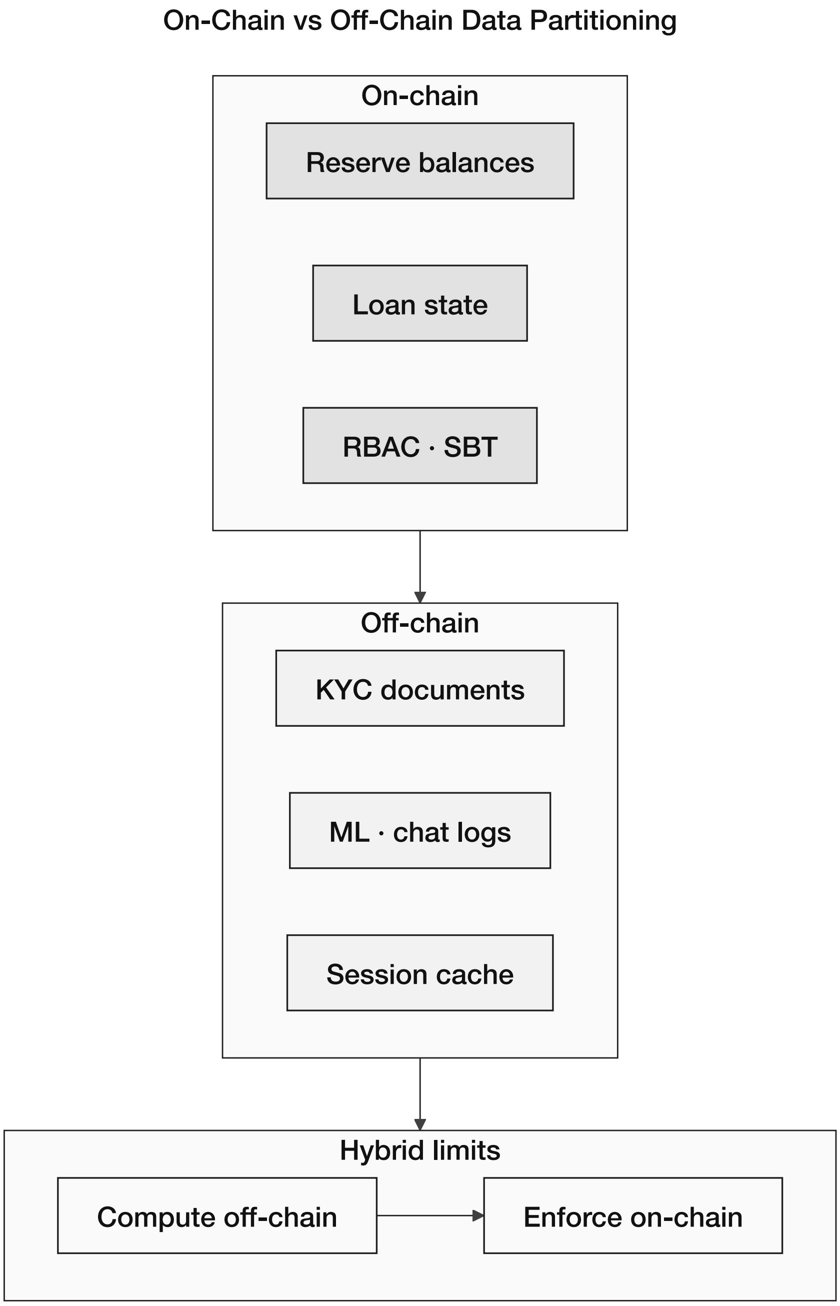

Figure 3.8: On-chain versus off-chain data partitioning

3.6 Operating Context

The Crypto World Bank is a programmable hierarchical banking platform on EVM Layer 2—not a single-country pilot—built for reserve enforcement, tiered capital flow, and on-chain auditability on public testnets (Polygon zkEVM Cardona, Ethereum Sepolia). Retail lending uses stablecoins (Section 1.8), L2 keeps installment and reserve transactions affordable, and optional account abstraction (Section 3.8.1) can onboard crypto-naive users without changing on-chain role logic.

3.7 Digital Identity System

Identity is split in two. On-chain, users sign in with a wallet (MetaMask or WalletConnect); that address gets a role—Owner, National Bank, Local Bank, Approver, or Retail Client—and every action is recorded on the ledger. Off-chain, the real-world checks live in PostgreSQL: government ID, income, KYC, and AML results, linked back to the wallet by hash. A wallet proves you control an address; it does not prove who you are legally. Production will need recovery if a key is lost, with on-chain role revocation and careful re-binding so no one can steal a tier by swapping wallets.

Later, zk proofs (Section 3.7.1) let the contract confirm "this user passed KYC" without putting documents on-chain. For the optional chat agent, clients can issue a short-lived session key (EIP-7702): capped amounts (e.g. 500 USDC), named tools only (`submit_loan_application`, `pay_installment`), 24-hour expiry, no transfers to random addresses—each use logged in `AGENT_ACTION_LOG` with the confirming chat turn as audit trail.

3.7.1 ZKP KYC and zkAML Compliance Architecture

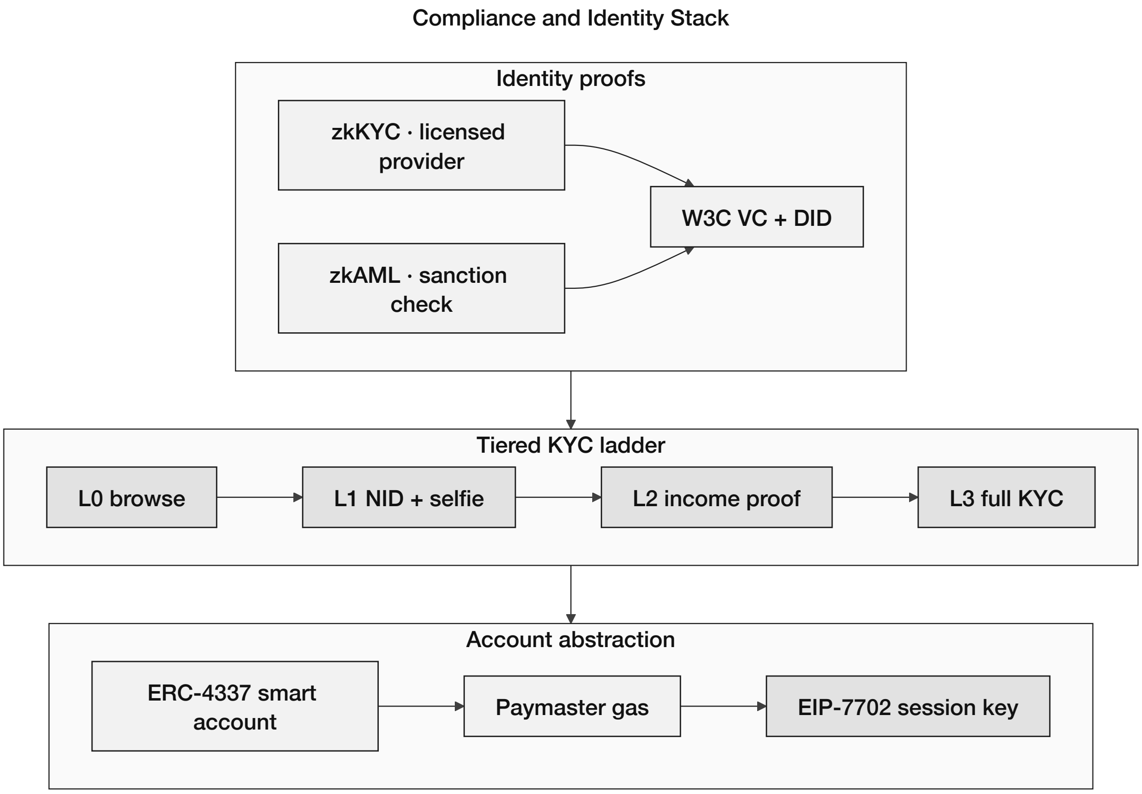

Figure 3.9: Compliance and identity stack

Figure 3.9 stacks identity in three layers: zkKYC and zkAML proofs from a licensed provider feed a W3C Verifiable Credential and DID, then the client climbs a tiered KYC ladder from browse-only (L0) through NID, income proof, to full KYC (L3). Below that, account abstraction gives retail users an ERC-4337 smart account with Paymaster-sponsored gas and an optional EIP-7702 session key so the AI agent can act only within scoped, revocable limits.

The compliance gateway uses two cooperating zero-knowledge circuits rather than a single KYC verifier: a **KYC circuit** and an **AML circuit**, both expressed as zk-SNARKs (Groth16 proofs via Circom 2.0 + snarkjs). KYC proves that the user owns a valid identity credential; AML proves that the user's wallet has not interacted with sanctioned addresses and stays within velocity limits. Decoupling the two circuits makes it possible to revoke AML certification without invalidating the KYC credential, and matches the actual structure of regulator-mandated compliance.

KYC circuit (identity). An off-chain KYC provider validates the user's government-issued ID and issues a signed credential $\mathit{credential} = \mathrm{Sign}_{\mathrm{KYC}}(\mathit{wallet\_address}, \mathit{country}, \mathit{age\_over\_18}, \mathit{kyc\_pas}$ The user generates a zk-SNARK proof that (a) they possess a valid signed credential from an approved KYC provider, (b) their age is over 18, and (c) their country is in the permitted jurisdiction list—without revealing the underlying credential, ID number, or personal data. The smart contract verifies the proof: `KYCVerifier.verify(proof, [wallet, country, age_over_18])` returns `true`. Piper et al. (2025) report proof generation in 1–4 s on consumer hardware; on-chain verification costs ≈200–300 k gas, within a one-time registration budget.

zkAML circuit (anti-money-laundering). The zkAML circuit (IACR ePrint 2025/465) proves three properties of a wallet without revealing its transaction graph: (1) the wallet has not transacted with any address on a public sanctions list (Chainalysis / TRM Labs); (2) the wallet's 30-day inflow does not exceed a governance-set velocity threshold; and (3) the wallet has not received funds from addresses flagged in the platform's on-chain risk registry within the preceding 90 days. The published benchmark for zkAML is 55 TPS on public networks with 226 ms proof-generation latency, both compatible with retail-tier flows on Polygon. The on-chain verifier is a sibling of `KYCVerifier` that exposes `AMLVerifier.verify(proof, [wallet, blockHeight])`. Both verifiers must return true before a wallet's CLIENT role is activated.

W3C DID / VC anchoring. The KYC credential issued off-chain is shaped as a W3C Verifiable Credential (VC) bound to a W3C Decentralized Identifier (DID) rather than a free-form signed blob. The DID is anchored on-chain (`did:ethr:0x...` per the EIP-1056 standard), and the smart contract checks possession of the VC by means of the zk-SNARK proof rather than by storing the VC itself. This places the platform's identity layer inside the Self-Sovereign Identity (SSI) framework and aligns it with the European Blockchain Services Infrastructure (EBSI) reference architecture.

Privacy posture. A 2025 ResearchGate study reports a 97% reduction in exposed user data versus conventional on-chain KYC; Decker (2025) gives the same finding under a different threat model. Phase IV implementation will target deployment of both Circom 2.0 circuits to Polygon zkEVM Cardona testnet, tested with synthetic credential data from a mock KYC provider.

3.8 User Taxonomy and Onboarding Flows

The platform defines nine actors (A1–A9) following the project systems-analysis convention : five primary roles that take action, and four external systems or authorities. Tables **??**–

?? and Figure 3.10 summarise who each user is, what they may do, and where their data lives.

User Type	Tier	KYC Level	Wallet Type	Data Residence
Anonymous Visitor	—	None	None	Session only (Redis)
Registered Retail User	Tier 4	Level 1 (gov. ID + selfie)	MetaMask or optional ERC-4337	Off-chain primary; on-chain role only
Group Client	Tier 4	Level 1 + group verification	ERC-4337 abstracted	Off-chain + on-chain group contract
Local Bank Operator	Tier 3	Full institutional KYC	MetaMask (institutional)	Off-chain + on-chain role binding
Local Bank Approver	Tier 3	Full institutional + background check	MetaMask + Safe co-signer	Off-chain + on-chain role binding
National Bank Admin	Tier 2	Full institutional + regulatory license	Safe multisig mandatory	Off-chain + on-chain role binding
World Bank Admin (Governance)	Tier 1	Founding team + supervisor attestation	Safe 3-of-5 multisig	On-chain governance only

Label	Actor	Role
Primary Actors (initiate actions)		
A1a	Retail Client (pre-KYC)	Browsing only; no banking transactions.
A1b	Retail Client (KYC Tier 1)	Bronze/Silver credit tier; small loans up to the KYC-1 cap.
A1c	Retail Client (KYC Tier 2)	Gold/Platinum/Diamond; larger limits, group lending.
A2	Local Bank Admin (Approver)	Loan approver; risk officer; reviews AML alerts.
A3	National Bank Admin	Capital allocator; compliance officer; SAR review; freeze authority.
A4	World Bank Admin (Governance)	Parameter governor; system operator via multi-sig.
A5	AI Agent (optional)	Conversational add-on; read tools always permitted; write tools require explicit human confirmation gate before execution.
Secondary Actors (external systems or authorities)		
A6	Regulatory Authority	Read-only audit access via encrypted data package.
A7	Chainlink DON	Oracle, price feeds, automation.
A8	Blockchain Validator	Polygon zkEVM validity proof generation.
A9	External Auditor	Chainlink PoR verification.

Action	Client	LB Admin	NB Admin	WB Admin	AI Agent
View own loan status	YES	YES*	YES*	YES*	READ
Submit loan application	YES	NO	NO	NO	WRITE†
Pay installment	YES	NO	NO	NO	WRITE†
Submit KYC documents	YES	NO	NO	NO	NO
Approve loan application	NO	YES	NO	NO	NO
Reject loan application	NO	YES	NO	NO	NO
Freeze client account	NO	YES	YES	YES	NO
Set borrowing rate	NO	NO	YES	YES	NO
Change reserve ratio	NO	NO	NO	YES	NO
Upgrade LB borrowing cap	NO	NO	YES	YES	NO
View audit logs (all)	NO	YES*	YES*	YES	NO
Generate SAR report	NO	YES	YES	YES	NO
View reserve summary (public)	YES	YES	YES	YES	READ

*† Agent write tools require human confirmation. * Own tier only.*

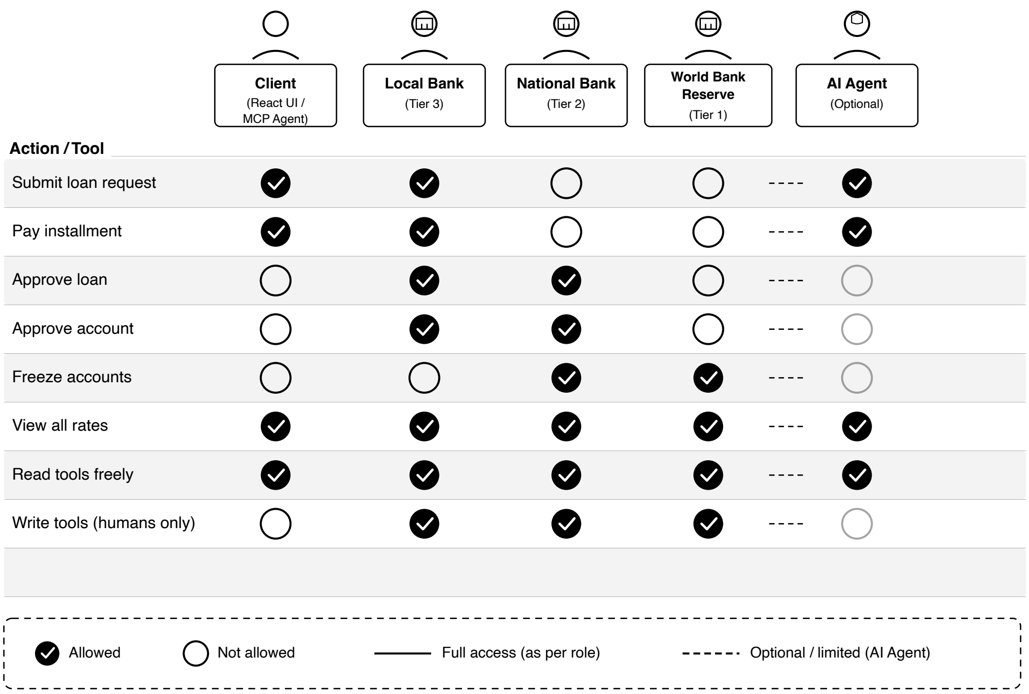

Figure 3.10: Actor permission matrix (primary actions)

3.8.1 ERC-4337 Account Abstraction for Retail Onboarding

Retail clients can sign up with email or Google and get a smart wallet—no seed phrase to remember—and the World Bank Reserve pays their gas fees on the testnet; users who prefer MetaMask can still use that instead.

Free gas is limited so it cannot be abused: only basic steps are sponsored before KYC, and stricter caps apply after identity is verified; if they use the optional chat agent, a short-lived session key lets it run only approved actions (with a spending limit and expiry).

3.8.2 Tiered Risk-Based KYC

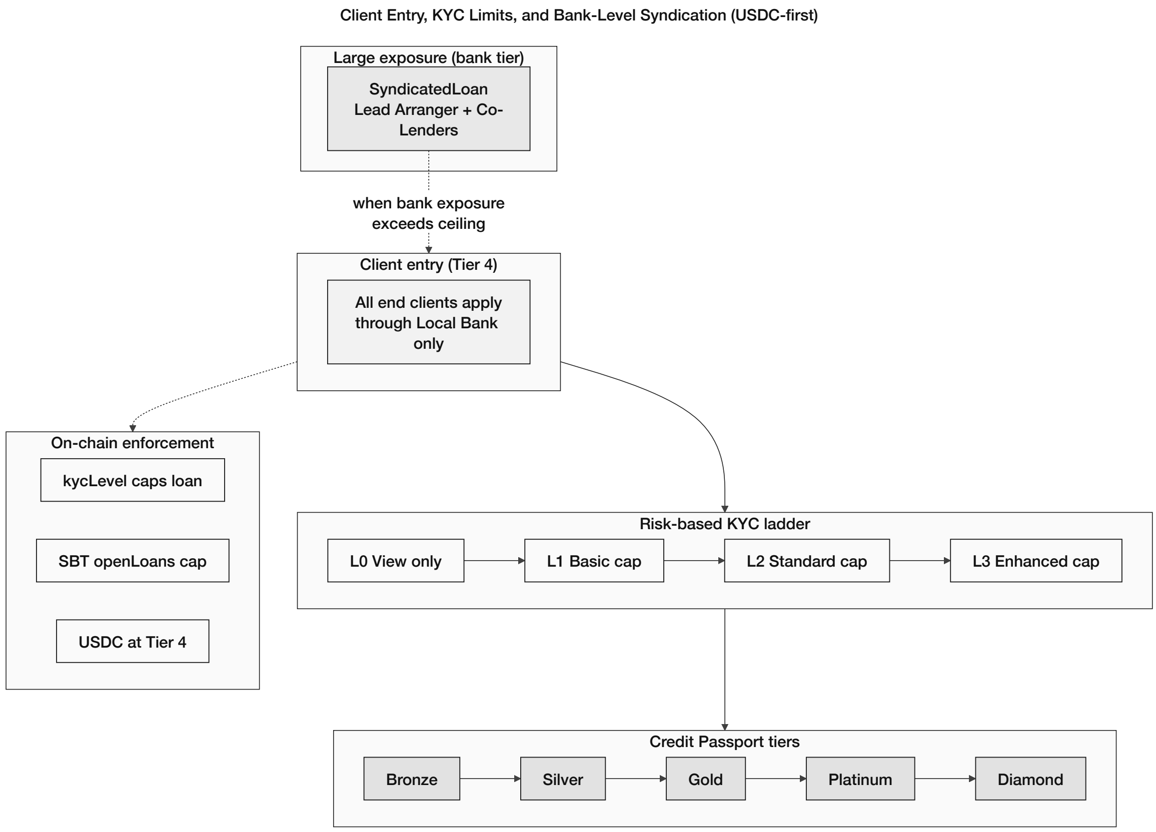

Figure 3.11: Client limits and bank-level syndication

KYC follows a four-step risk ladder (Table 3.18): higher limits require more documents and longer review. After approval the contract sets `kycVerified`, `kycLevel`, and a one-year `kycExpiry`; only document hashes and zk proofs go on-chain—never raw ID or biometrics.

Table 3.18: Tiered, risk-based KYC ladder

Risk Level	Required Documents	Borrow Limit	Verify Time
Level 0 (Unverified)	None	View only	Instant
Level 1 (Basic)	Government ID photo + selfie liveness check	Up to 0.1 ETH-equivalent (USDC)	2–5 min (automated)
Level 2 (Standard)	Government ID + proof of income + bank statement	Up to 2 ETH-equivalent	24 h (human review)
Level 3 (Enhanced)	Full document set + video interview	Up to 10 ETH-equivalent	48–72 h

3.8.3 Five-Stage Retail Onboarding Funnel

Retail clients are onboarded to the system in five steps, as shown in the following graph (Figure 3.12): you can browse the system without an account, or you can open a wallet or smart account; then the client has to complete KYC, which will take about five minutes, and take his first loan after accepting all the terms; then he will unlock all the advanced features of the smart contract and will be able to navigate into the user dashboard.

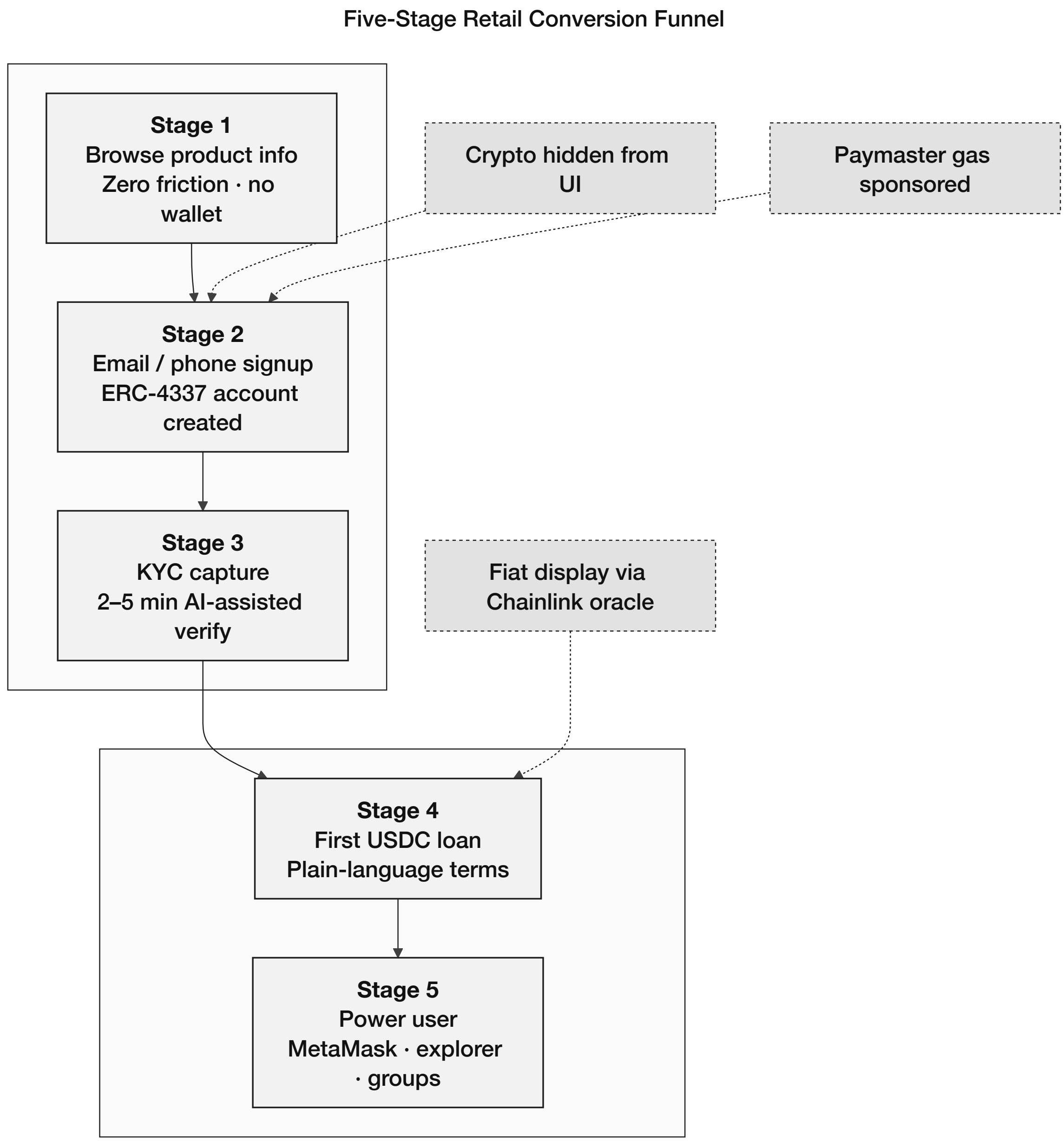

Figure 3.12: Five-stage retail onboarding funnel

3.8.4 Optional Conversational Interface (Add-On)

The conversational AI is an optional feature that a client may want to use for guidance on how to interact with the system. The conversation AI uses information through RAG and the information given in the chat by the clients.

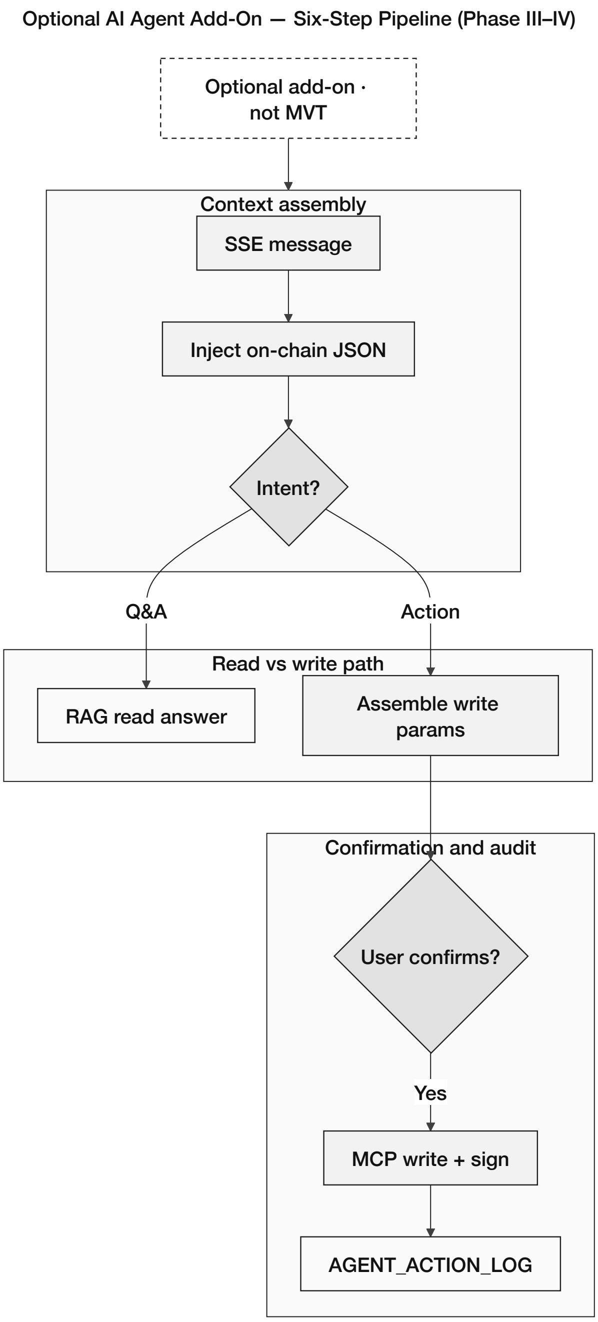

Figure 3.13: Optional AI agent add-on (Phase III–IV, excluded from MVT)

3.9 Kinked Interest Rate Model

Borrowing cost follows a kinked utilization model (Compound/Aave pattern) with optimal utilization $U^* = 80\%$ for retail pools. Utilization is $U = L/(L+B)$, where L is lent amount and B is available liquidity.

Below the kink ($U \leq U^*$), the rate rises gently:

$$r_b(U) = r_0 + \frac{U}{U^*} \cdot r_1, \quad U \leq U^* \qquad \text{(Formula 18)}$$

At $U = 0$ the rate is base r_0; at $U = U^*$ it reaches $r_0 + r_1$ (r_1 = below-kink slope).

Above the kink ($U > U^*$), the rate rises steeply to incentivise repayment and new deposits:

$$r_b(U) = r_0 + r_1 + \frac{U - U^*}{1 - U^*} \cdot r_2, \quad U > U^* \qquad \text{(Formula 19)}$$

r_2 is the jump multiplier above the kink. On-chain guards enforce `r2 >= MINIMUM_JUMP_MULTIPLIER` and `U_star <= MAXIMUM_KINK_POINT` before any parameter update.

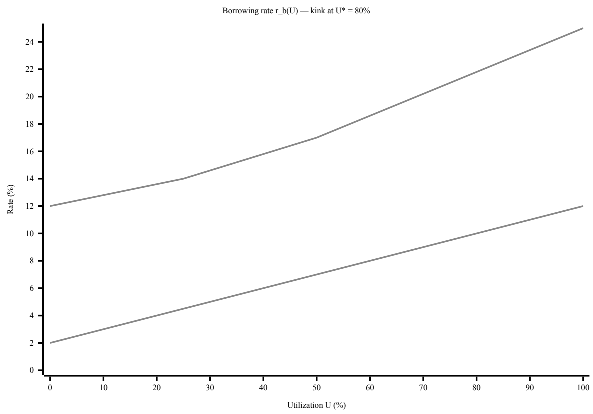

Figure 3.14: Kinked utilization-based borrowing rate

3.10 Liquidation Engine

The **LiquidationEngine** contract monitors health factor (HF) and pays third-party liquidators a 5–8% bonus when $HF < 1.0$. HF models vary by loan type:

Tier 4 over-collateralized retail.

$$HF = \frac{C \times LT}{D} \tag{3.1}$$

Collateral value C (oracle-priced) times liquidation threshold LT, divided by debt D; liquidate when $HF < 1.0$.

Tier 4 group lending.

$$HF_{\text{group}} = \frac{\text{total\_group\_collateral} \times LT}{\text{total\_group\_outstanding\_debt}} \tag{3.2}$$

Pool-level health score; liquidation loss is shared across all group members.

Tier 4 credit-based (no collateral). HF is not applicable. Default triggers SBT `riskTier` downgrade and `lastDefault` timestamp [69]; no collateral to seize.

Tier 1–3 institutional.

$$HF_{\text{inst}} = \frac{\text{on\_chain\_reserve} - \text{min\_reserve}}{\text{outstanding\_borrowing}} \tag{3.3}$$

Parent-tier reserve surplus above the mandatory minimum acts as effective collateral.

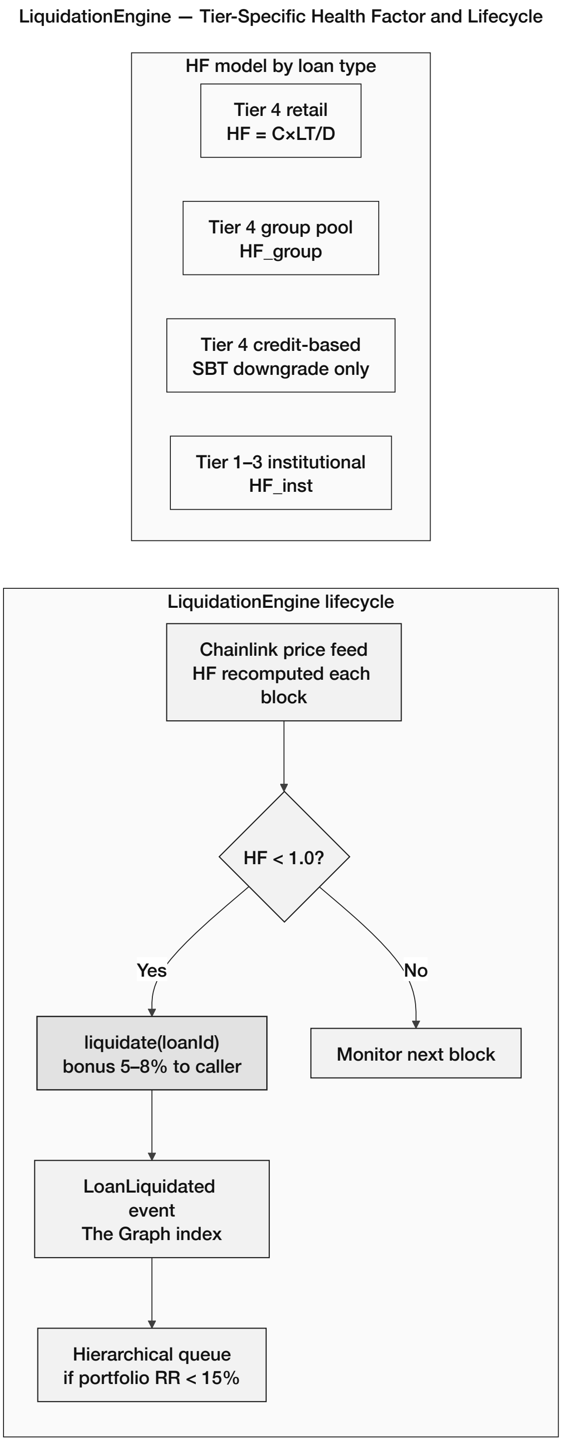

Figure 3.15: LiquidationEngine tier models and lifecycle

3.11 SavingsVault and FixedDeposit Modules

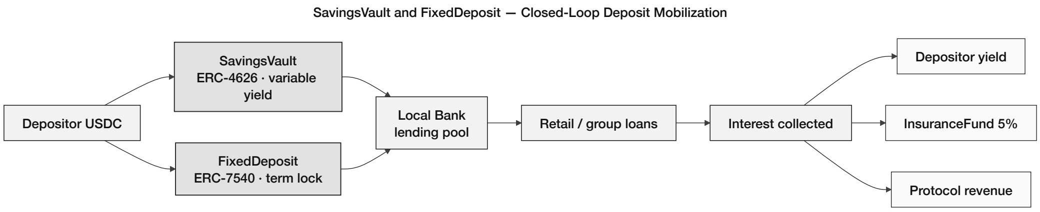

Figure 3.16: SavingsVault and FixedDeposit closed loop

3.12 On-Chain Credit Passport (Soulbound Token)

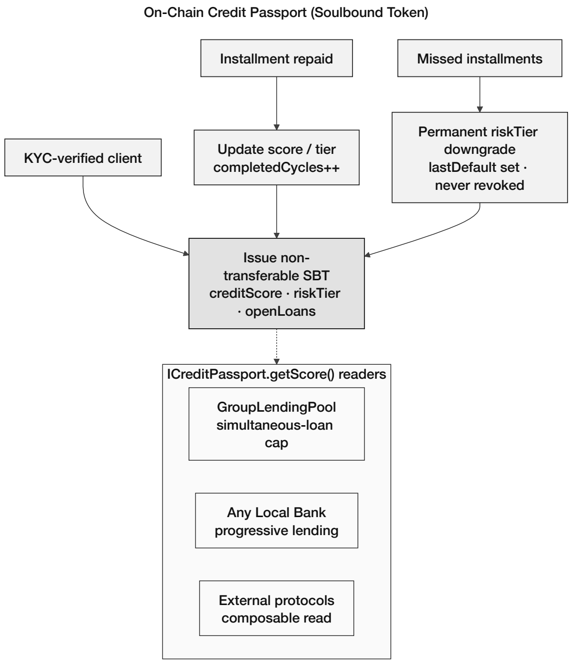

Figure 3.17: On-chain Credit Passport SBT lifecycle

3.13 Cross-Chain Bridge Architecture

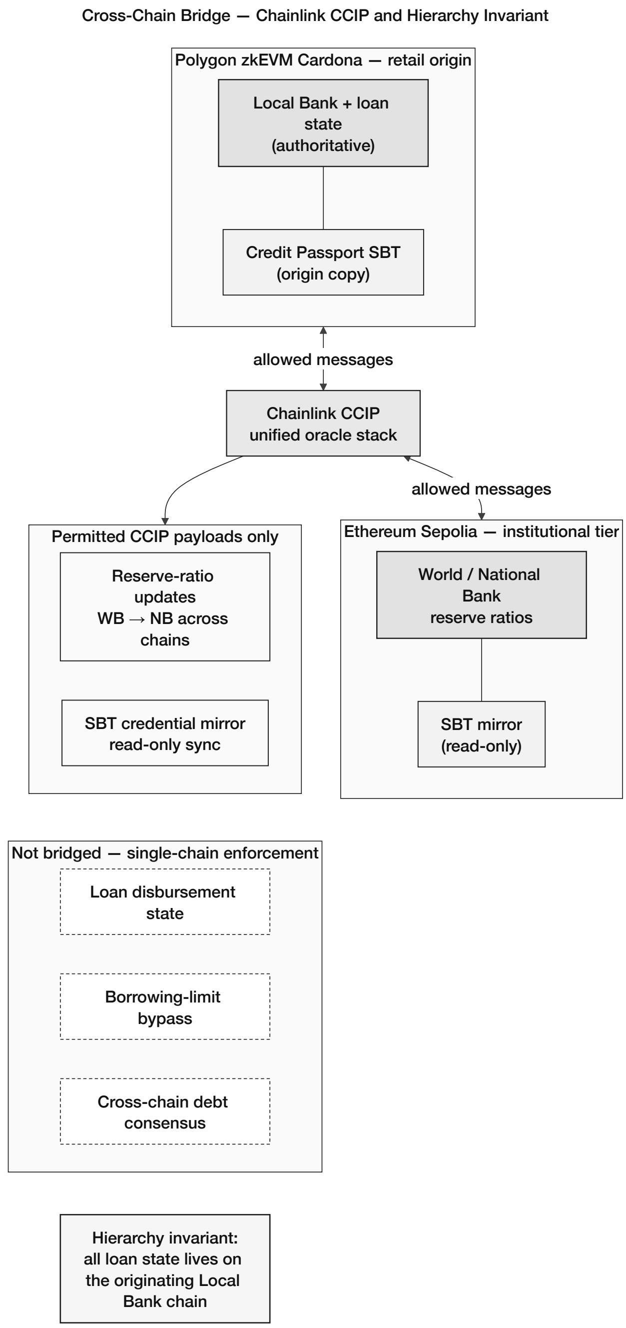

Figure 3.18: Chainlink CCIP cross-chain hierarchy invariant

3.14 Multi-Entity and Cross-Tier Capital Operations

Full formula derivations and step-by-step explanations for this section will be expanded after pre-thesis 1 in the next version of this document. For now, the following formulas are referenced by name only: Utilization Rate (U), Kinked Borrowing Rate (Formulas 18–19), Interbank Rate (r_{IB}), Upward Deposit Rate Spread ($r_{\text{up}} < r_{\text{down}} - \delta$), FX Conversion / Treasury Swap, Platform Net Interest Allocation (NetInterest), Credit Velocity (CV), Reserve Ratio (RR), and Health Factor (HF) variants for retail, group, and institutional tiers.

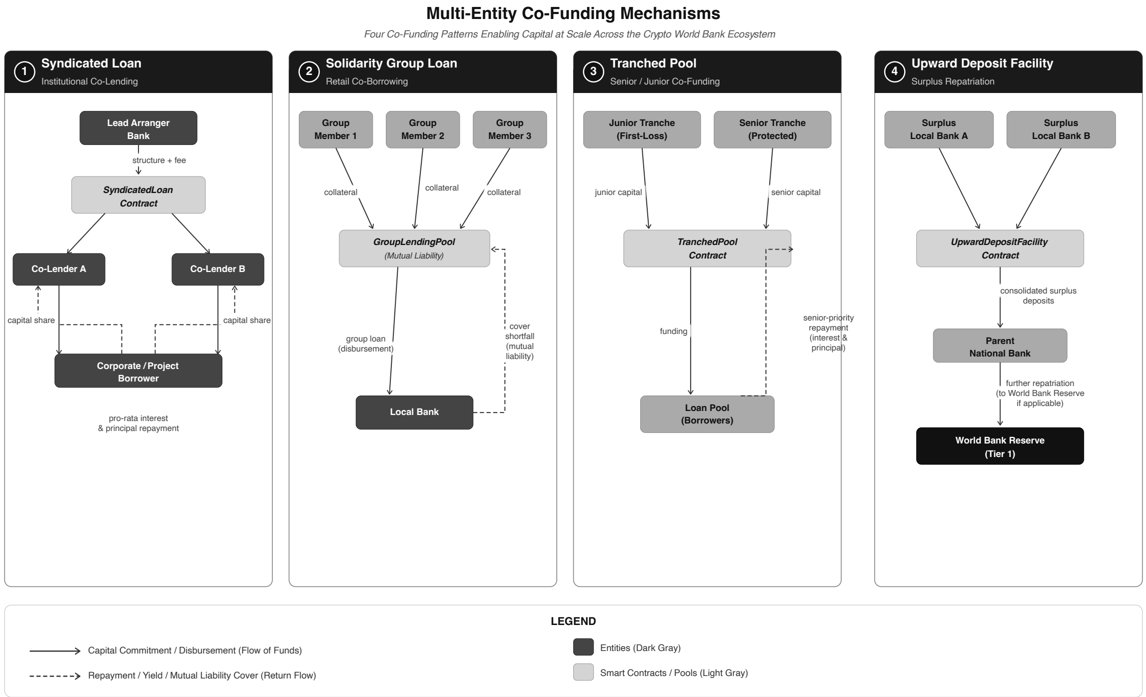

Figure 3.19: Multi-entity and cross-tier capital operations

3.15 System Modeling

3.15.1 Use Case Diagram

Figure 3.20 shows nine actors (A1–A9) and 20 use cases; see Section 3.8 for the full user taxonomy.

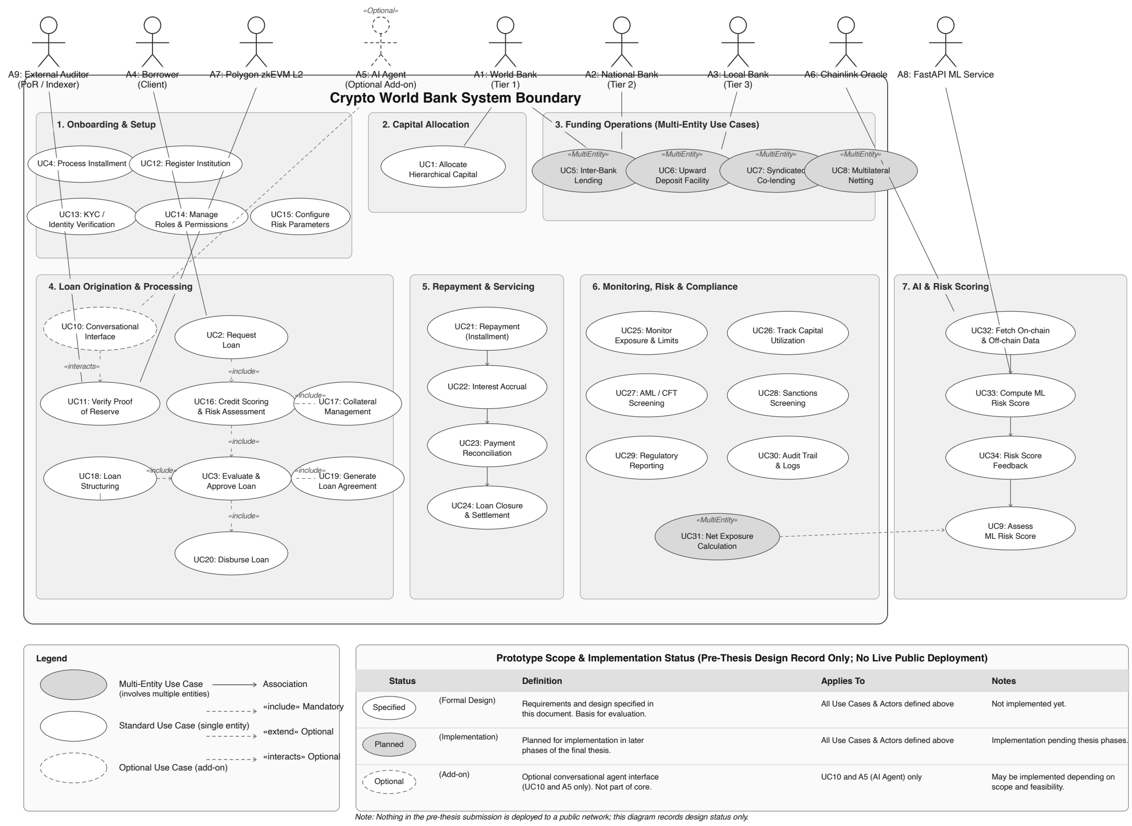

Figure 3.20: Use-case diagram (nine-actor taxonomy)

3.15.2 Activity Diagrams

This part of the architecture design shows how each activity flow is managed in the activity diagrams. It shows the start of an action and why those actions lead to what end, and how the decisions make clients do certain actions with the system.

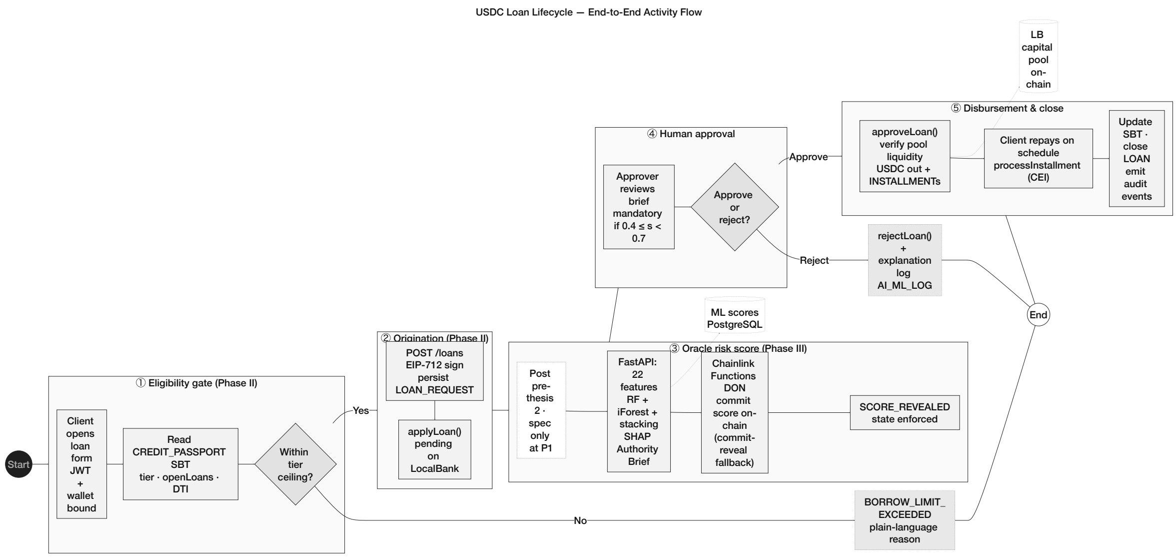

Figure 3.21: USDC loan lifecycle activity flow

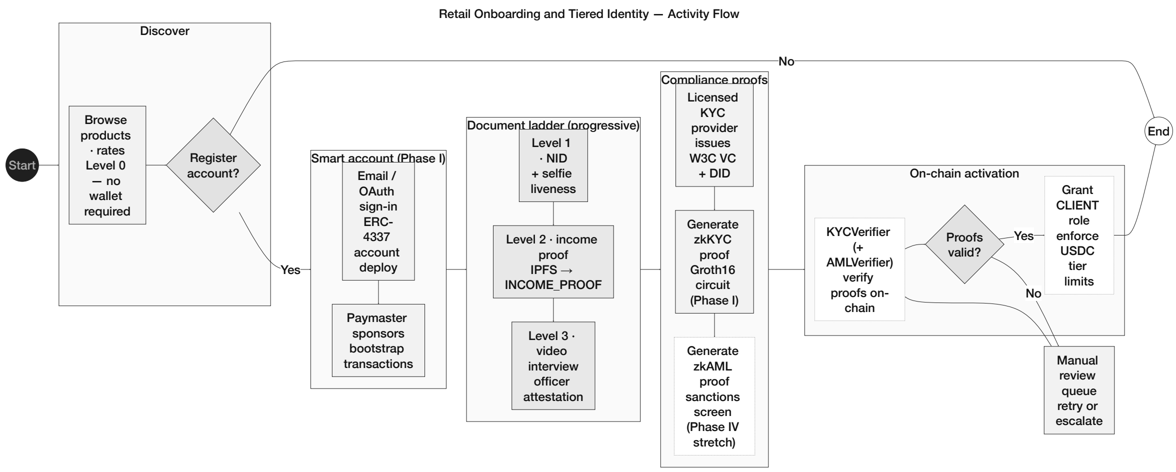

Figure 3.22: Retail onboarding and identity activity flow

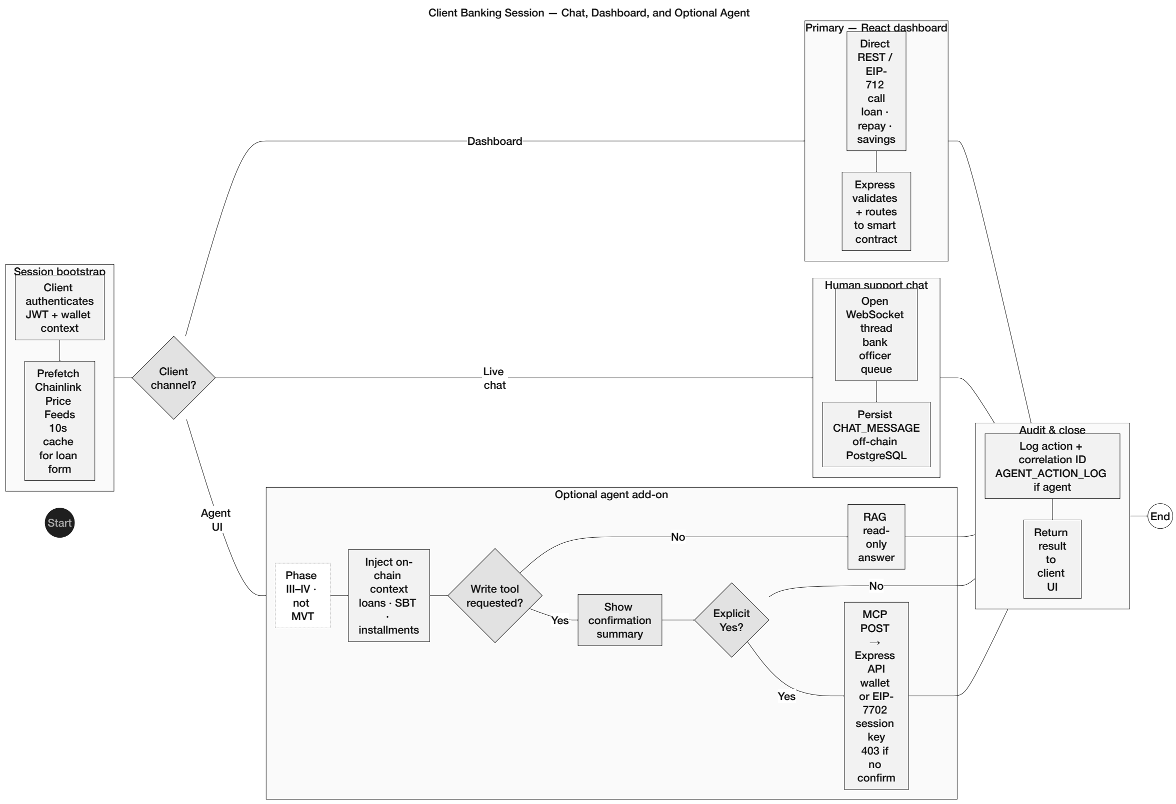

Figure 3.23: Client banking session activity flow

3.15.3 Data Flow Diagrams

Here is a visual representation of how the Level-0 context diagram and Level-1 decomposition work in different sectors of the system.

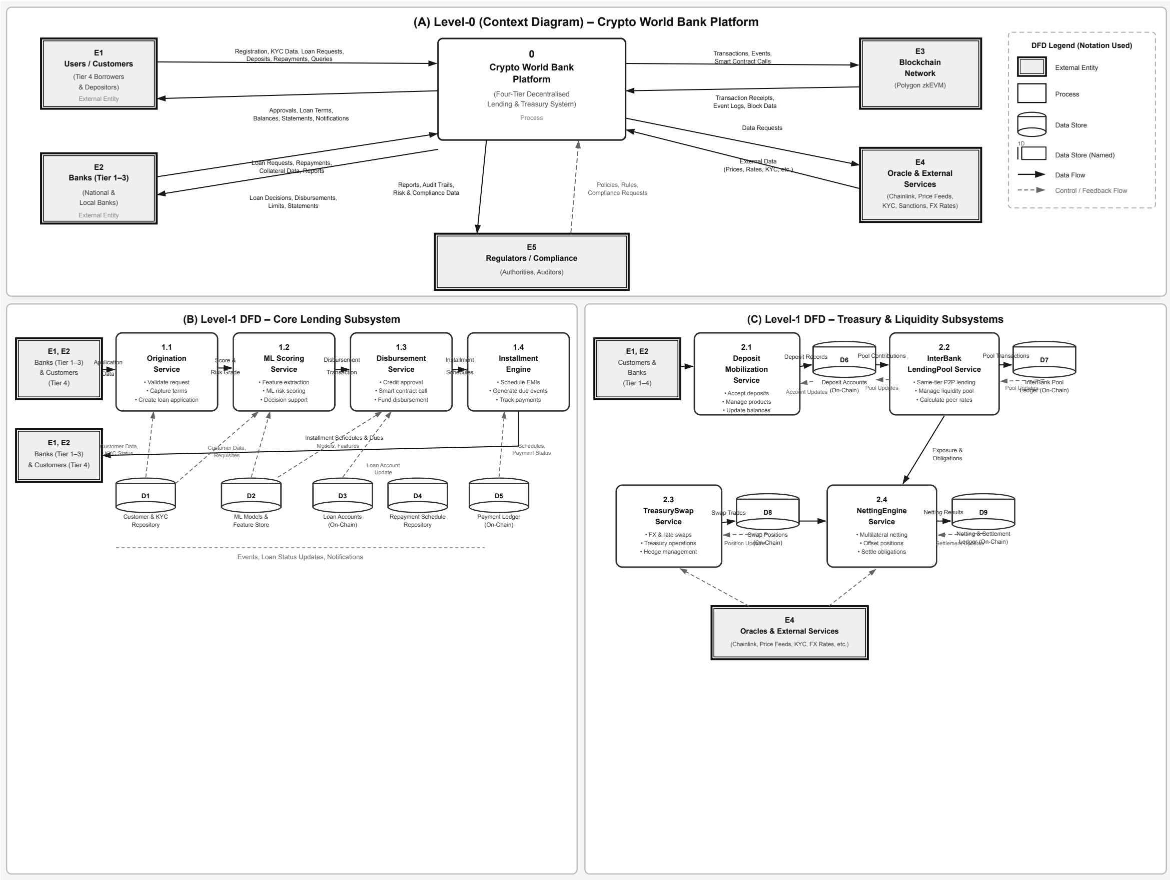

Figure 3.24: Data-flow diagrams

3.15.4 Sequence Diagrams

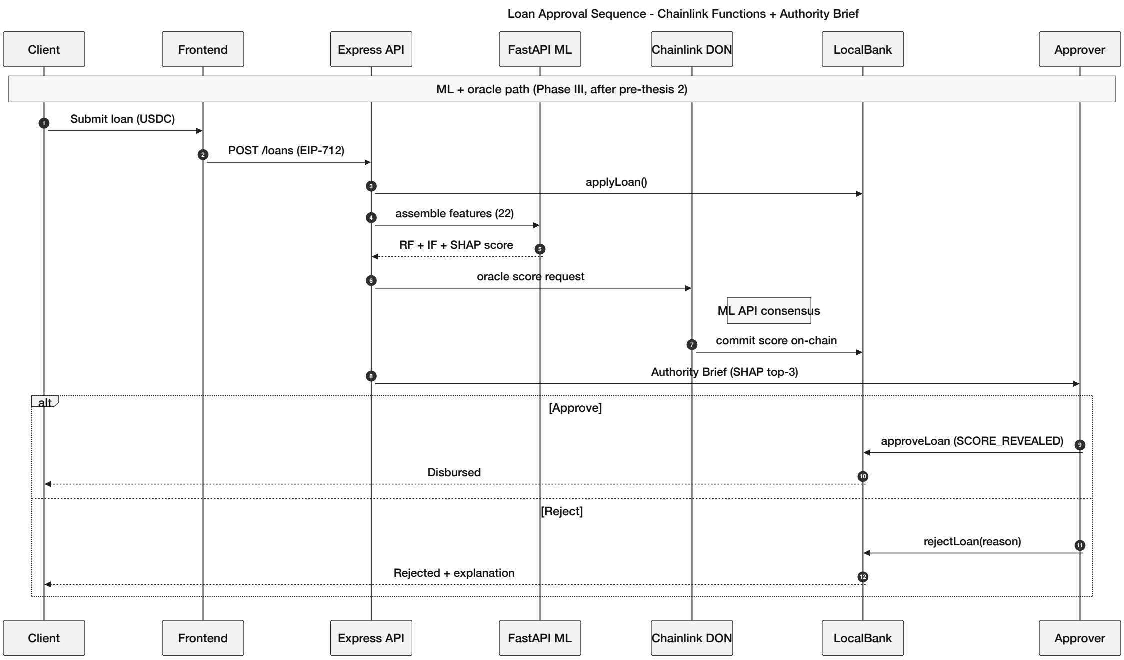

Figure 3.25: Loan approval sequence diagram

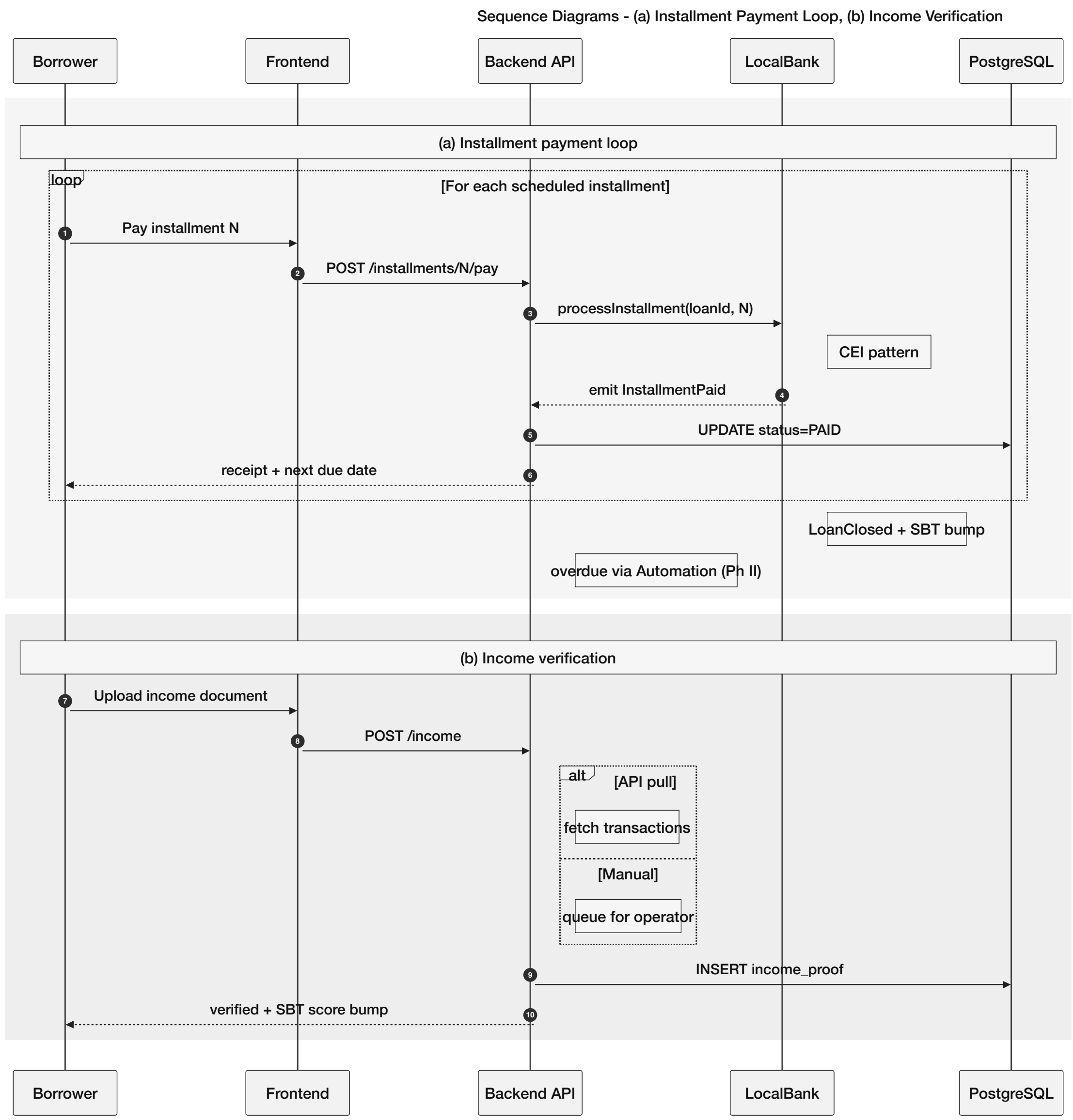

Figure 3.26: Sequence diagrams

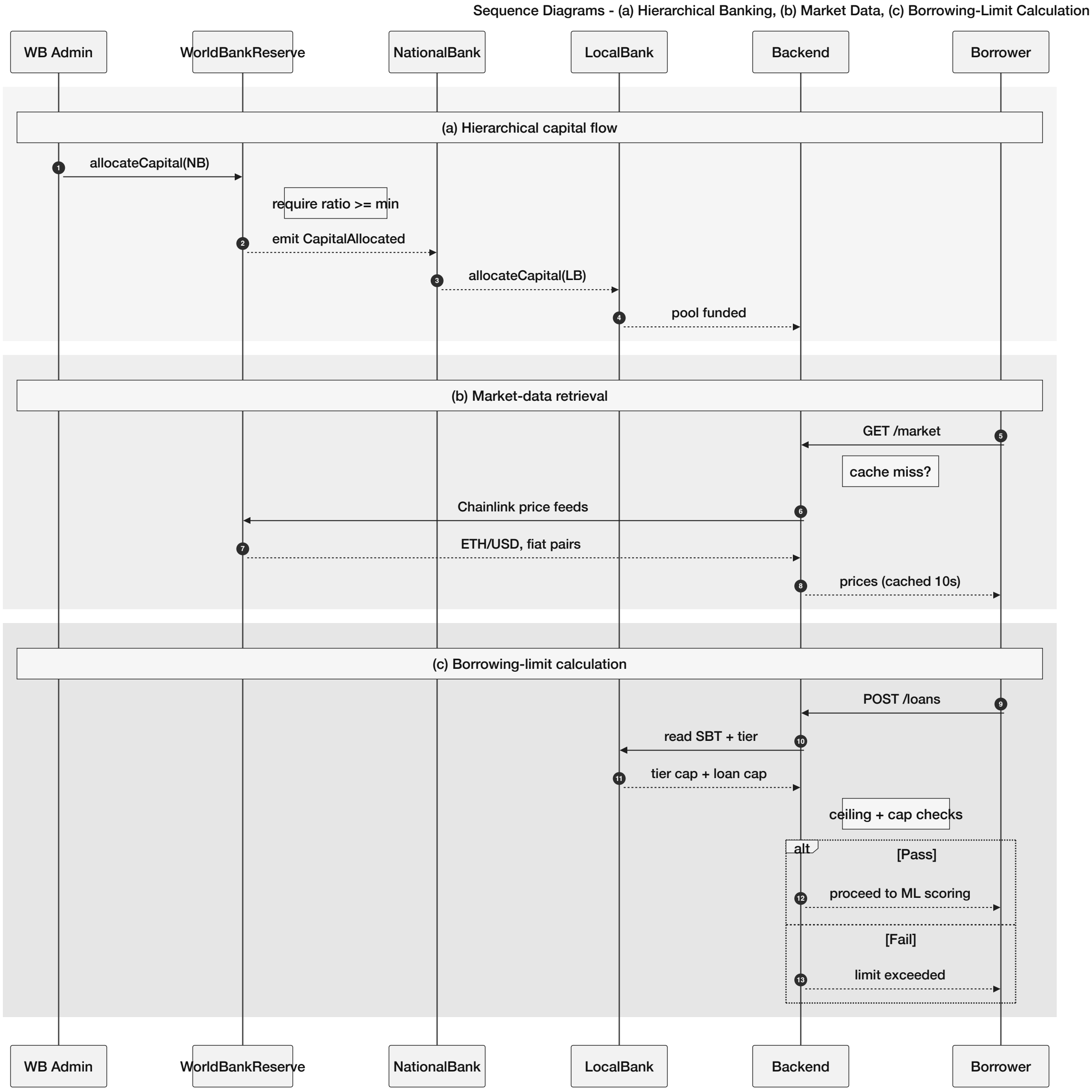

Figure 3.27: Sequence diagrams

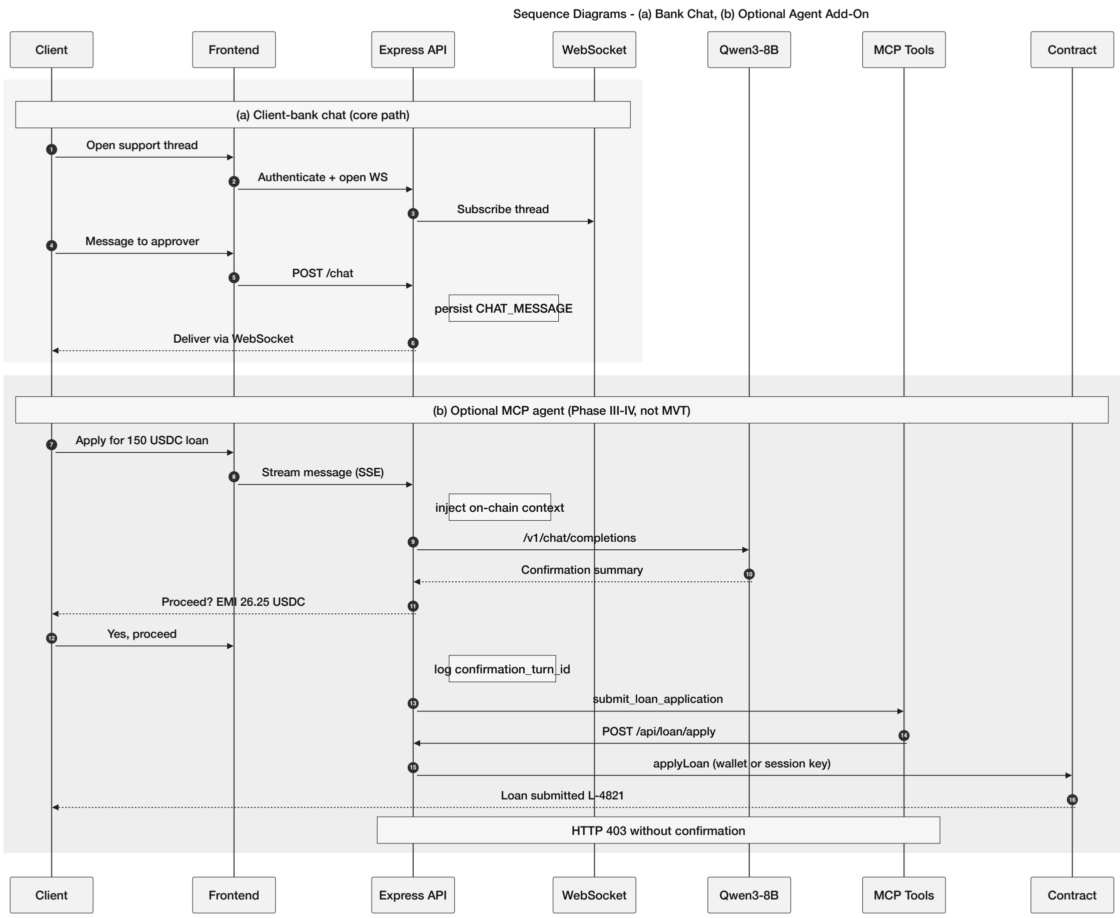

Figure 3.28: Sequence diagrams

3.15.5 Four-Tier Capital Flow

Figure 3.29 illustrates the hierarchical capital flow and cascading repayment structure.

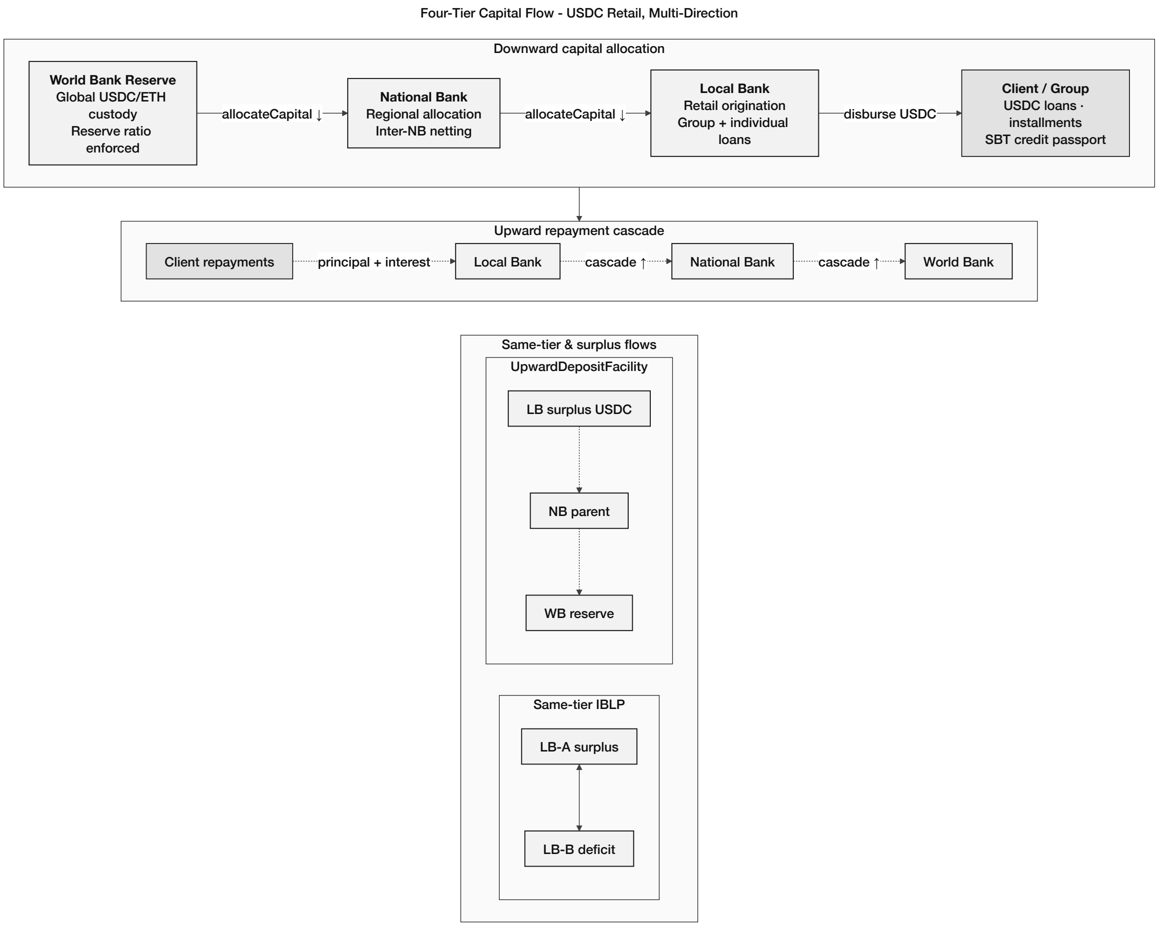

Figure 3.29: Four-tier hierarchical capital flow

3.16 Banking Product Suite

The formulas for this section will be expanded and implemented for the whole project as we move on to Phase II and Phase III in our development timeline; fuller explanations will follow in later versions of this document.

3.16.1 Savings Products

$$A = P\left(1 + \frac{r}{n}\right)^{nt} \tag{3.4}$$

$$\text{NetInterest} = \text{InterestCollected} - \text{DepositorYield} - \text{InsuranceAllocation} \tag{3.5}$$

3.16.2 Checking and Transactional Accounts

$$EMI = \frac{P \cdot r \cdot (1+r)^n}{(1+r)^n - 1} \tag{3.6}$$

3.16.3 Group / Solidarity Lending

$$\text{Share}_i = \frac{\text{LoanTotal}}{N_{\text{members}}} \tag{3.7}$$

$$\text{DTI} = \frac{\text{existingDebt} + \text{requestedAmount}}{\text{estimatedMonthlyIncome}}, \quad \text{reject if DTI} > 0.40 \tag{3.8}$$

3.17 Governance Framework

3.17.1 Governance Execution Paths and Emergency Controls

Parameter changes and upgrades route through a 24–48 h `TimelockController` (5 min in lab demo). Emergency path: 4-of-7 Security Council Safe may pause, upgrade, or revoke roles within 2 h (mandatory 48 h disclosure); cannot change system parameters.

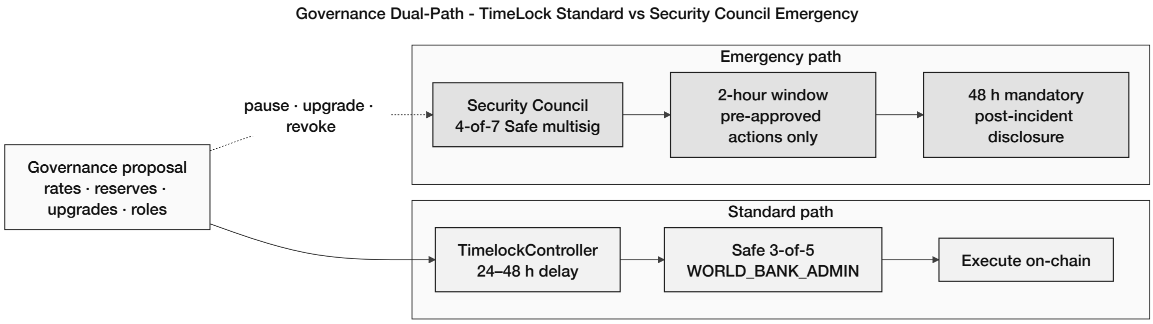

Figure 3.30: Governance dual-path

3.17.2 Network Membership Governance

Table 3.19: Network membership governance

Governance Aspect	Implementation
Member on-boarding	World Bank owner registers National Banks; National Banks register Local Banks; Local Banks designate approvers — all enforced on-chain.
Member off-boarding	Deactivation flags in smart contracts; cascading access revocation.
Regulatory oversight	Audit log emission via smart contract events; planned read-only regulator dashboard.
Permission structure	Hierarchical: Owner → National Bank → Local Bank → Approver → Retail Client; enforced by on-chain role check modifiers.
Network operations	Pause/unpause mechanism for emergency response; emergency withdrawal for critical situations.

3.17.3 Business Network Governance

Table 3.21: Business network governance

Governance Aspect	Implementation
Business charter	Defined in project documentation; operational parameters coded in smart contract constants.
Common services	Reserve management, loan lifecycle orchestration, event-driven notification system.
Business SLA	Testnet phase: best-effort availability. Production phase: 99.5% target API-layer uptime (team-controlled; multi-region deployment). Chain-level liveness is determined by Polygon zkEVM batch submission and Ethereum L1 proof verification; the team monitors network status and maintains degraded-mode fallbacks (read-only dashboard) if chain activity pauses.
Regulatory compliance	Architecture designed for audit trail generation; data partitioning supports GDPR-style data subject requests.

3.17.4 Technology Infrastructure Governance

Table 3.23: Technology infrastructure governance

Governance Aspect	Implementation
Distributed IT structure	Client-side frontend (decentralised delivery); blockchain layer (fully decentralised); backend API (centralised, horizontally scalable).
Technology assessment	Continuous evaluation of EVM alternatives (L2 rollups, sidechains) for cost and performance optimisation.
On-chain / off-chain data services	Clearly partitioned (see Table 3.16); event listeners synchronise state between layers.
Risk mitigation	Smart contract pause mechanism; ReentrancyGuard; input validation; planned formal security audit.

3.17.5 Regulatory Compliance Considerations

Prototype runs on public testnets only; architecture emits audit logs for sandbox review; production targets regulatory sandbox programmes. zkKYC design: Section 3.16.3. Planned SAR handling is summarised in Figure 3.31.

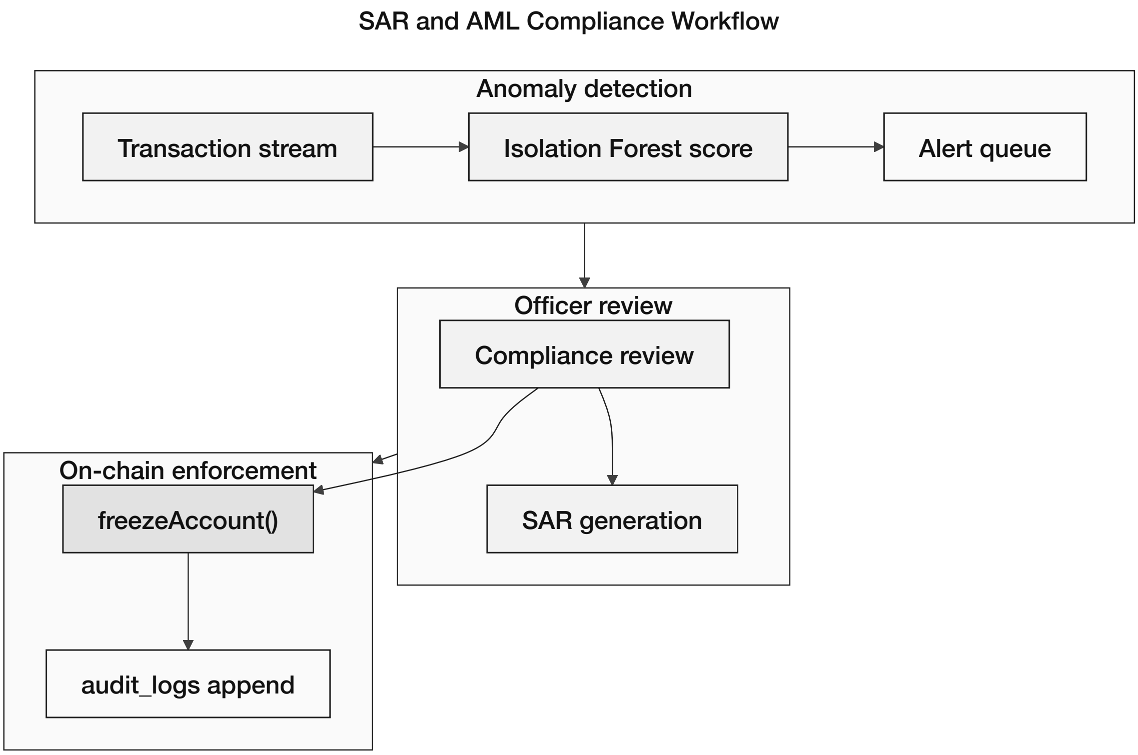

Figure 3.31: SAR and AML compliance workflow

3.17.6 Asset Tokenization

- **Specified (Phase I):** Native blockchain currency (ETH) as reserve and loan currency on testnet.
- **Planned extension:** ERC-20 stablecoins (USDC, USDT) for lending operations; tokenized collateral instruments where collateral is insufficient.

3.18 Five-Layer Defense-in-Depth Security Architecture

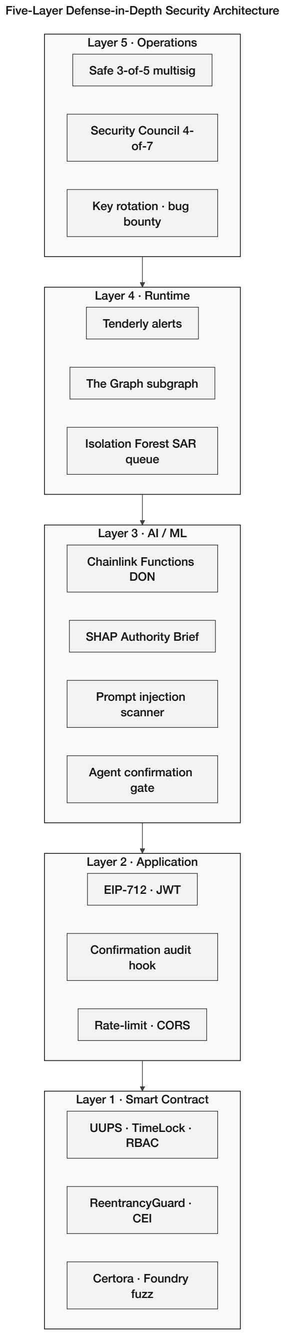

Figure 3.32: Five-layer defense-in-depth security architecture

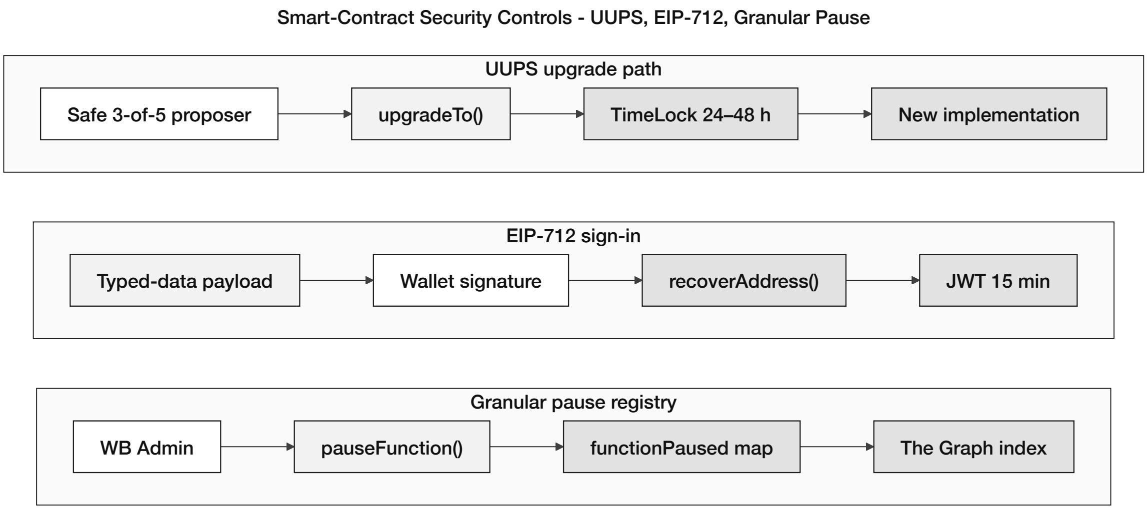

Figure 3.33: Smart-contract security controls

Security is not contracts alone—it is five stacked layers (Figure 3.32; Table 3.25). An attacker must get past operations, monitoring, ML checks, the API, and only then the chain. Section 3.19 maps threats to each layer.

Table 3.25: Five-layer defense-in-depth model

Layer	Focus	CWB approach
L5	Operations	Multisig admin keys (Safe 2-of-3 / 3-of-5), hardware wallet, key rotation
L4	Monitoring	Tenderly alerts, The Graph event history, simulate before deploy
L3	ML / fraud	RF + IF score, SHAP explanations, on-chain oracle commit
L2	Application	Wallet-signed JWT, rate limits, input checks, prompt scanning
L1	Contracts	Reentrancy guards, RBAC, pausable functions, Slither + Foundry tests

3.18.1 Smart Contract Security Controls

Core lending contracts (Figure 3.33) follow standard hardening plus a few banking-specific rules:

- **CEI + reentrancy guards.** State updates before external calls on `disburseLoan`, `processInstallment`, and `allocateCapital` [7].
- **Governance safety.** UUPS upgrades need a Safe 3-of-5; roles expire by block; functions can be paused individually.
- **Economic guards.** Chainlink feeds with circuit breakers; flash-loan exposure limited until Phase II–III TWAP checks.

- **Agent and API (L2–L5).** Failed confirmations lock the agent session; disbursement addresses whitelisted at KYC; prompt injection scanned before LLM context; admin keys always multisig.

3.19 Threat Model and Security Controls

Threats span contracts, APIs, ML, monitoring, and admin keys—not just Solidity bugs. Table 3.26 maps each vector to a layer and mitigation.

Table 3.26: Threat model and security controls mapping

Threat	Layer	Attack Vector	Mitigation
Reentrancy	L1	Recursive external calls	CEI; `ReentrancyGuard`; Foundry tests
Oracle / flash-loan abuse	L1	Price or atomic-loan manipulation	Chainlink feeds; TWAP; pause on anomaly
Governance capture	L1+L5	Privileged role abuse	Multisig; TimeLock; role expiry
Sybil / fake groups	L1+L3	Many wallets bypass limits	RBAC; KYC; ML graph checks
JWT / API abuse	L2	Stolen session or bulk requests	Short JWT; rate limits; input validation
ML / LLM evasion	L3	Fraud or wrong compliance answers	SHAP review; RAG; human approver
Monitoring blind spots	L4	Small attacks below thresholds	Multi-trigger alerts; event index
Admin key loss	L5	Compromised signer	Multisig; hardware wallet; tx simulation

3.20 Final Specifications and Requirements

Table 3.27 consolidates the functional, non-functional, and constraint requirements implied by the architecture above and the MVT boundary (Section 4.1). At pre-thesis 1 these are *specified*; post–pre-thesis 2 they are *verified* on testnet.

Table 3.27: Final specifications and requirements

Category	Requirement (summary)
Functional	Four-tier capital flow; loan request → approval → disbursement → repayment; RBAC per tier; reserve-ratio enforcement; optional deposits, group lending, multi-entity flows (Ch. 3).
Non-functional	Public testnet deployment; sub-cent L2 gas per loan lifecycle; EIP-712 + JWT API auth; SHAP-backed ML decision support; append-only audit logs.
Constraints	Testnet tokens only (no real funds); no native governance token in prototype; ML training and live scoring after pre-thesis 2; agent optional (Could).
Compliance targets	MiCA/GENIUS stablecoin use; EU AI Act human oversight for credit scoring (design); FATF Travel Rule for inter-tier transfers >USD 1,000 (Section 3.17.5).

3.21 Societal Impact

This section will be expanded in Phase III (pre-thesis 2).

3.22 Environmental Impact

This section will be expanded in Phase III (pre-thesis 2).

3.23 Ethical Issues

This section will be expanded in Phase III (pre-thesis 2).

Chapter 4

Methodology

This project and its planning have been structured in accordance with the Blockchain Olympiad Bangladesh: Guideline and Evaluation Scheme and the final-year project evaluation rubrics.

Core vs. optional. The thesis centres on on-chain banking—contracts, lending, oracle scoring, and verification. The conversational agent (Section 4.4) is a Could-have add-on; deposits and loans work without it. Table 4.1 breaks down priorities by phase.

Table 4.1: Core vs. optional capabilities

Capability	Priority	Primary interface / phase
Four-tier smart contracts	Must (Ph. I–II)	React UI + wallet; Hardhat/Foundry tests
On-chain loan lifecycle	Must (Ph. II)	React forms; `LoanController` state machine
Chainlink Functions oracle	Must (Ph. III)	FastAPI → DON → on-chain score commit
RF + Isolation Forest + SHAP	Must (Ph. III)	Approver Authority Brief; client decline reasons
Authority Brief UI	Should (Ph. II–III)	Bank approver dashboard (React) — *not* agent-only
MCP conversational agent	Could (Ph. III–IV)	Alternate path to same Express.js API
LLM evaluation protocol	Could (Ph. IV)	Red-team + task accuracy (Section 4.13.1)

4.1 Minimum Viable Thesis (MVT) Checklist

The tables below are the **graded pass line** for the final thesis—everything else is either spec in this report or Future Work. Three claims matter: hierarchical architecture (C1), oracle-mediated explainable ML (C3), and group lending specification (C2).

Module	Status	Examiner expectation
Four-tier smart-contract hierarchy	**Primary (C1)**	Must deploy + test on public testnet; reserve and RBAC demonstrated
Oracle-mediated RF + IF + SHAP pipeline	**Primary (C3)**	Specified in pre-thesis 1; must train, benchmark, and show Authority Brief + audit log after pre-thesis 2 (Section 4.3.1)
Solidarity `GroupLendingPool` spec	**Primary (C2)**	Specification + minimal PoC sufficient; full mutual-liability lifecycle is Phase II stretch
GNN (GraphSAGE) ablation	Future Work	Design only at pre-thesis; optional ablation if time permits
Federated learning across banks	Future Work	Requires 500+ tx/bank; not MVT
zkAML second circuit	Future Work	KYC circuit only in MVT scope
CCIP cross-chain bridge	Assumed infra	Single-chain testnet sufficient for MVT; CCIP is integration, not research novelty
Certora + Foundry invariant suite	Stretch (MVT+)	One reserve invariant OR one Foundry fuzz test strongly recommended
MCP conversational agent	Optional add-on	Excluded from MVT; Phase III–IV Could
ERC-4337 full Paymaster stack	Stretch	WalletConnect path sufficient for MVT demo
Multi-entity lend / fund / FX suite (IBLP, SyndicatedLoan, UpwardDeposit, TreasurySwap)	**Primary arch. (C1 ext.)**	Fully specified Ch. 3 (§3.14); Phase II–III; demo of one flow recommended
Syndicated / tranched / NettingEngine	Stretch (Phase III)	Specified; full suite beyond MVT minimum

#	Deliverable	Phase	Done when (acceptance criterion)
1	World / National / Local Bank contracts deployed on Polygon zkEVM Cardona	I	Addresses in Appendix C manifest; role-gated functions callable
2	Loan request → approver review → disbursement → one installment repayment (E2E)	II	Screen recording + tx hashes on block explorer
3	Reserve-ratio / borrowing-limit enforcement visible on-chain	II	Hardhat or Foundry test proves revert on violation
4	FastAPI ML service: RF + IF + stacking → composite score s	II–III	`POST /score` returns JSON in $<$50 ms on laptop
5	SHAP + anomaly deviation report + Authority Brief UI	II–III	Approver sees brief on sample loan; decline shows top-3 SHAP reasons
6	`AI_ML_SECURITY_LOG` row per inference	III	DB row contains `shap_json`, `model_version`, `loan_id`
7	Commit-reveal or centralized relay score commit before approval	III	Contract in `SCORE_REVEALED` before `approveLoan` succeeds
8	BCCC-DeFiFraudTrans benchmark table (precision, recall, F1, AUC)	II–III	Table in Section 4.3.1 populated with measured values
9	300-client Foundry simulation OR documented gas-cost table	IV	Appendix C manifest + reserve dynamics output
10	Open-source repo README with MVT run instructions	IV	Panel can reproduce steps 1–5 from README
Optional (not graded for MVT pass): agent chat, GNN ablation, Certora proof, zkAML, CCIP bridge.			

Table 4.2: Pre-thesis 1 deliverable boundary

Stage	What examiners should expect
Pre-thesis 1 (this report)	Formal specification: architecture (Ch. 3), literature (Ch. 2), methodology and phase plan (Ch. 4), feasibility framing (Ch. 5). Smart-contract interfaces, ERD, diagrams, MVT checklist. ML pipeline *documented only*—no trained models or benchmark table with measured metrics.
Pre-thesis 1 (demo)	**React platform frontend** is available for live demonstration (wallet connect, tier navigation, loan/savings UI shells, approver views). Core banking does not depend on the agent.
Pre-thesis 1 (optional demo)	**Optional conversational agent** (Qwen3-8B + MCP) may be shown if schedule permits; it is a Could-have add-on, not required for pre-thesis 1 pass.
After pre-thesis 2	Deploy contracts on Cardona; E2E loan lifecycle; train RF/IF on BCCC; wire oracle and Authority Brief; populate MVT items 4–8 with evidence; Phase IV simulation and proofs as stretch.

4.2 Development Methodology

We adopted a **lightweight Agile/Scrum** methodology tailored for an academic prototype with a fixed two-month development window. The iterative approach enables incremental delivery of demonstrable features while accommodating evolving requirements inherent in research-oriented development. Figure 4.1 illustrates our Agile process.

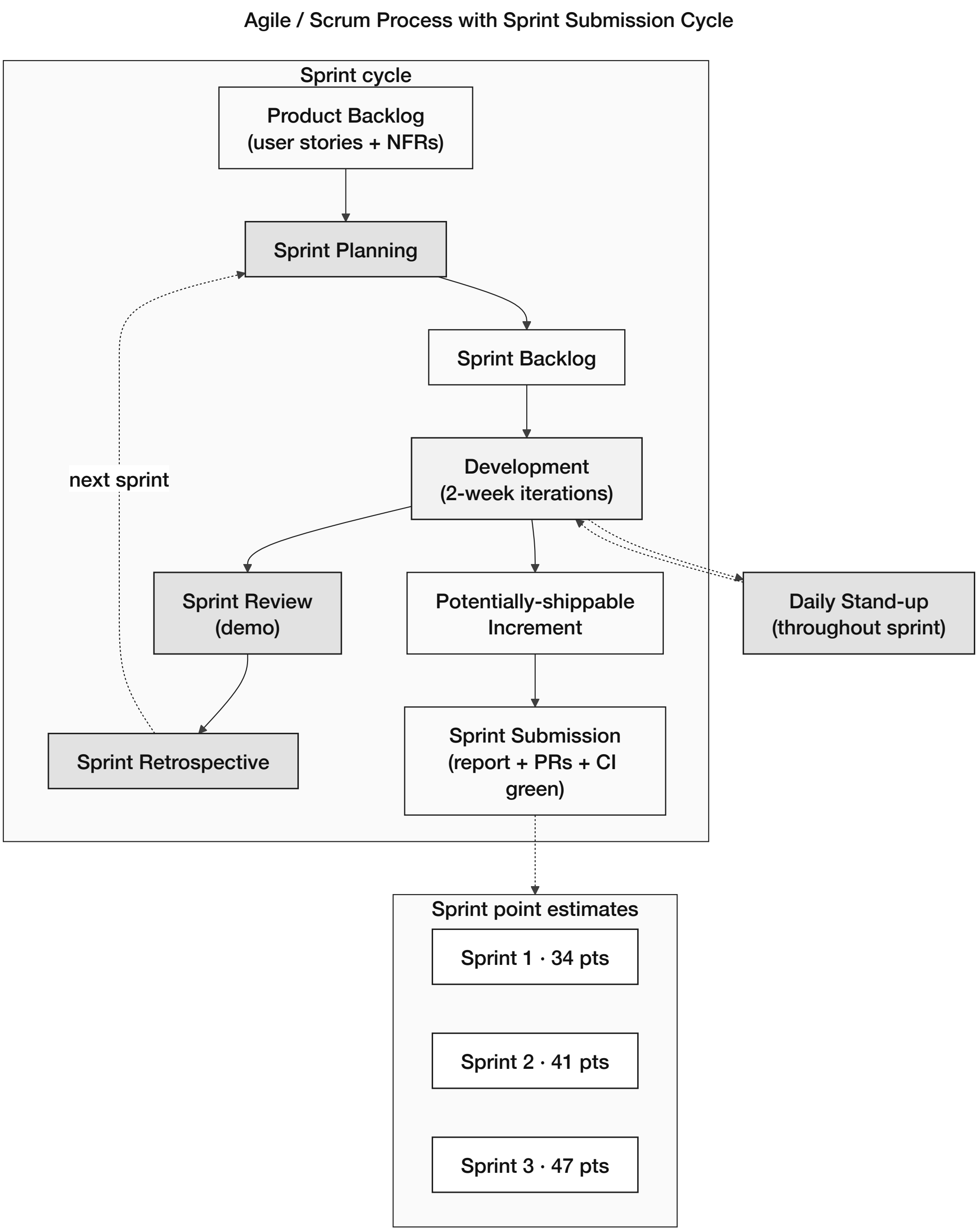

Figure 4.1: Agile / Scrum development process

The following points summarize how Agile will be applied:

- **Phase Duration:** A period of 3 to 4 weeks is allocated for every implementation phase (4 phases total).
- **Team Size:** A group of two developers will be assigned (thesis project). The work will be focused on the thesis project.
- **Weekly Sync:** A weekly gathering will occur to assess progress. Issues will be identified during these meetings. Plans will also be developed in these sessions alongside progress checks.
- **Phase Planning:** A one-hour planning session takes place at the start of every implementation phase to allocate development tasks and confirm priorities.

- **Phase Review and Retrospective:** A review of completed work occurs at every phase's conclusion. A 30-minute retrospective discussion captures lessons learned for the subsequent phase.

4.3 Planned AI/ML Support and Risk-Score Wiring

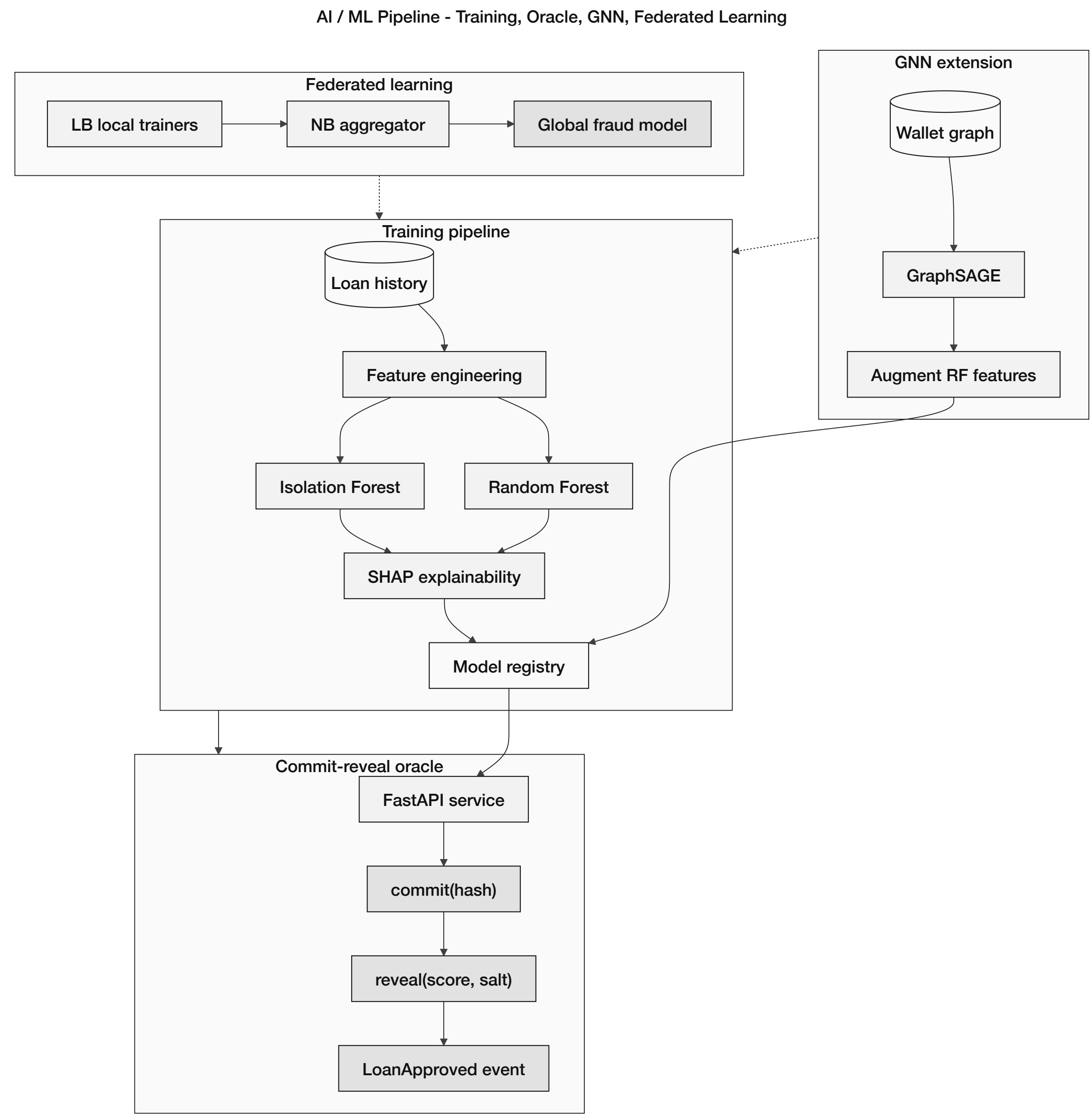

Figure 4.2: AI/ML pipeline wiring

Each loan gets one risk score, committed on-chain before anyone can approve it (Figure 4.2; oracle detail in Section 3.3.2). This report describes the wiring only—training and live scoring start after pre-thesis 2.

1. **Features.** FastAPI builds 22 inputs per application: 18 from wallet behaviour (tx rhythm, repayment history, graph distance to flagged addresses) and 4 from the form (amount, KYC level, document freshness).

2. **Scoring.** Random Forest outputs a fraud probability p_f; Isolation Forest flags unusual behaviour. A stacking meta-learner blends them into one calibrated score $s \in [1]$ (Section 4.3.2).
3. **Oracle.** The service commits $h = \mathrm{keccak256}(s \,\|\, \mathrm{nonce})$ via `commitRiskScore`, then reveals (s, nonce). Approval is blocked until `SCORE_REVEALED`.
4. **Thresholds.** $s < 0.4$ approver may approve; $0.4 \leq s < 0.7$ manual review; $s \geq 0.7$ on-chain decline. An approver still signs every disbursement—ML advises, it does not auto-lend.

4.3.1 Datasets, Training Workload, and Pre-Submission Deliverables

Here we spell out what data the models will train on and what gets delivered when. This report is the plan—actual training starts after pre-thesis 2. Figure 4.2 shows the pipeline; Section 4.3.2 covers how we explain scores.

Table 4.3: ML datasets

Dataset	Role	Usage in this project
BCCC-DeFiFraudTrans-2025 [83]	Primary	Main training set for fraud detection benchmarks after pre-thesis 2
Elliptic Bitcoin Dataset	Baseline	Optional comparison only; different chain model than CWB
Palaiokrassas multi-chain set [3]	Reference	Guides behavioural feature choices
Synthetic Foundry manifest (Phase IV)	Demo	End-to-end UI test only; not a fraud benchmark
On-chain CWB log (future)	Retrain	Real Local Bank traffic for production tuning

Each loan application feeds **22 features** into the scorer (Table 4.4). After pre-thesis 2 we train on BCCC, blend Random Forest and Isolation Forest into one score, and log every run for audit.

Table 4.4: Feature mapping (22 inputs)

#	Feature group	What it captures	Source
1–5	Transaction rhythm	How often and how fast the wallet moves funds	BCCC + indexer
6–10	Repayment track record	Past loans and defaults	CWB loan tables
11–15	Network risk	Links to flagged wallets and group membership	Graph features
16–18	Institutional context	Pool usage, tier, wallet age	On-chain reads
19–22	Application fields	Amount, KYC level, document freshness	Form + identity service

At loan time the backend scores the application; approvers see an Authority Brief before signing off. Scores are logged and committed on-chain before disbursement (Section 3.3.2).

Table 4.5: ML deliverables by submission stage

Deliverable	Pre-thesis 1	After pre-thesis 2
Dataset + feature documentation	Done in this report	Update as needed
Model training + metrics file	Plan only	Train and report results
SHAP explanation examples	Plan only	Required for RQ3
Scoring API + Authority Brief UI	Design / mock	Live on loan workflow
Audit log per inference	Schema in Ch. 3	One row per loan scored
On-chain score commit	Design only	MVT item 7
Chainlink Functions / GNN / federated learning	Future work	Stretch goals

Post pre-thesis 2 we report precision, recall, F1, AUC, and SHAP reasons on declines. Figure 4.3 is the *planned* layout—nothing is trained yet. BCCC is a flat-DeFi baseline, not a perfect match for our four-tier flows; we score once at origination to keep oracle cost down.

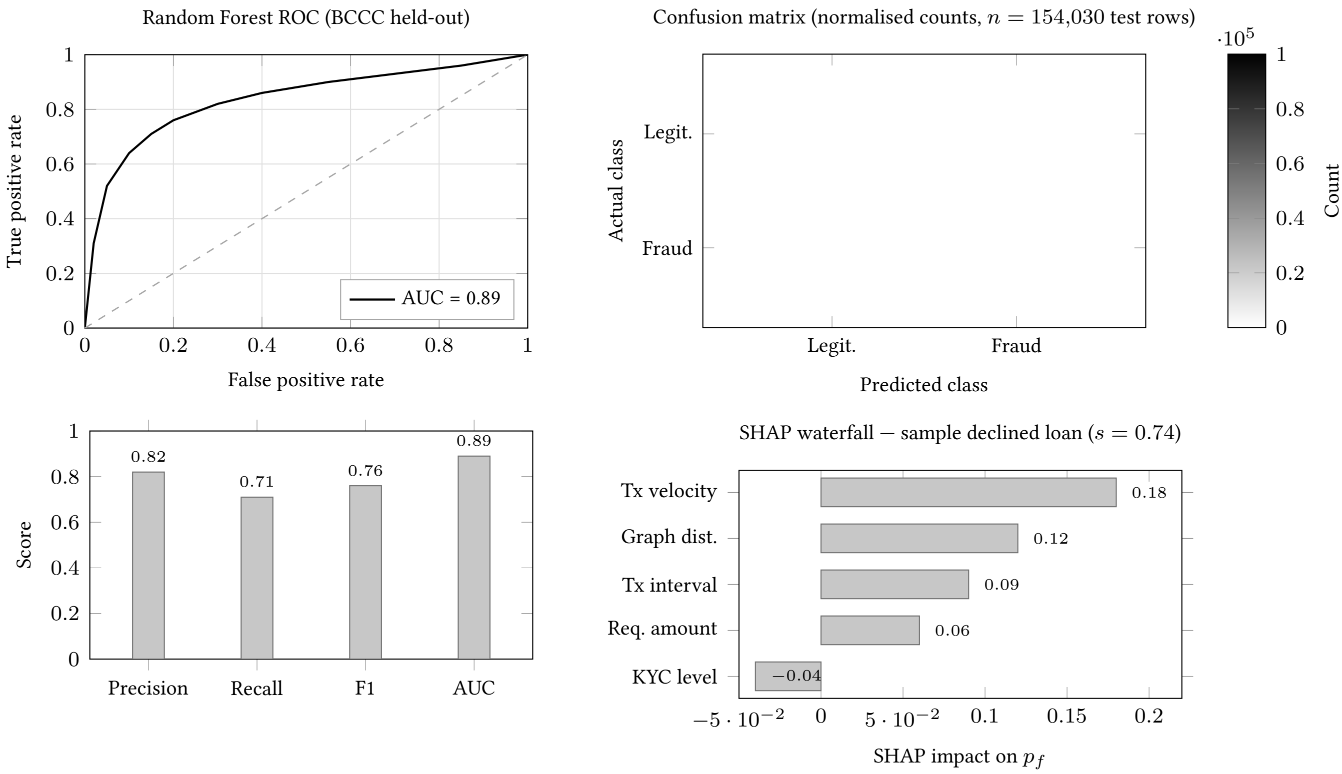

Figure 4.3: Planned ML evaluation layout (post pre-thesis 2)

4.3.2 Explainability and Anomaly Detection Methodology

Humans need to see *why* a score was high, not just the number. This path is separate from the optional agent (Section 4.4).

Random Forest gives p_f with **SHAP TreeExplainer** attributions per feature [4]. Isolation Forest catches behaviour that looks nothing like normal clients—trained on non-fraud history only—and scores anomalies via path length (List of Formulas). Because IF scores are not probabilities, a stacking meta-learner maps (p_f, s_{IF}) to the single s the contract reads.

IF has no native SHAP, so outliers get a **feature-deviation report**: top features vs. training median and percentile, e.g. *"repayment interval variance at the 98th percentile; two hops from a flagged wallet."*

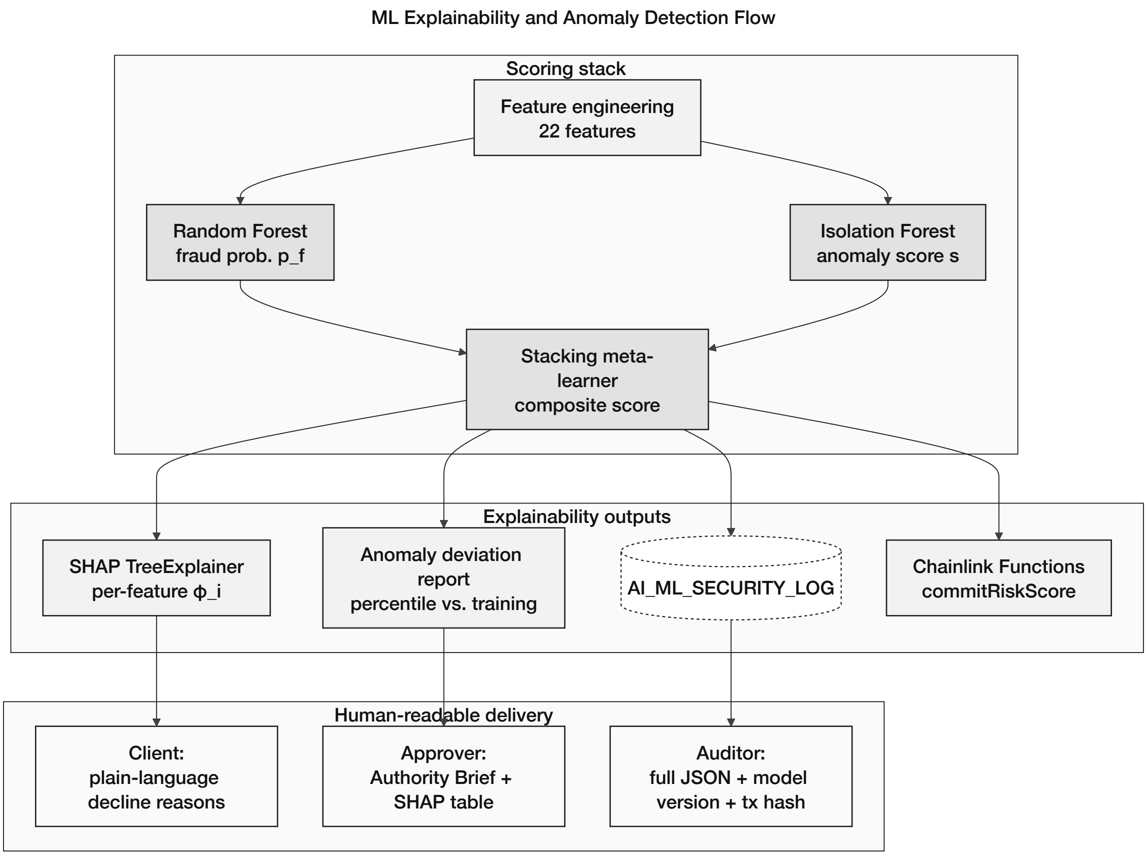

Figure 4.4: Explainability and anomaly-detection flow

Audience	UI / channel	Content stored and displayed
Retail client	React loan status page	Top-3 SHAP reasons in plain language on decline; optional anomaly summary if score $\geq$ 0.7
Bank approver	Authority Brief dashboard	SHAP waterfall + anomaly deviation table + composite s; one-click approve / decline
Compliance auditor	`AI_ML_SECURITY_I`	Full SHAP vector, IF score, feature JSON, model version hash, oracle tx hash

The Authority Brief on the approver dashboard shows s, SHAP, and anomaly lines (example in Section 4.4). Every inference logs to `AI_ML_SECURITY_LOG` with model version and oracle tx hash. ML advises; an approver still signs disbursement.

4.4 Optional Conversational Agent Add-On (Phase III–IV)

Clients use the React dashboard by default. The conversational agent is an **optional add-on** (Phases III–IV Could)—same API and contracts as the UI, never a gate on deposits or loans. Table 4.6 lists its tools; approver ML explainability (Section 4.3.2) works without it.

The agent runs a six-step loop: (1) message in via SSE; (2) Qwen3-8B gets live account context in the system prompt; (3) Q&A answers from RAG, no tools; (4) actions pick an MCP write tool and show a confirmation summary; (5) user must say yes explicitly—no write without it; (6) tool POSTs to Express.js, optional EIP-7702 signing, then status update to the client. Harness detail is in Appendix B.

Table 4.6: MCP banking tools (17)

Tool	Type	Maps to / Description
`get_account_state`	READ	SBT tier, loan count, savings balance
`get_credit_score`	READ	Current score, tier, next threshold
`get_loan_status`	READ	All active loans + next installment
`get_installment_schedule`	READ	Full repayment calendar
`get_pool_utilisation`	READ	Current Local Bank pool capacity
`get_interest_rate`	READ	Rate for client's credit tier
`get_market_data`	READ	ETH/USD, fiat/USD pairs, 30-day volatility
`get_requirements`	READ	Documents + KYC needed for a given loan amount
`get_session_history`	READ	Search prior sessions (PostgreSQL full-text index)
`submit_loan_application`	WRITE†	POST /api/loan/apply
`submit_deposit`	WRITE†	POST /api/savings/deposit
`submit_fixed_deposit`	WRITE†	POST /api/savings/fixed-deposit
`pay_installment`	WRITE†	POST /api/installment/pay
`join_group_pool`	WRITE†	POST /api/group/join
`submit_kyc_upgrade`	WRITE†	POST /api/kyc/upgrade
`schedule_payment_reminder`	WRITE†	POST /api/reminder/set
`submit_group_applicatio`	WRITE†	POST /api/group/apply

† Agent write tools require human confirmation before execution.

Every loan application gets an Authority Brief on the approver dashboard (UI or agent):

```
AUTHORITY BRIEF -- Loan Application #4821
```

```
-------------------------------------------------
Client tier: Silver (score: 512)
Requested amount: 150 USDC
Duration: 6 months
Collateral: 75 USDC (50% LTV)

ML Risk Score: 0.23 (LOW RISK)
SHAP breakdown:
 + On-time repayment rate: +0.41 (reduces risk)
 + Loan-to-income ratio: +0.18 (within safe range)
 - Wallet age: -0.09 (account < 6 months)
 - No prior completed loans: -0.27 (no repayment history)

ML system recommendation: APPROVE (decision support only)
Suggested interest: Base rate - 0.5% (Silver tier)

[ APPROVE ] [ REQUEST MORE INFO ] [DECLINE]
```

4.5 Graph Neural Network Extension

Fraud often lives in *who connects to whom*, not one transaction. As a stretch goal we would add GraphSAGE: wallets as nodes, transfers and tier links as edges, same 18 behaviour features as the Random Forest. The test is whether bank-hierarchy edges beat a flat graph on BCCC [83]. Future Work only—if we have time, one before/after F1 or AUC table.

4.6 Federated Learning Across Banking Tiers

Local Banks would train on their own borrowers; the National Bank aggregates updates (FedAvg + differential privacy) and ships back a shared model—no raw data crosses borders. We turn federation on only after a bank hits **500 transactions**; until then everyone uses the public baseline. Future Work, not MVT.

4.7 Formal Verification of Reserve Invariants (Certora)

Beyond unit tests we plan two Certora proofs on reserve rules (Phase IV MVT+):

- Reserve never under-collateralised: $\text{totalReserve} \geq \text{minReserveRatio} \cdot \text{totalDeposited}$ on every path.
- No over-allocation: the World Bank cannot allocate more to National Banks than it holds.

CVL sketches sit in Appendix B; scope limits in Section 5.7.

4.8 Foundry Invariant and Fuzz Test Suite

Hardhat covers integration and deployment; Foundry adds fuzzing and invariant checks. Together they catch different failure modes.

Three invariants under Foundry's stateful fuzzer.

- Solvency: outstanding loans never exceed total reserve.
- Role segregation: one address cannot be both borrower and approver.
- Capital flow: a single cross-tier move cannot break the minimum reserve ratio.

Each invariant is fuzzed for 10,000 runs in Phase IV; any counterexample becomes a replay test in CI.

4.9 On-Chain Economic Feasibility Simulation

Phase IV runs a **reproducible stress test** on Polygon zkEVM Cardona: 300 synthetic clients, six banks, seed 42, 5% mid-run defaults. A Foundry script walks the full lifecycle (apply → approve → disburse → repay) and writes a JSON manifest—tx hashes, reserve ratios, gas—for Appendix C and Table **??**. Results are reported as ranges; microfinance calibration stays in Future Work (Section 6.1).

4.10 Real-Time Dashboard Pipeline and Runtime Monitoring

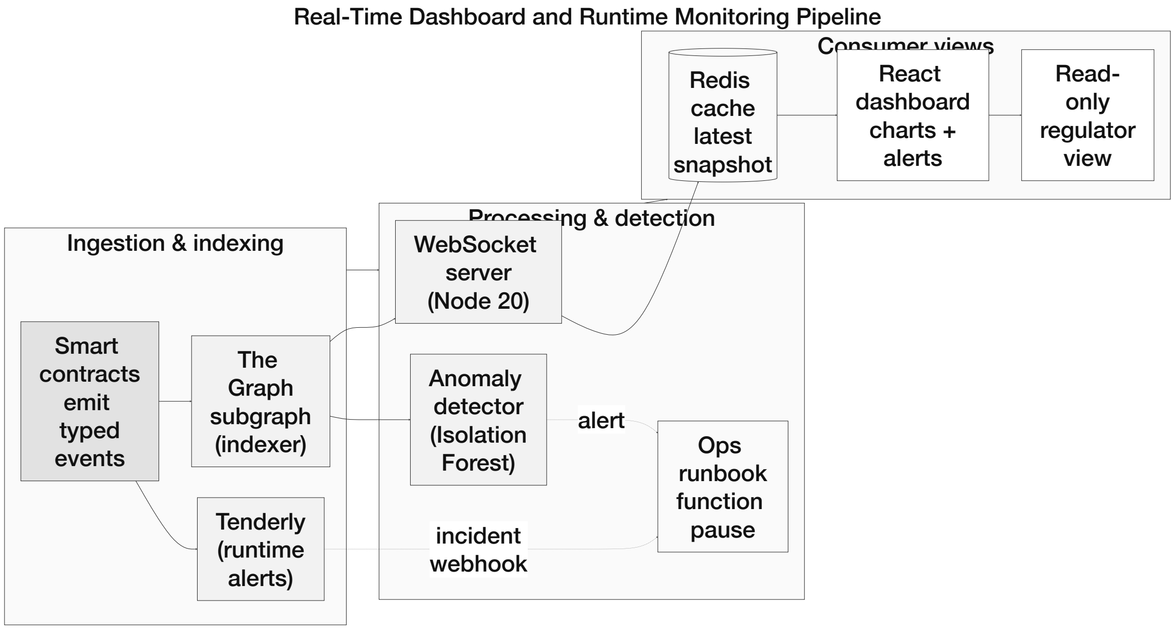

Figure 4.5: Real-time dashboard and runtime monitoring pipeline

Three pieces connect chain events to the UI (Figure 4.5). A **The Graph subgraph** indexes contract events (`LoanRequested`, `Repayment`, `ReserveRatioUpdated`, and related emissions) so the React app can query history in milliseconds instead of polling RPC. A **WebSocket listener** pushes the same events to the browser via Socket.IO for live updates. **Tenderly** watches testnet contracts for large disbursements, repeat loan spam, reserve-ratio drops, failed calls, and governance pauses—and simulates txs before deployment demos.

4.11 Transaction-State Machine for Frontend UX

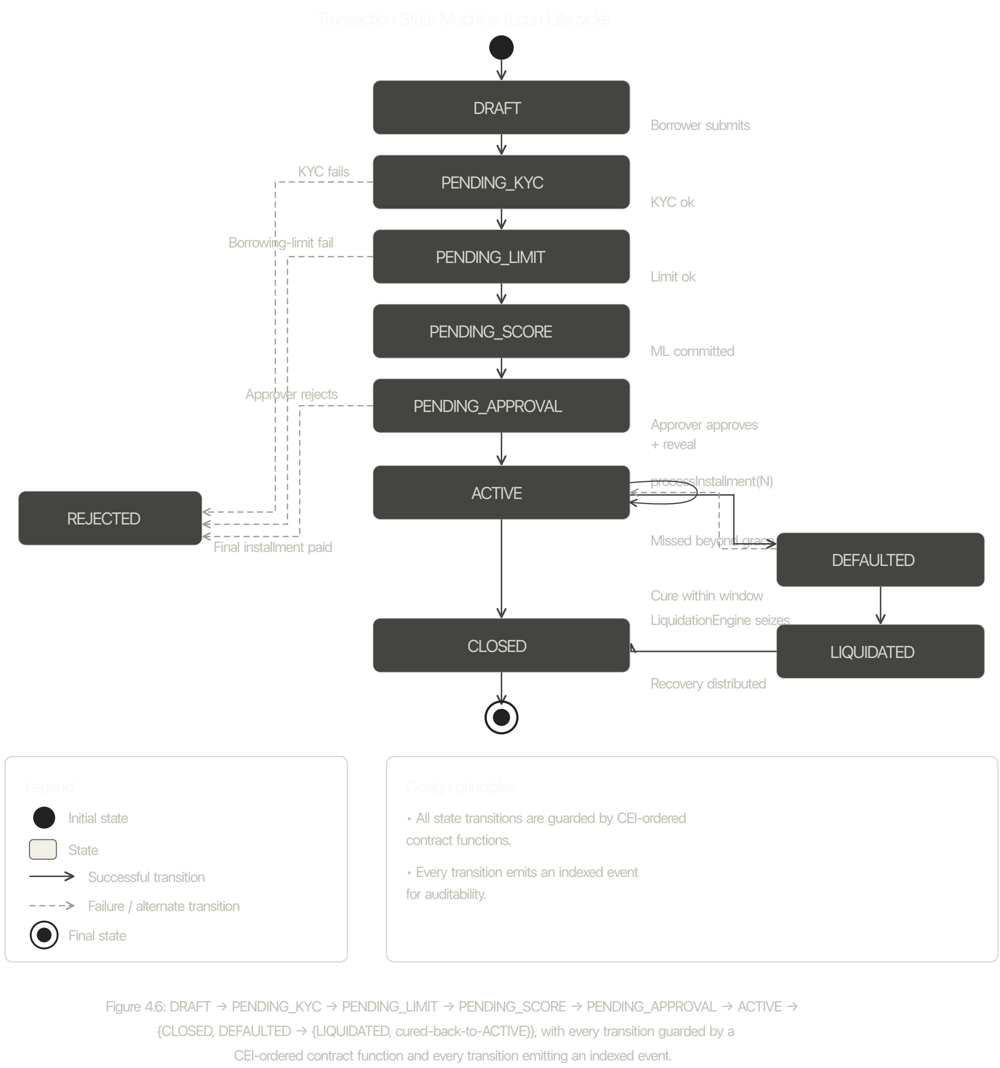

Figure 4.6: Transaction state machine of the loan lifecycle

The frontend tracks every loan tx through five states (Figure 4.6): `idle` → `pending_signature` → `submitted` → `confirming` → `success / error`. Each state has its own UI feedback. Wallet rejection (MetaMask 4001), contract revert, and network errors get separate messages so users know what failed and can retry.

4.12 EIP-712 Authentication and API Hardening

Users sign an EIP-712 message `(wallet, timestamp, nonce)` at login; the backend recovers the address and issues a 15-minute JWT (refresh tokens rotate on use). Each request carries a correlation ID from React through Express.js and FastAPI to the on-chain tx for end-to-end tracing. `slowapi` rate-limits by endpoint (5/min auth, 60/min reads, 10/min writes); logout blacklists the JWT in Redis. See Section 3.18 for the full Layer 2 control set.

ML scores are decision support only—not auto-approval. Class imbalance, drift, and cold-start remain; GNN ablation, federated learning, and LLM red-teaming (Section 4.13.1) bound but do not remove them.

4.13 Evaluation Methodology

We tie each research question to something we can actually demonstrate on the prototype:

- **RQ1:** Four-tier flows and roles vs. flat DeFi (Aave, Compound, MakerDAO).
- **RQ2:** Testnet latency and gas; reserves and state changes visible on-chain.
- **RQ3:** After pre-thesis 2, report precision, recall, F1, and AUC on BCCC [83], plus SHAP explanations and audit logs. GNN ablation and testnet retrain are stretch goals.
- **RQ4:** Full demo path (wallet → loan → approval → repayment) and Foundry invariant pass rate (Section 4.8). Deployment evidence lands in Phase IV.
- **RQ5:** Deposit and group-lending specs; Phase IV simulation (Section 4.9) stress-tests reserves under default.

4.13.1 LLM Assistant Evaluation Protocol

If the optional agent ships (Phase IV Could), we run three checks: **Q&A** on 200 policy prompts (exact numbers, similarity on prose); **tool use** on 50 scenarios (correct tool, params, confirmation summary—$\geq$95%, always human-gated); and a **bypass red-team** of 20 adversarial prompts that must never trigger a write without explicit user consent.

4.14 Implementation Phase Plan and Deliverables

The build runs in four phases (I–IV), roughly three to four weeks each. Task registers below list effort by phase; Figures 4.7–4.10 show the roadmap and toolchain.

4.14.1 Phase I: Foundation and Infrastructure (Weeks 1–4)

Goal: Contract skeleton, database migration, React shell with wallet connect.

Table 4.7: Phase I task register

Task	Priority	Development Task	Effort (days)
DT-I.01	Must	World Bank contract — reserve management, national bank registration	5
DT-I.02	Must	National Bank contract — borrow from World Bank, lend to Local Banks	5
DT-I.03	Must	Local Bank contract — borrow from National Bank, lend to users	5
DT-I.04	Must	Role-based access control — WorldBankAdmin, NationalBankAdmin, LocalBankAdmin, BankUser, RetailClient	3
DT-I.05	Should	Gas cost management — initiator-pays pattern; Polygon low-fee design	2
DT-I.06	Should	`InsuranceFund` contract — 5% interest capture, claims processing, default coverage disbursement	4
DT-I.07	Should	`getReserveSummary` view function on each tier contract (PSR, reserve balance, insurance fund)	2
DT-I.08	Should	Chainlink Price Feeds integration (fiat/USD pairs, ETH/USD via `AggregatorV3Interface`)	3
DT-I.09	Should	Chainlink Proof of Reserve — WorldBankReserve attestation	2
DT-I.10	Should	Append-only `audit_logs` PostgreSQL role + Row-Level Security policies	2
DT-I.11	Should	Polygon zkEVM Cardona migration — update RPC endpoints, chain IDs, factory addresses	1
DT-I.12	Must	Full 20-entity database schema migration — all tables including SESSIONS, AGENT_ACTION_LOG (with `injection_scan_flag`, `toolset_name`), and all banking entities	4
DT-I.13	Must	Core indexes and referential integrity constraints	2
DT-I.14	Must	React frontend shell — Vite build, routing, layout components	3
DT-I.15	Must	WalletConnect integration — wallet connection, address display, network switching	2
DT-I.16	Should	Sessions and assets tables integration in frontend (dashboard stubs)	2
DT-I.17	Should	Credit tier and interest rate schedule display	2
Phase I total estimated effort (Must-have):			*29 days*
Phase I total estimated effort (all priorities):			*48 days*

4.14.2 Phase II: Core Banking and On-Chain Lifecycle (Weeks 5–9)

Goal: Full loan lifecycle on testnet—request, approval, installments, bank registration, SBT limits. Agent and multi-entity work defer to later phases.

[0]Must = MVT core; Should = important; Could = if time allows. Effort = person-days; streams run in parallel.

Task	Priority	Development Task	Effort (days)
DT-II.01	Must	Loan request submission with blockchain transaction	5
DT-II.02	Must	Loan approval and rejection workflow with bank approver	5
DT-II.03	Must	Installment payment system with automated schedule generation	8
DT-II.04	Must	Borrowing limit enforcement (on-chain authoritative check per client SBT)	5
DT-II.05	Should	Client–bank real-time chat system	5
DT-II.06	Should	Income verification document upload and review	5
DT-II.07	Must	Hierarchical bank registration (National → Local)	5
DT-II.08	Should	Bank user management and approver designation	3
DT-II.09	Should	Loan history and transaction tracking pages	5
DT-II.10	Could	QR code generation for wallet addresses	2
DT-II.11	Should	Responsive UI polish and error handling	2
DT-II.12	Should	Authority Brief UI (SHAP breakdown for bank approver dashboard)	5
DT-II.13	Should	Chainlink Automation for installment due-date checks and overdue flagging	5
DT-II.14	Should	Credit tier schedule in Credit Passport SBT (Bronze → Diamond)	3
DT-II.15	Should	`interest_rate_tier` table integration with Kinked Rate Model	2
DT-II.16	Could	MCP tool server — 17 banking tools wired to Express.js API (optional add-on)	8
DT-II.17	Could	Agent chat interface (Qwen3-8B + tool calling + human-gate)	8
DT-II.18	Could	EIP-7702 session key management (scope enforcement + TTL)	5
DT-II.19	Could	`agent_action_log` table (append-only, FK to `sessions`)	3
Phase II total estimated effort (Must-have):			*33 days*
Phase II total estimated effort (all priorities):			*66 days*

4.14.3 Phase III: AI/ML Oracle Pipeline and Optional Agent (Weeks 10–13)

Goal: Train models, deploy the scoring API, wire Chainlink Functions and the Authority Brief. Agent harness is Could-have.

Task	Priority	Development Task	Effort (days)
DT-III.01	Must	Chainlink Functions oracle (primary ML score commitment): replaces commit-reveal relay; deploys `commitRiskScore` via Chainlink Functions DON	6
DT-III.02	Must	Random Forest + Isolation Forest + SHAP pipeline wiring: FastAPI service computing 22 features, stacking meta-learner, calibrated composite score $s \in [1]$; client-facing SHAP plain-language explanation via Authority Brief UI	7
DT-III.03	Should	Three-tier prompt constructor in Express.js (optional agent add-on)	5
DT-III.04	Could	Session lineage + `get_session_history` MCP tool	5
DT-III.05	Could	Context compression path + toolset scoping + Hook 2	9
DT-III.06	Could	Prompt injection scanning + confirmation audit hook (Hook 1)	5
DT-III.07	Should	The Graph subgraph + WebSocket event listener for Reserve Transparency Dashboard	4
DT-III.08	Should	SAR workflow: Isolation Forest $\rightarrow$ aml-alert $\rightarrow$ `freezeAccount`; compliance review queue integration	4
DT-III.09	Should	Reserve Transparency Dashboard: live queries to The Graph subgraph	3
DT-III.10	Could	GraphSAGE GNN ablation	5
DT-III.11	Could	Federated learning aggregator	7
Phase III total estimated effort (Must-have):			*13 days*
Phase III total estimated effort (all priorities):			*55 days*

4.14.4 Phase IV: Verification, Evaluation, and Finalization (Weeks 14–16)

Goal: On-chain simulation, Certora/Foundry checks, testnet evidence, optional LLM eval, final thesis revision.

Task	Priority	Development Task	Effort (days)
DT-IV.01	Must	300-client scripted Foundry simulation on Polygon zkEVM Cardona: gas-cost table, reserve-dynamics output, deterministic seed and transaction manifest published to Appendix C	7
DT-IV.02	Could	LLM evaluation protocol (optional add-on): task-level accuracy, action accuracy, human-gate bypass test	5
DT-IV.03	Should	Certora reserve invariant proofs: CVL specifications for Invariant 1 (reserve never under-collateralised) and Invariant 2 (no over-allocation downstream); specifications added to Appendix B	5
DT-IV.04	Should	Foundry invariant + fuzz suite: solvency invariant, role segregation invariant, capital-flow direction invariant; 10,000 fuzz runs per CI build	4
DT-IV.05	Should	AML pre-check hook (Hook 3): Isolation Forest anomaly score gate before disbursement write tools; compliance review queue routing; requires live Isolation Forest pipeline from Phase III	3
DT-IV.06	Could	Groth16 zkKYC circuit (Circom 2.0): government-ID possession proof without revealing PII; circuit design and input/output specification	5
DT-IV.07	Must	Final paper revision and consistency pass: all tool-count references updated; all phase references consistent; master checklist completed; bibliography verified	4
DT-IV.08	Must	Red-team testing and documentation: 20-payload injection red-team set; 20 bypass-attempt set; 10 session key scope enforcement scenarios; results documented in evaluation subsections	3
Phase IV total estimated effort (Must-have):			*21 days*
Phase IV total estimated effort (all priorities):			*38 days*

Phase	**Focus**	**Weeks**	**Tasks**	**Key deliverables**
I	Foundation and infrastructure	1–4	DT-I.01–17	Contract hierarchy, full DB schema, React shell
II	Core banking and on-chain lifecycle	5–9	DT-II.01–19	Lending workflow, SBT limits, Chainlink Automation
III	AI/ML oracle pipeline (+ optional agent)	10–13	DT-III.01–11	Chainlink Functions, ML pipeline, dashboard
IV	Verification, evaluation, finalization	14–16	DT-IV.01–08	Certora, Foundry, simulation, testnet deploy

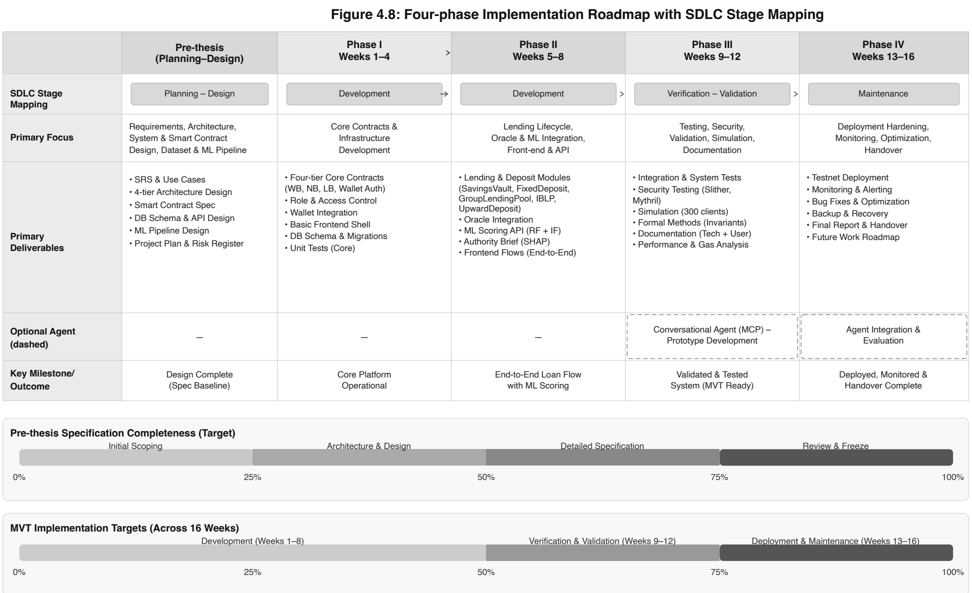

Figure 4.7: Four-phase roadmap with SDLC mapping and progress

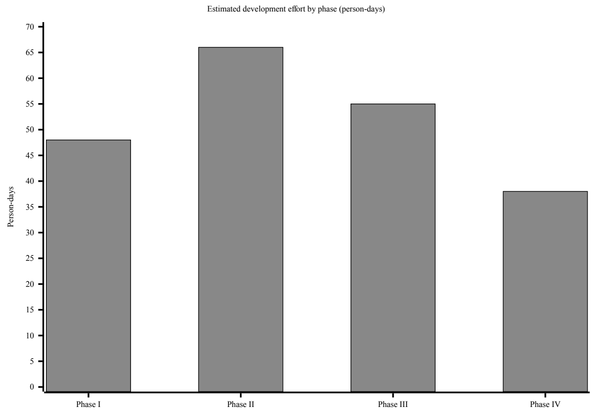

Figure 4.8: Development effort by implementation phase

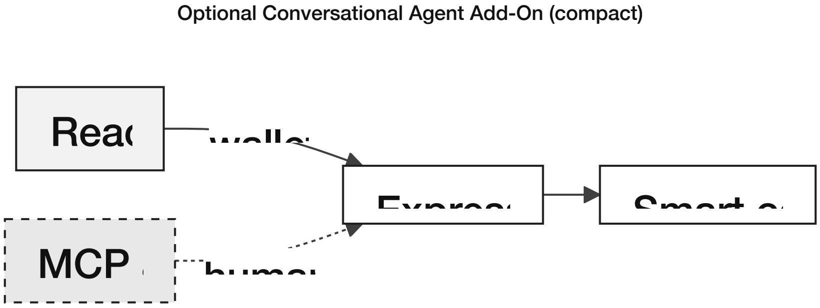

Figure 4.9: Optional conversational agent add-on

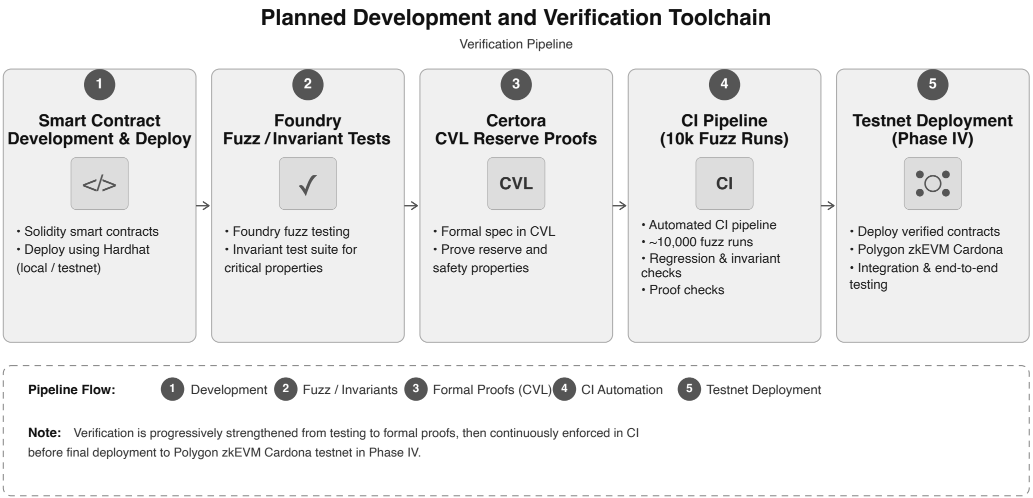

Figure 4.10: Planned development and verification toolchain

4.15 SDLC Stage Mapping

We map our four implementation phases to the seven stages of the Software Development Life Cycle (SDLC). Figure 4.7 illustrates this mapping.

Table 4.8: SDLC stage mapping

#	SDLC Stage	Project Activity	Deliverable
1	Planning	Feasibility studies; professor consultations; guideline review	Feasibility Report; Project Plan
2	Requirements	System analysis; use case definitions; constraints identification	Use case diagrams; UC-1 through UC-5
3	Design	Three-layer architecture; DB schema (3NF); smart contract interfaces	Architecture diagrams; ERD; DFD
4	Development	Phase I–IV: smart contracts, frontend, backend, AI/ML	Source code; DApp prototype
5	Testing	Hardhat unit tests; Foundry fuzz; integration testing; AI/ML evaluation	Test reports; model metrics
6	Deployment	Planned testnet deployment (Phase IV); frontend (Vercel); backend (Render)	Testnet prototype manifest
7	Maintenance	Monitoring; model retraining; bug fixes; iteration	Updated docs; retrained models

4.16 Design Decisions and Alternatives

For each major design decision, we evaluated alternatives and justified selections based on technical criteria, ecosystem maturity, and project constraints (Table 4.9; Figure 4.11).

Table 4.9: Design decisions and alternatives considered

Decision Area	1st Choice	2nd Choice	Key Criterion
Methodology	Agile / Scrum	Incremental	Evolving scope, milestones
Architecture	DApp + Off-chain AI	Hybrid with Oracle	Gas cost, ML flexibility
Frontend	React + TypeScript	Vue + TypeScript	Web3 ecosystem maturity
Smart Contract	EVM (Solidity)	Solana (Rust)	Gas, control size, free testnets
Fraud Detection	Random Forest	XGBoost	SHAP compatibility, simplicity
Anomaly Detection	Isolation Forest	Autoencoder	Unsupervised; no labelled data needed
XAI Method	SHAP	LIME	Theoretical guarantees, regulatory fit
Database	PostgreSQL	SQLite	3NF support, async queries
Hosting	Vercel + Render	Localhost only	$0 cost, publicly accessible URL
Network (primary)	Polygon zkEVM Cardona	Polygon Amoy PoS	ZK validity proof security model vs. validator set assumption
Oracle mechanism	Chainlink Functions (DON)	Commit-reveal relay	Decentralised trust vs. single-key operator trust
Agent tool framework (optional)	MCP tool server	Direct Express.js API	Defined schema = safety boundary; Phase III–IV Could
Agent session auth (optional)	EIP-7702 session keys	ERC-4337 paymasters	Key-level scope; optional add-on
Prompt construction (optional)	Three-tier assembly	Single ad-hoc string	Agent-only; Phase III Could
Session memory (optional)	Lineage model	10-turn window	Agent-only cross-session continuity
Tool exposure per turn (optional)	Named toolsets	Full 17-tool list	Agent attack-surface reduction
Write-tool enforcement (optional)	Confirmation audit hook	In-pipeline check	Agent write-path only

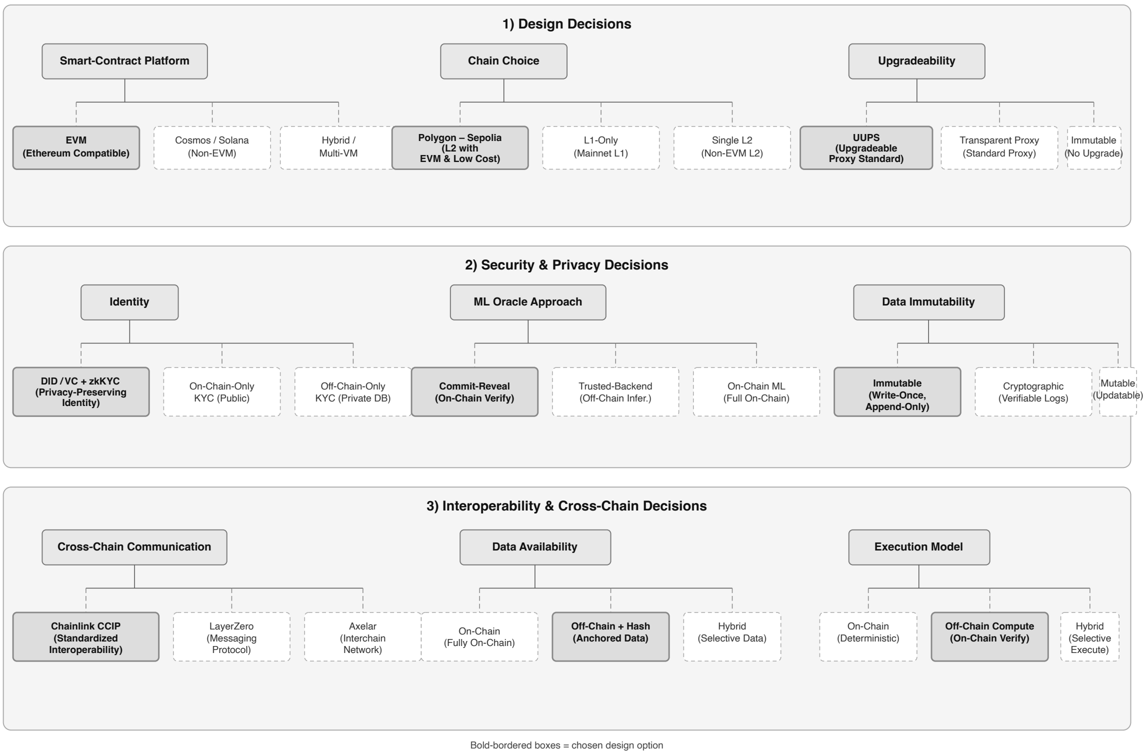

Figure 4.11: Key design decisions and alternatives considered

4.16.1 Justification of Selected Technologies

Selections follow Table 4.9 in Section 4.16. One-line decision logic per row.

1st choices:

- **Agile/Scrum** — evolving scope; short sprints fit thesis window.
- **DApp + off-chain AI** — ML off-chain (cost); disbursement rules on-chain.
- **React + TypeScript** — strongest Wagmi/ethers wallet stack.
- **EVM (Solidity)** — OpenZeppelin, Slither/Mythril, free testnets.
- **MUI** — banking tables/forms without custom UI build.
- **Random Forest** — exact SHAP TreeExplainer; tabular + imbalanced labels [4].
- **Isolation Forest** — unsupervised; scarce fraud labels.
- **SHAP** — stable attributions vs. LIME [4].
- **PostgreSQL** — ACID; window queries for rolling limits; JSON columns.
- **Vercel + Render** — free public demo for examiners.
- **Polygon zkEVM Cardona** — ZK proofs for reserve claims; Amoy fallback.
- **Chainlink Functions** — DON consensus on score; OK latency for loan approval.
- **MCP tool server** — 17-tool schema caps agent; auditable.
- **EIP-7702 session keys** — tool list + cap + TTL in key; ERC-4337 for retail gas only.

2nd choices (retained, not selected):

- **Waterfall** — fixed scope; poor fit for research prototype.
- **On-chain ML oracle** — latency, cost, third-party dependency.
- **Vue** — thinner Web3 libraries.

- **Solana** — Rust curve; weaker audit toolchain in 8-week window.
- **Tailwind** — more UI time than MUI.
- **XGBoost** — approximate SHAP.
- **Autoencoder** — more data/tuning than IF here.
- **LIME** — unstable repeat explanations [4].
- **SQLite** — no concurrent writes; weak window functions.
- **Localhost only** — no remote examiner demo.
- **Amoy PoS** — faster dev; weaker trust than ZK.
- **Commit-reveal relay** — Ph. I–II fallback; trusts operator key.
- **Direct Express API** — no schema cap on agent.
- **ERC-4337 for agent ops** — scope in app layer, not key.

4.17 Design Patterns

Our system uses five design patterns:

- **Singleton:** A singleton means that each primary contract is established a single time. Only one version is utilized by all ensuring a unified source of truth. Not everyone can create multiple versions of the contract.
- **Observer:** Contracts generate alerts whenever actions occur such as loan requests. These alerts are processed by a service that updates the database. Notifications are not ignored; they receive attention.
- **Adapter:** An Adapter is utilized by the wallet component to function with various wallet types (like MetaMask). Different wallets that adhere to the EIP-1193 standards can be easily integrated.
- **Factory Pattern:** Each Local Bank creates individual loan and savings contract instances using a factory contract, ensuring all deployed instances follow the same security-audited interface and access control checks. The factory pattern prevents unauthorized contract deployment that could bypass the hierarchical governance structure.
- **Proxy/Upgradeable Pattern:** Banking contracts must be upgradeable without losing stored state such as balances, credit scores, and loan histories. OpenZeppelin's transparent proxy pattern separates the contract's logic from its storage, allowing governance-approved upgrades to the logic layer while preserving all stored financial data. This pattern is essential for a long-lived banking system that must adapt to evolving regulatory requirements and security patches.
- **Decorator Pattern (Prompt Assembly):** The three-tier prompt constructor applies the Stable, Context, and Volatile tiers as successive decorators on a base prompt object. Each tier adds its content without modifying the structure established by prior tiers, enabling independent caching and testing of each layer without rebuilding the full prompt.
- **Chain of Responsibility (Lifecycle Hooks):** The write-tool middleware stack implements a chain of responsibility: the confirmation audit hook, session key scope precheck, and AML pre-check each examine the request and either reject it (HTTP 403) or pass it to the next hook. New policy controls are added as new chain links without modifying existing route logic or the model pipeline.
- **Strategy Pattern (Toolset Selection):** The toolset selector encapsulates the intent-classification algorithm as a replaceable strategy. The current strategy is a lightweight keyword classifier; it can be replaced with a fine-tuned intent model without chang-

ing the prompt constructor or the MCP tool server, adhering to the Open/Closed Principle.

4.18 Software Testing Strategy

The testing strategy covers four layers of the system, each with defined acceptance criteria.

- **Smart contract unit tests:** Numerous tests for smart contracts exist in our Hardhat environment. Over 12 tests are conducted to verify actions related to managing reserves, sign-ups, loans and access permissions. No scenarios such as depositing zero or executing disallowed actions are missed in these assessments. **Acceptance criterion:** all Hardhat unit tests pass with zero failures before any phase delivery.
- **Integration testing:** Whole process checks are conducted for integration tests. Wallet connection, loan request, approval and repayment processes are evaluated on the network. No critical steps are excluded from the process. **Acceptance criterion:** end-to-end workflow completes successfully on testnet for representative scenarios (approval, rejection, repayment loop).
- **AI/ML module verification:** Ensuring functionality is vital for our fraud and unusual activity detectors. Results are generated by the integration of the detectors with the overall system. **Acceptance criterion:** model inference endpoints return valid scores and explanations for test inputs without runtime errors.
- **Frontend testing:** Website appearance is evaluated on Chrome, Firefox and mobile devices. Tests are conducted to ensure that functionality is maintained across various platforms. **Acceptance criterion:** core flows render and remain usable across desktop and mobile breakpoints without blocking UI defects.

Chapter 5

Evaluation and Discussion

5.1 Performance Evaluation

Pre-thesis 1 delivers specifications and planned checks, not live benchmarks. Component and economic feasibility are in Section 5.4; RQ-level evaluation follows Section 4.13 after pre-thesis 2 (testnet deployment, gas measurement, ML metrics, MVT demo). On L2, projected gas per loan stays below interest earned; default stress is in Table 5.7.

5.2 Analysis of Design Solutions

Stack choices were scored on ecosystem maturity, zero-cost prototype constraints, and auditability (Table 4.9, Section 4.16). The selected path pairs EVM Solidity on zkEVM L2, PostgreSQL + FastAPI off-chain, and Chainlink Functions (with commit-reveal as an interim relay) for ML scores. The optional MCP agent and EIP-7702 session keys remain Phase III–IV extensions outside the MVT.

5.3 Final Design Adjustment

The final pre-thesis design locks four adjustments before implementation: (i) **stablecoin-first** retail lending (USDC/USDT, Section 1.8) to avoid ETH liability volatility; (ii) **L2 deployment** for micro-loan gas economics; (iii) **ML as decision support only**—human approvers and on-chain gates, no auto-disbursement; (iv) **MVT scope** limited to the three-tier lending core, database schema, and dashboard—syndicated/tranched modules, full regulatory filing, and the conversational agent are deferred (Table **??**).

5.4 Feasibility Analysis

5.4.1 Technical Feasibility

Each layer is specified against mature tooling; nothing requires bespoke infrastructure beyond standard testnets and open-source stacks (Table 5.1).

Table 5.1: Technical feasibility assessment

Component	Assessment	Evidence
Smart contracts	Feasible (specified)	Solidity; Hardhat; OpenZeppelin; Phases I–II
Frontend DApp	Feasible (specified)	React 18 + Wagmi/RainbowKit
Blockchain deployment	Feasible (planned)	Polygon zkEVM Cardona + Sepolia
AI/ML integration	Feasible with constraints	RF + IF + SHAP; Chainlink Functions Phase III
Optional agent	Feasible (deferred)	MCP + Qwen3-8B; Phase III–IV; outside MVT
Database backend	Feasible	PostgreSQL 20-entity schema (3NF); FastAPI

5.4.2 Economic Feasibility

The academic prototype can run at **zero direct cost**; scaled hosting is a separate pilot concern (Table 5.2). L2 gas keeps small loans viable versus mainnet; spread revenue and default scenarios are modelled in Section 5.5.

Table 5.2: Economic feasibility — zero-cost prototype

Cost Category	Estimate	Notes
Blockchain deployment	\$0	Public testnets
Frontend / backend hosting	\$0	Vercel / Render free tiers or localhost
AI/ML training	\$0	Local machine or Colab free tier
Development tools	\$0	Hardhat, VS Code, Git (open-source)
Total prototype	**\$0**	Pilot ops ≈\$500–\$2,000/month at scale

5.4.3 Risk Feasibility

Main delivery risks and mitigations are summarised in Table 5.3; smart-contract and regulatory items dominate severity.

Table 5.3: Risk taxonomy and mitigation

Risk Category	Description	Severity	Mitigation
Partner non-cooperation	Key partners decline to join	Medium	Testnet pilots; open-source replication
Smart contract vulnerability	Logic exploit	High	OpenZeppelin; audits; pause mechanism
Regulatory adversity	Jurisdiction blocks retail crypto	Medium	Testnet-only prototype; sandbox path
AI/ML model degradation	Fraud model drift	Low	Retraining; human-in-the-loop; SHAP review

5.4.4 Regulatory Feasibility

Retail stablecoin lending must align with MiCA, GENIUS Act, and EU AI Act expectations; CWB uses authorised stablecoins and keeps PII off-chain (Table 5.4). Full conformity files and national licensing remain future work; pre-thesis scope is testnet demonstration only.

Table 5.4: Regulatory design controls (selected)

Statute / Article	Requirement	CWB Control
MiCA Art. 16	Authorised EMT issuer	Uses USDC/USDT; does not issue
MiCA Art. 36	1:1 reserve backing	Issuer attestations before pool acceptance
EU AI Act (credit scoring)	High-risk AI from Aug. 2026	Human approver gate; SHAP + audit log
GDPR Art. 17	Right to erasure	PII off-chain only; hashes on-chain

5.5 Statistical Analysis

The tables below model spread revenue, ETH-price sensitivity, and default stress at full-deployment scale (planning assumptions only; not observed platform data). Interest tiers follow the 3%/5%/8% hierarchy (Figure 5.2).

Table 5.5: Revenue projection assumptions

Parameter	Value
Reference ETH price	$2,500 (planning mid-point, Feb 2026)
World Bank → National Bank	3% APR
National Bank → Local Bank	5% APR
Local Bank → Client	8% APR
Average loan term	12 months
Default rate provision (base)	8%
Origination fee	0.25% per disbursement

Table 5.6: ETH-price sensitivity of spread revenue

ETH Price	Loan Base	Spread Revenue	Incl. Fees (est.)
$2,000	$1.376B	$110.1M	$120–128M
$2,500 (base)	$1.720B	$137.6M	$150–159M
$3,000	$2.064B	$165.1M	$180–191M

Table 5.7: Default-rate sensitivity

Scenario	Default rate	Annual loss	Basis
Optimistic	3.7%	$7.4M	Mature MFI benchmarks
Base case	8%	$16M	New platform, limited history
Stress	15%	$30M	Thin credit history, volatile collateral

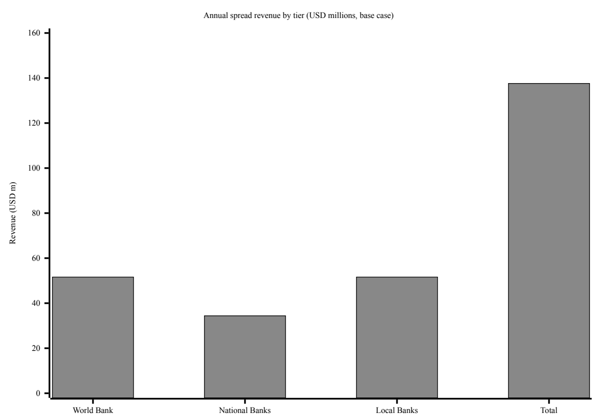

Figure 5.1: Annual revenue projection (10-year maturity)

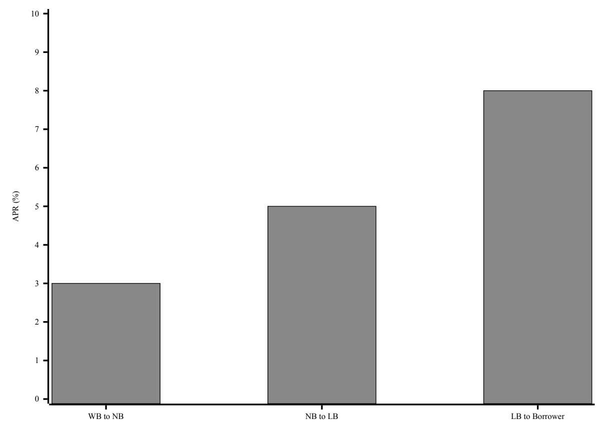

Figure 5.2: Hierarchical interest-rate spread (APR)

5.6 Comparisons and Relationships

Table 5.8: Market segments with supporting data

Segment	Description	Estimated Scale
TAM	Global DeFi lending; remittances; MSME gap [8, 15, 20]	\$55B – 5T+
SAM	Institutional hierarchical lending; emerging-market credit	\$5B – 15B
SOM	Testnet pilots; sandbox partnerships	\$50 – 200M

Table 5.9: Target customer segment profile

Characteristic	Description
Primary users	Tier 1–3 bank operators; Tier 4 retail via Local Banks
Loan range	Micro/SME at Local Bank tier; larger exposures via syndication
Key pain points	Opaque reserves; flat DeFi pools; weak explainability in on-chain lending

Table 5.10: Partner categories and roles

Partner Category	Role	Incentive
Financial regulators	Sandbox oversight	On-chain audit trails
Banking institutions	National/Local Bank nodes	Reserve access; lower settlement friction
Oracle providers	Chainlink DON, indexing	Unified oracle stack
Academic partners	ML / formal-verification review	Datasets; co-publication

No current protocol combines four-tier hierarchy, explainable ML governance, and programmable group lending. Table 5.11 contrasts representative projects on the dimensions that matter for CWB.

Table 5.11: Competitive feature comparison

Feature	Typical competitors	Crypto World Bank
Hierarchical lending tiers	Flat pools (Aave, Compound)	Four-tier World → Client flow
Governance-controlled rates	Algorithmic pool rates only	Tier-bounded contract parameters
AI-assisted risk scoring	Rare in DeFi lending	RF + SHAP + anomaly path to oracle
Retail + institutional access	Often one or the other	Retail via Local Bank; banks at upper tiers

5.7 Discussion

This pre-thesis specifies architecture and planned evaluation; deployment evidence, trained models, and formal proofs arrive after pre-thesis 2. Five examiner-facing limits are recorded explicitly:

- **Novelty is compound, not absolute**—Maple, Goldfinch, and Morpho cover parts of institutional credit; the claim is the four-tier reserve stack plus explainable oracle ML plus group liability in one open design.
- **BCCC is a baseline only**—flat-pool fraud labels may not transfer to hierarchical CWB flows without testnet retraining.
- **Testnet scope**—workshop-grade evidence needs benchmarks and reproducible repos; MVT deployment supplies the empirical layer.
- **Group lending**—on-chain cosigning is not social solidarity; BRAC/Grameen statistics inform design only.
- **Oracle and economics**—scores attach at origination; collusion and model drift remain open; Certora proofs state assumptions explicitly.

The prototype runs on public testnets with test tokens only. Jurisdiction-specific pilots (e.g. regulatory sandboxes) are future work; the architecture is global L2 institutional DeFi, not a single-country rollout claim.

Chapter 6

Conclusion

The Crypto World Bank addresses a **multi-party coordination and trust** problem at the heart of hierarchical development finance. Existing DeFi lending protocols collectively manage over $55 billion in TVL [8], yet they rely on flat, single-tier pool designs that cannot model how capital actually moves between multilateral institutions, national banks, local lenders, and clients. This work proposes a four-tier institutional hierarchy with cross-tier, same-tier, and upward lending flows, governed by transparent rules for network membership, business operations, and technology infrastructure.

The platform is grounded in the six core functions of a bank—deposit mobilization, credit allocation, payment and settlement, risk intermediation, liquidity management, and ancillary services—but encodes them as publicly verifiable on-chain rules rather than discretionary enforcement by a trusted central authority. Reserve ratios, interest adjustments, group consent, and the full loan book are enforced and auditable in real time. A further distinctive capability is **multi-entity co-lending and co-funding**: at the retail level, solidarity groups can pool collateral and borrow together under mutual liability, drawing on the structural logic of Grameen Bank and BRAC; at the institutional level, multiple banks can co-fund a single loan or share risk across senior and junior tranches—mechanisms that no existing DeFi protocol combines in one open architecture.

This thesis contributes four scoped advances—hierarchical lending architecture, programmable group lending, explainable AI-assisted credit governance, and privacy-preserving compliance identity—as detailed in Section 1.5:

- A four-tier DeFi architecture that mirrors multilateral development-finance capital flows, with reserve enforcement at every tier (Section 5.7)
- Programmable joint-liability group lending with over-indebtedness controls (Section 3.16.3)
- Oracle-mediated fraud and credit scoring with explainable outputs for approvers and clients (Sections 4.3.2, 4.3.1)
- A privacy-preserving identity and compliance pathway combining zero-knowledge proofs with decentralized identifiers (Section 3.7.1)

These contributions are positioned against the institutional-DeFi landscape in Section 2.2.6 and against the MiCA / GENIUS Act / EU AI Act frontier in Section 5.4.4.

Analysis of 20+ competing projects in DeFi lending, institutional credit, banking infrastructure, and financial inclusion confirms that no current system combines multi-level lending, interbank mechanisms, and graded borrower access in one decentralized architecture. Institutional adoption—from the World Bank FundsChain initiative [35] to JPMorgan Kinexys and R3 Corda's $17 billion in tokenized assets —shows that blockchain-based financial infrastructure is both technically feasible and institutionally demanded. The Crypto World Bank extends this trajectory from settlement and fund tracking into an open, hierarchically governed lending system at the intersection of institutional finance, decentralized lending, and financial inclusion—a combination still largely unaddressed in both academic literature and commercial practice.

At the pre-thesis stage, this report delivers a fully specified architecture, market and partnership framing, and phased implementation roadmap. The hierarchical lending workflow, tiered governance, capital-flow design, and planned explainable analytics layer are specified in depth; deposit products, group lending, multi-entity cross-tier operations, retail payment rails, formal verification, and runtime monitoring remain at the design level for final-thesis implementation per the Minimum Viable Thesis checklist (Section 4.1).

6.1 Future Work

The remaining work falls into three directions: completing the final-thesis deliverable, extending into everyday banking, and pursuing longer-term research.

Final thesis priorities.

- **Complete hierarchical lending.** End-to-end cross-tier fund transfers, interest accrual, installment scheduling, and cascading repayment across all four tiers (Appendix B).
- **Stablecoin-first deployment.** USDC at retail tiers with MiCA / GENIUS Act mapping validated against deployed configuration (Sections 1.8, 5.4.4).
- **Deposit and credit products.** Savings, fixed deposits, group lending with over-indebtedness controls, and liquidation handling (Sections 3.11, 3.16.3, 3.10).
- **Explainable analytics integration.** Connect fraud and credit scoring into live loan approvals via the oracle pipeline (Sections 4.3.2, 4.3.1).
- **Simulation, verification, and monitoring.** Testnet simulation, reserve invariant proofs, and live transparency dashboards (Sections 4.9, 4.7, 4.10).

Retail and everyday banking (post-MVT). Before the platform can support everyday spending with stablecoins—not only lending and installment servicing—it must close gaps in remittance, merchant acceptance, and cash-like deposit/withdrawal identified in Section 1.8 and grounded in the literature synthesis of Section 2.2.5:

- **Client-to-client remittance.** KYC-gated, compliance-checked transfers between registered clients under Local Bank sponsorship (Section 3.8.2).
- **Fiat on/off-ramp integration.** Licensed payment-service APIs so clients can fund balances and withdraw to local currency—the binding constraint identified by Du et al. [110] and CPMI.
- **Merchant checkout.** Invoice-based payment requests and settlement for registered merchants, evaluated against the CLEAR framework of Li et al..
- **Consumer protection.** Deposit insurance, dispute resolution, and retail FX conversion before off-ramp or cross-border remittance (Section 3.16).

No prior work composes these retail rails with hierarchical development-finance lending in one architecture (Section 2.2.5).

Longer-term research.

- **Graph-based fraud detection.** Hierarchical lending graph ablation on BCCC [83] (Section 4.5).
- **Federated learning across banks.** Privacy-preserving fraud detection benchmarked against FedFraud (Section 4.6).
- **Privacy-preserving compliance.** Zero-knowledge AML and KYC circuits with FATF Travel Rule support (Section 3.14).
- **Formal verification and domain adaptation.** Multi-tier reserve proofs and ML model transfer from flat-pool to hierarchical transaction data.
- **Regulatory and retail evaluation.** EU AI Act conformity for credit scoring and CLEAR-framework assessment of retail payment modules.

Appendix A

Database Schema Reference

Physical-schema detail for Chapter 3. B-tree indexes support rolling borrowing-limit windows, overdue installments, and agent audit queries (Table A.1). Core functional dependencies are in Table A.2; on-chain rules remain authoritative for limits and loan state.

A.1 Indexing Strategy

Table A.1: Indexing strategy

Index	Table / Column(s)	Rationale
`idx_loan_borrower`	`LOAN_REQUEST(borrower_id)`	Borrower loan history.
`idx_loan_status`	`LOAN_REQUEST(status)`	Active vs. closed loans.
`idx_installment_d`	`INSTALLMENT(due_date, status) WHERE status = 'PENDING'`	Overdue installment detection.
`idx_txn_created`	`TRANSACTION(created_at)`	Rolling borrowing-limit windows.
`idx_txn_borrower`	`TRANSACTION(borrower_id)`	Per-borrower history.
`idx_market_symbol`	`MARKET_DATA(symbol, recorded_at)`	Price feed time series.
`idx_loan_bank_sta`	`LOAN(local_bank_id, status, created_at DESC)`	Bank loan queues.
`idx_loan_client_ac`	`LOAN(client_id, status) WHERE status = 'ACTIVE'`	Open-loan count per client.
`idx_agent_client_`	`AGENT_ACTION_LOG(client_id, created_at DESC)`	Agent action history.
`idx_agent_status`	`AGENT_ACTION_LOG(status) WHERE status = 'PENDING'`	Pending agent actions.
`idx_aiml_score`	`AI_ML_LOG(anomaly_score DESC) WHERE anomaly_score > 0.5`	AML alert queue.
`idx_session_parent`	`SESSIONS(parent_session_id) WHERE parent_session_id IS NOT NULL`	Session lineage.

A.2 Functional Dependencies

Table A.2: Representative functional dependencies

Relation	Functional Dependency	Notes
LOAN_REQUEST	`request_id` → all attributes	Application primary key.
LOAN	`loan_request_id` → `loan_id`	1:1 request to disbursement.
BORROWING_LIMIT	`borrower_id` → limits, timestamps	1:1; enforced on-chain.
BORROWER	`wallet_address` → `borrower_id`, …	Unique wallet in prototype.
INSTALLMENT	`loan_id, installment_number` → amount, due, status	Composite key.
SESSIONS	`session_id` → client, scope, expiry	MCP session + EIP-7702 scope.
AGENT_ACTION_LO	`action_id` → session, tool, status, tx hash	Write-tool audit trail.
INTEREST_RATE_TIER	`tier_id` → rates, kink params	Normalised rate schedule.

Appendix B

Optional Agent Harness Reference

Optional Phase III–IV add-on only (Section 4.4). One shared Qwen3-8B model; personalisation comes from per-user context injected each turn:

```
{
 "client_id": "0xABC...", "credit_tier": "Silver",
 "active_loans": [{"loan_id": "L-4821", "next_due": "2026-06-15"}],
 "conversation_history": ["...last 10 turns..."]
}
```

Prompt assembly uses three tiers: **stable** (persona + tool schema), **context** (rates, KYC matrix), and **volatile** (live on-chain state + recent turns). Child sessions store a compression summary when the context window fills. Write tools require logged user confirmation; middleware checks session-key scope and AML flags before POST.

Table B.1: Agent toolsets

Toolset	Tools visible
`read_only`	Nine read tools (default).
`loan_actions`	Reads + loan / installment / group writes.
`account_managemen`	Reads + deposit, KYC, reminder writes.

Appendix C

Technology Stack

The optional in-product assistant uses a local Qwen3-8B endpoint behind the same Express.js API as the React UI. Figures C.1 and C.2 show the request path; Table C.1 lists the full stack.

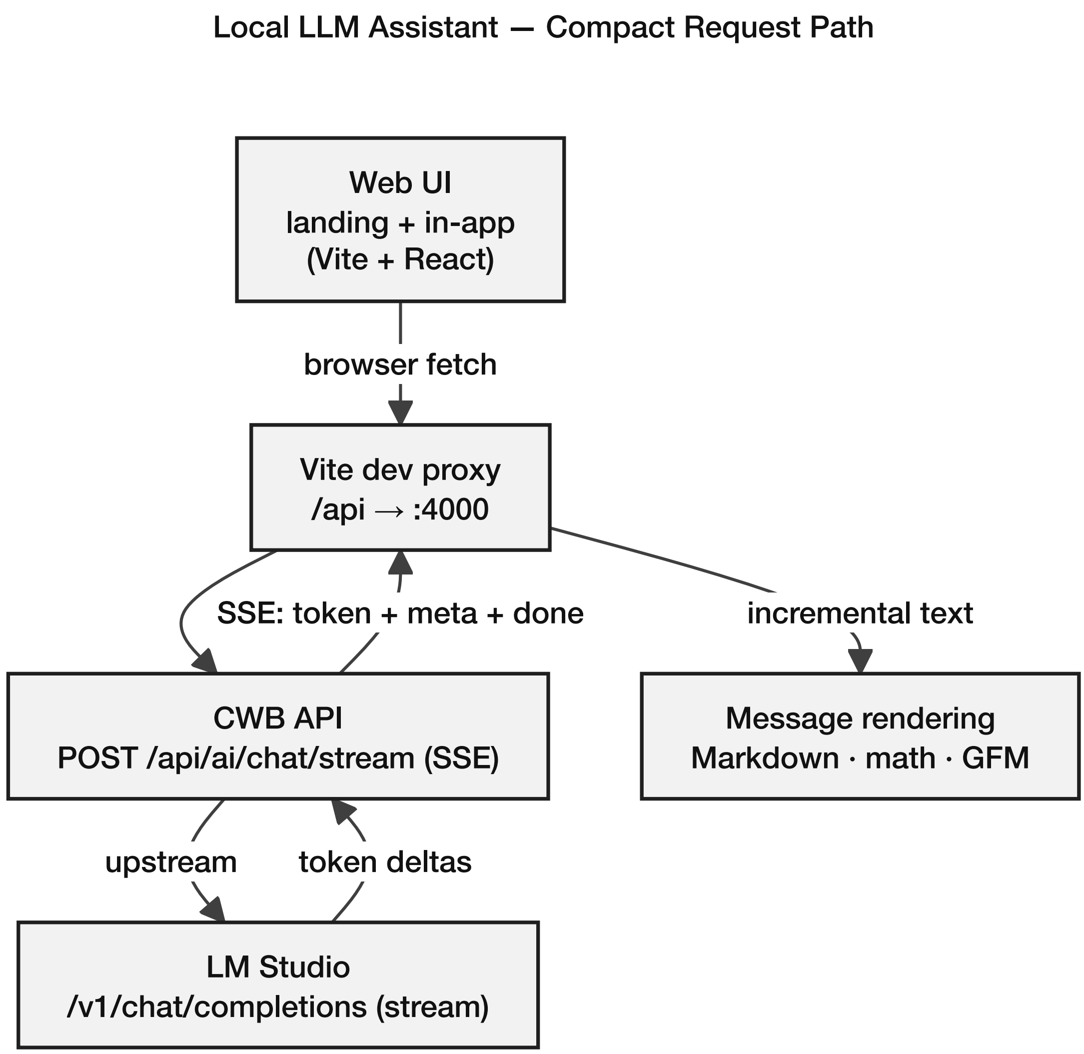

Figure C.1: Optional AI banking agent request path (add-on)

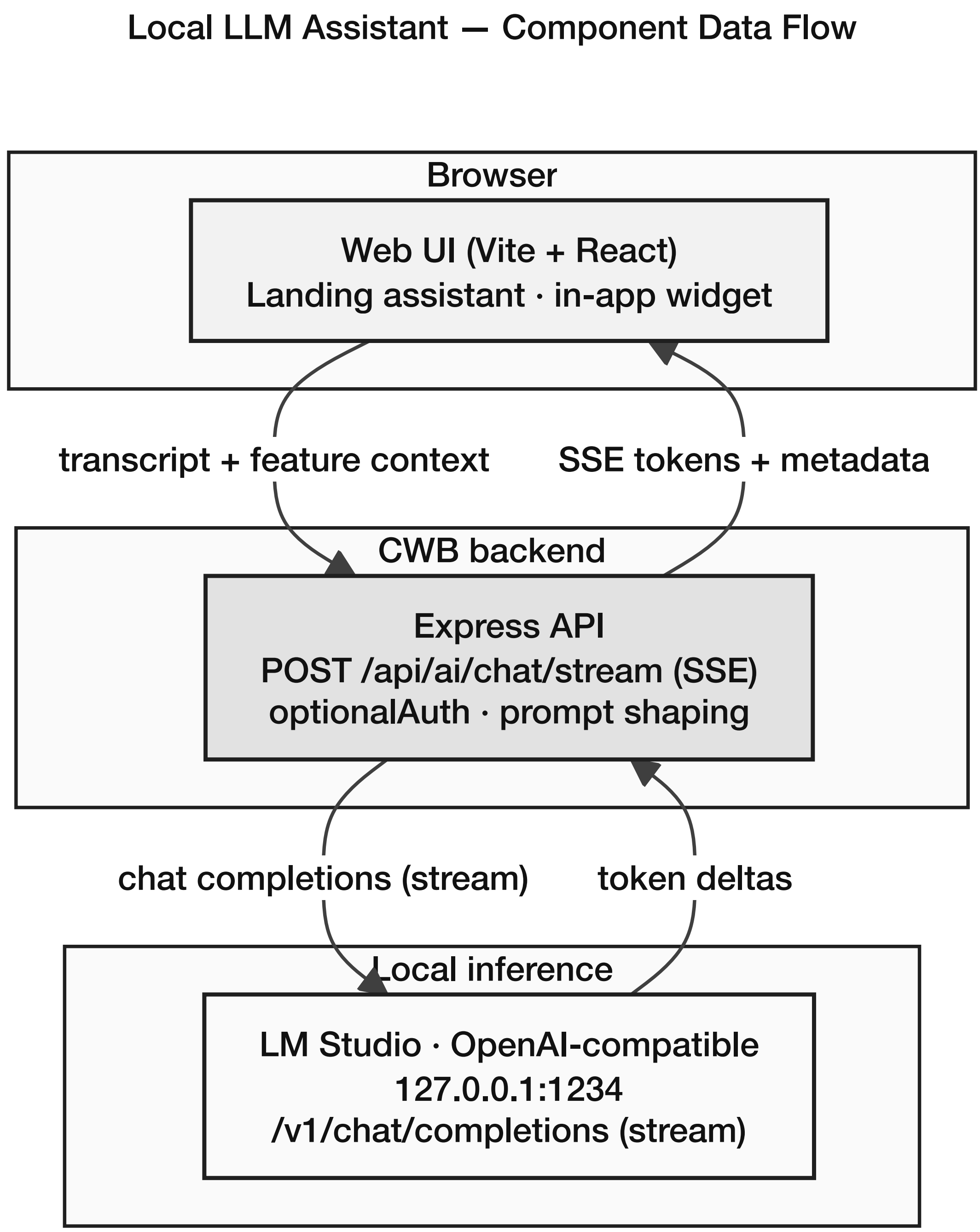

Figure C.2: Optional AI agent data flow

Table C.1: Technology stack summary

Layer	Technology	Notes
Smart Contract	Solidity 0.8.20, OpenZeppelin	ReentrancyGuard, AccessControl
Frontend	React 18, TypeScript, MUI	Wagmi / RainbowKit / Viem
Backend	Express.js, PostgreSQL, FastAPI	REST + ML scoring service
Network	Polygon zkEVM Cardona	Sepolia fallback
AI (optional)	Qwen3-8B, MCP (17 tools)	Human confirmation on writes
Oracles	Chainlink Functions, Feeds, PoR, Automation	ML scores + rates + upkeep
Indexing	The Graph subgraph	Dashboard / audit queries

Appendix D

Smart Contract Capabilities

Fifteen modular contracts: three core (Phase I), six banking products (Phase II), six multi-entity modules (Phases II–III). Appendix D shows the Tier 1 public interface.

Table D.1: Contract inventory by phase

Phase	Contract	Role
I	WorldBankReserve, NationalBank, LocalBank	Reserve, allocation, loan lifecycle, `freezeAccount`
II	SavingsVault, FixedDeposit, GroupLendingPool, FXModule, InsuranceFund, CurrentAccount	Deposits, groups, FX, insurance, transactions
II–III	InterBankLendingPool, UpwardDepositFacility, SyndicatedLoan	Same-tier liquidity, upward flow, syndication
III	TranchedPool, TreasurySwap, NettingEngine	Tranching, treasury FX, multilateral netting

Appendix C: Planned Testnet Deployment Manifest

Planned Polygon zkEVM Cardona manifest (Phase IV). Addresses are filled after deployment under `deployments/cardona/`; nothing is live at pre-thesis submission.

Table D.2: Planned Cardona testnet deployment manifest

Contract	Network	Address / manifest path
WorldBankReserve	Polygon zkEVM Cardona	`deployments/cardona/WorldBankReserve.json`
NationalBank (Scaffold)	Polygon zkEVM Cardona	`deployments/cardona/NationalBank.json`
LocalBank (Scaffold)	Polygon zkEVM Cardona	`deployments/cardona/LocalBank.json`
LendingController	Polygon zkEVM Cardona	`deployments/cardona/LendingController.json`

Appendix D: WorldBankReserve Contract Interface

Tier 1 public API (Phase I target). `nonReentrant` guards value transfers; `AccessControl` maps RBAC roles; events feed the off-chain indexer and ML pipeline.

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

import "@openzeppelin/contracts/security/ReentrancyGuard.sol";
import "@openzeppelin/contracts/access/AccessControl.sol";

/// @title WorldBankReserve
/// @notice Tier 1 reserve contract: manages global capital allocation
/// to registered National Bank addresses.
contract WorldBankReserve is ReentrancyGuard, AccessControl {

 bytes32 public constant NATIONAL_BANK_ROLE = keccak256("NATIONAL_BANK_ROLE");
 bytes32 public constant AUDITOR_ROLE = keccak256("AUDITOR_ROLE");

 uint256 public totalReserve;
 uint256 public minimumReserveRatio; // e.g. 1000 = 10.00%
 mapping(address => uint256) public allocatedTo; // NationalBank => amount

 event CapitalAllocated(address indexed nationalBank, uint256 amount);
 event RepaymentReceived(address indexed nationalBank, uint256 amount);
 event ReserveRatioUpdated(uint256 newRatio);

 /// @notice Allocate capital downward to a registered National Bank (CEI pattern)
 function allocateCapital(address nationalBank, uint256 amount)
 external onlyRole(DEFAULT_ADMIN_ROLE) nonReentrant;

 /// @notice Record repayment from a National Bank (CEI pattern)
 function recordRepayment(uint256 amount)
 external onlyRole(NATIONAL_BANK_ROLE) nonReentrant;

 /// @notice Query current utilization ratio (scaled 1e4)
 function utilizationRate external view returns (uint256);

 /// @notice Returns true if reserve ratio constraint is satisfied
 function isReserveAdequate external view returns (bool);

 /// @notice Returns a four-value reserve summary for Proof of Reserve
 /// verification and the Reserve Transparency Dashboard.
 /// @return totalDeposited Gross deposits ever made to this reserve tier
 /// @return totalLoaned Outstanding principal lent to National Banks
 /// @return reserveRatio (totalDeposited - totalLoaned) * 1e4 / totalDeposited
 /// (e.g. 5000 = 50.00%)
 /// @return insuranceFundBalance Current balance of the InsuranceFund contract
 function getReserveSummary external view returns (
 uint256 totalDeposited,
 uint256 totalLoaned,
 uint256 reserveRatio,
 uint256 insuranceFundBalance
 );
}
```

Listing D.1: WorldBankReserve.sol – Tier 1 public interface. Full implementation in project repository.

Appendix E: Internal Industry Research Notes

Internal notes in `Documentation/2nd Phase (Bokhtiar)/Improvements/`; cited formally as – where used in the body.

Table D.3: Internal industry research corpus

ID	Document	Thesis use
IR-1	`Binance_FTX_Full_Report.md`	Exchange vs. bank model; token-risk lesson
IR-2	`blockchain platforms research.csv`	Exchange platform comparison
IR-3	`Binance Business Analysis.md`	Revenue taxonomy reference
IR-4	`binance_software_engineering_architecture.md`	Nine-actor / sequence-diagram conventions
IR-5	`binance_architecture_research.md`	Off-chain stack patterns (background)

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